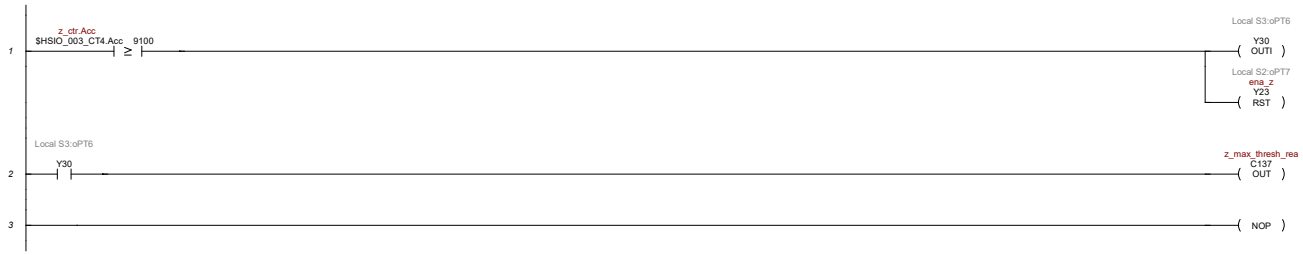
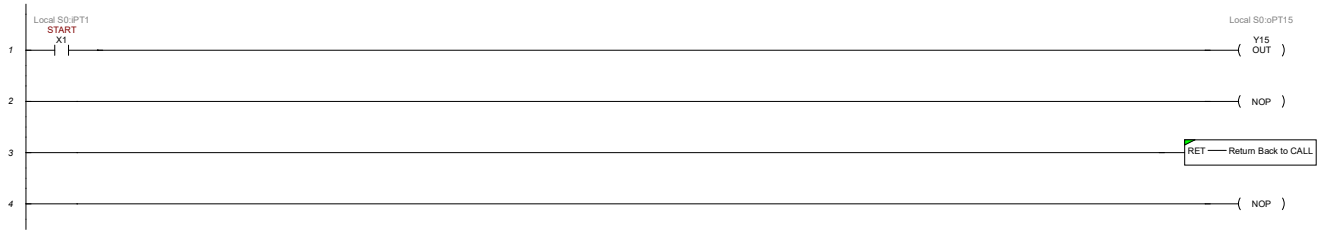
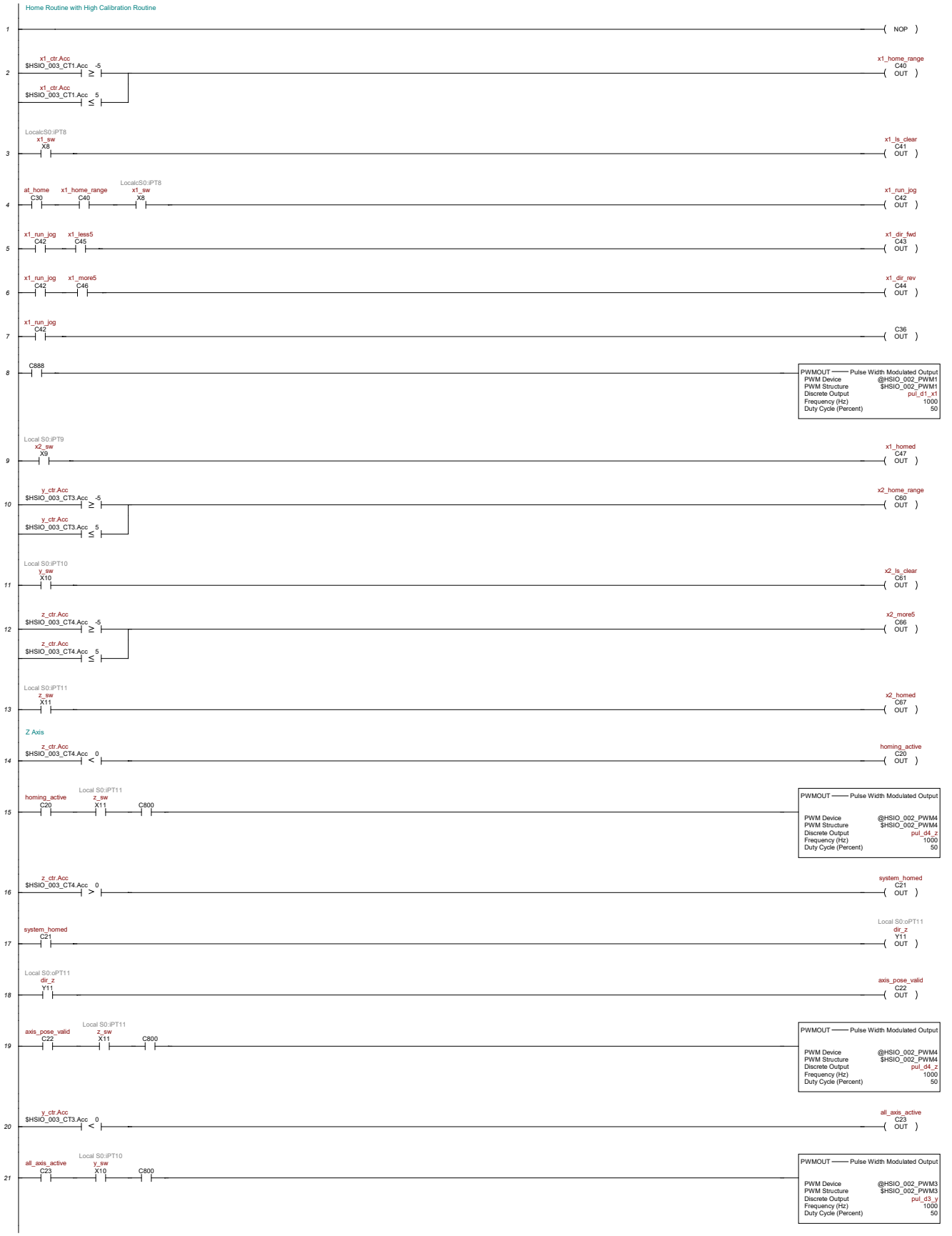
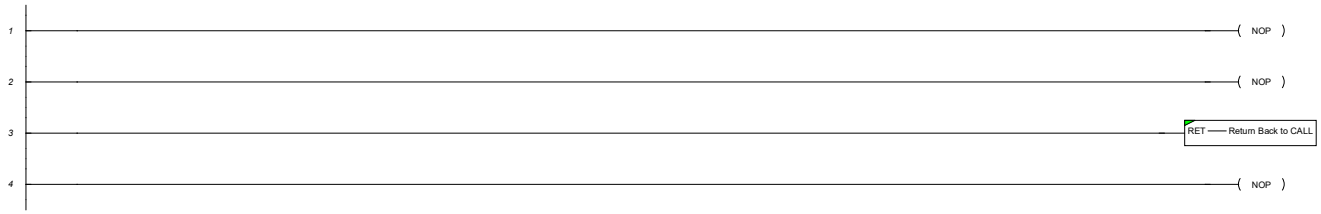
Project Path: C:\Users\Public\Documents\Do-more\Designer2_11\Projects\~DmD11.dmd
Save Date: 09/04/26 14:14:07
Creation Date: 09/04/26 14:14:07
PLC Type: BX-DM1E-x
Class ID: Do-more BRX Series
Link Name: GANTRY_BRX
Description: This is the calibration routine for the Gantry Robotics Project
Version: 1.0
Company: UIdaho
Department: Computer Science
Programmer: Davis Onyeoguzoro

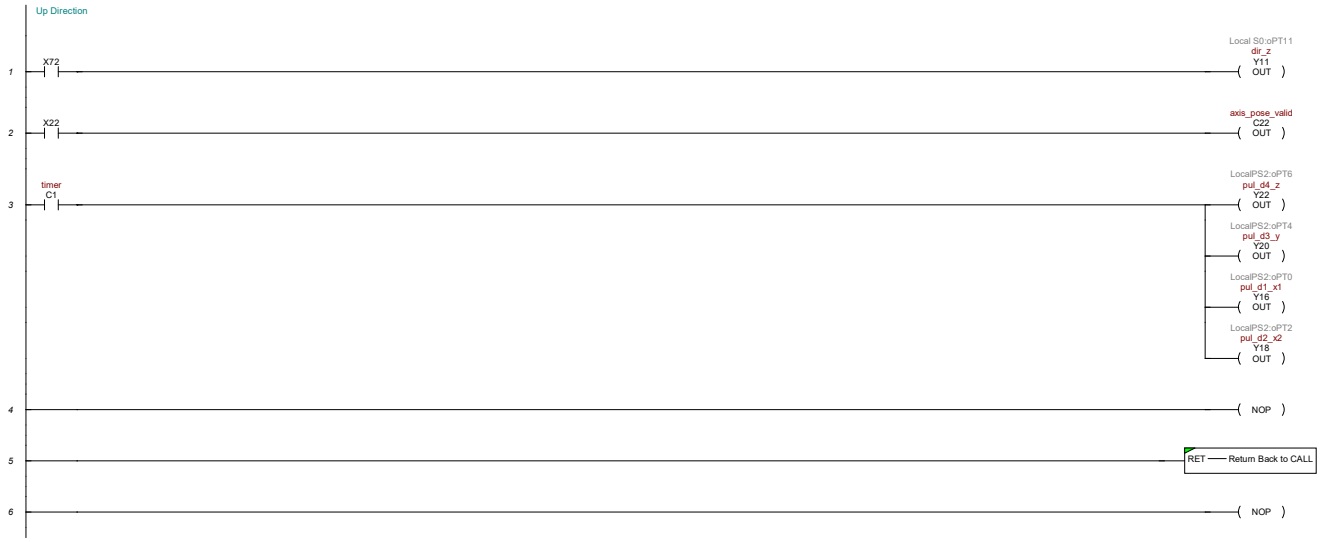

1
z_ctr.Acc
$HSIO_003_CT4.Acc
≥
9100
Local S3:oPT6
Y30
OUTI
Local S2:oPT7
ena_z
Y23
RST
2
Local S3:oPT6
Y30
z_max_thresh_rea
C137
OUT
3
NOP


Local S0:iPT1
START
X1
Local S0:oPT15
Y15
OUT
NOP
RET
Return Back to CALL
NOP

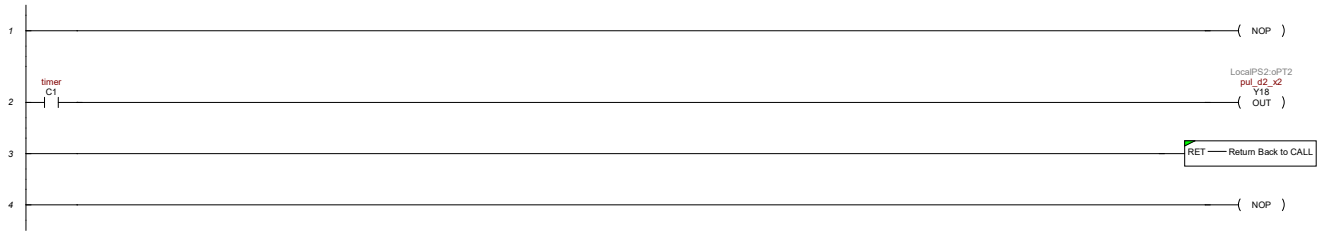

Home Routine with High Calibration Routine
NOP
x1_ctr.Acc
$HSIO_003_CT1.Acc -5
x1_home_range
C40
OUT
x1_ctr.Acc
$HSIO_003_CT1.Acc 5
LocalcS0:iPT8
x1_sw
X8
x1_ls_clear
C41
OUT
at_home
C30
x1_home_range
C40
LocalcS0:iPT8
x1_sw
X8
x1_run_jog
C42
OUT
x1_run_jog
C42
x1_less5
C45
x1_dir_fwd
C43
OUT
x1_run_jog
C42
x1_more5
C46
x1_dir_rev
C44
OUT
x1_run_jog
C42
C36
OUT
C888
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 1000
Duty Cycle (Percent) 50
Local S0:iPT9
x2_sw
X9
x1_homed
C47
OUT
y_ctr.Acc
$HSIO_003_CT3.Acc -5
x2_home_range
C60
OUT
y_ctr.Acc
$HSIO_003_CT3.Acc 5
Local S0:iPT10
y_sw
X10
x2_ls_clear
C61
OUT
z_ctr.Acc
$HSIO_003_CT4.Acc -5
x2_more5
C66
OUT
z_ctr.Acc
$HSIO_003_CT4.Acc 5
Local S0:iPT11
z_sw
X11
x2_homed
C67
OUT
Z Axis
z_ctr.Acc
$HSIO_003_CT4.Acc 0
homing_active
C20
OUT
homing_active
C20
Local S0:iPT11
z_sw
X11
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM4
PWM Structure $HSIO_002_PWM4
Discrete Output pul_d4_z
Frequency (Hz) 1000
Duty Cycle (Percent) 50
z_ctr.Acc
$HSIO_003_CT4.Acc 0
system_homed
C21
OUT
system_homed
C21
Local S0:oPT11
dir_z
Y11
OUT
Local S0:oPT11
dir_z
Y11
axis_pose_valid
C22
OUT
axis_pose_valid
C22
Local S0:iPT11
z_sw
X11
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM4
PWM Structure $HSIO_002_PWM4
Discrete Output pul_d4_z
Frequency (Hz) 1000
Duty Cycle (Percent) 50
y_ctr.Acc
$HSIO_003_CT3.Acc 0
all_axis_active
C23
OUT
all_axis_active
C23
Local S0:iPT10
y_sw
X10
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 1000
Duty Cycle (Percent) 50


22
y_ctr.Acc
$HSIO_003_CT3.Acc > 0
any_axis_active
C24
OUT
23
any_axis_active
C24
Local S0:oPT10
dir_y
Y10
OUT
24
Local S0:oPT10
dir_y
Y10
Local S0:iPT10
y_sw
X10
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 1000
Duty Cycle (Percent) 50
25
N1 < -5
mode_home
C25
OUT
26
mode_home
C25
Local S0:iPT9
x2_sw
X9
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 1000
Duty Cycle (Percent) 50
27
N1 > 5
mode_start
C26
OUT
28
mode_start
C26
Local S0:oPT9
dir_x2
Y9
OUT
29
Local S0:oPT9
dir_x2
Y9
Local S0:iPT9
x2_sw
X9
C800
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 700
Duty Cycle (Percent) 50
30
NOP
31
RET — Return Back to CALL
32
NOP


1
NOP
2
NOP
3
RET —— Return Back to CALL
4
NOP


Up Direction
1
X72
Local:$0:oPT11
dir_z
Y11
OUT
2
X22
axis_pose_valid
C22
OUT
3
timer
C1
LocalPS2:oPT6
pul_d4_z
Y22
OUT
LocalPS2:oPT4
pul_d3_y
Y20
OUT
LocalPS2:oPT0
pul_d1_x1
Y16
OUT
LocalPS2:oPT2
pul_d2_x2
Y18
OUT
4
NOP
5
RET — Return Back to CALL
6
NOP

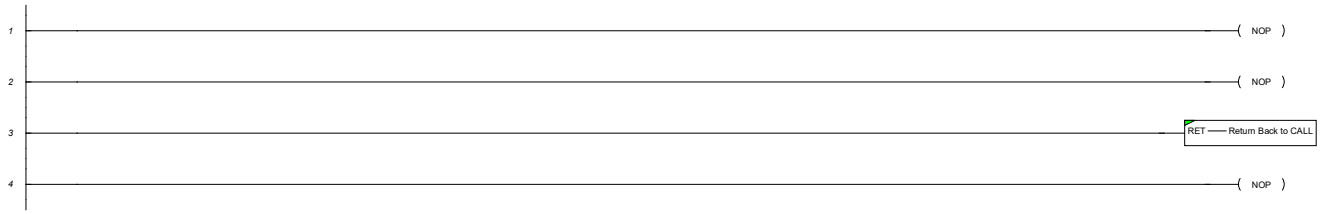

1
NOP
2
timer
C1
LocalPS2:oPT0
pul_d1_x1
Y16
OUT
3
x1_ctr.Acc
$HSIO_003_CT1.Acc
>
0
homing_active
C20
OUT
4
homing_active
C20
Local S2:oPT1
ena_x1
Y17
OUT
5
NOP
6
NOP
7
RET —— Return Back to CALL
8
NOP

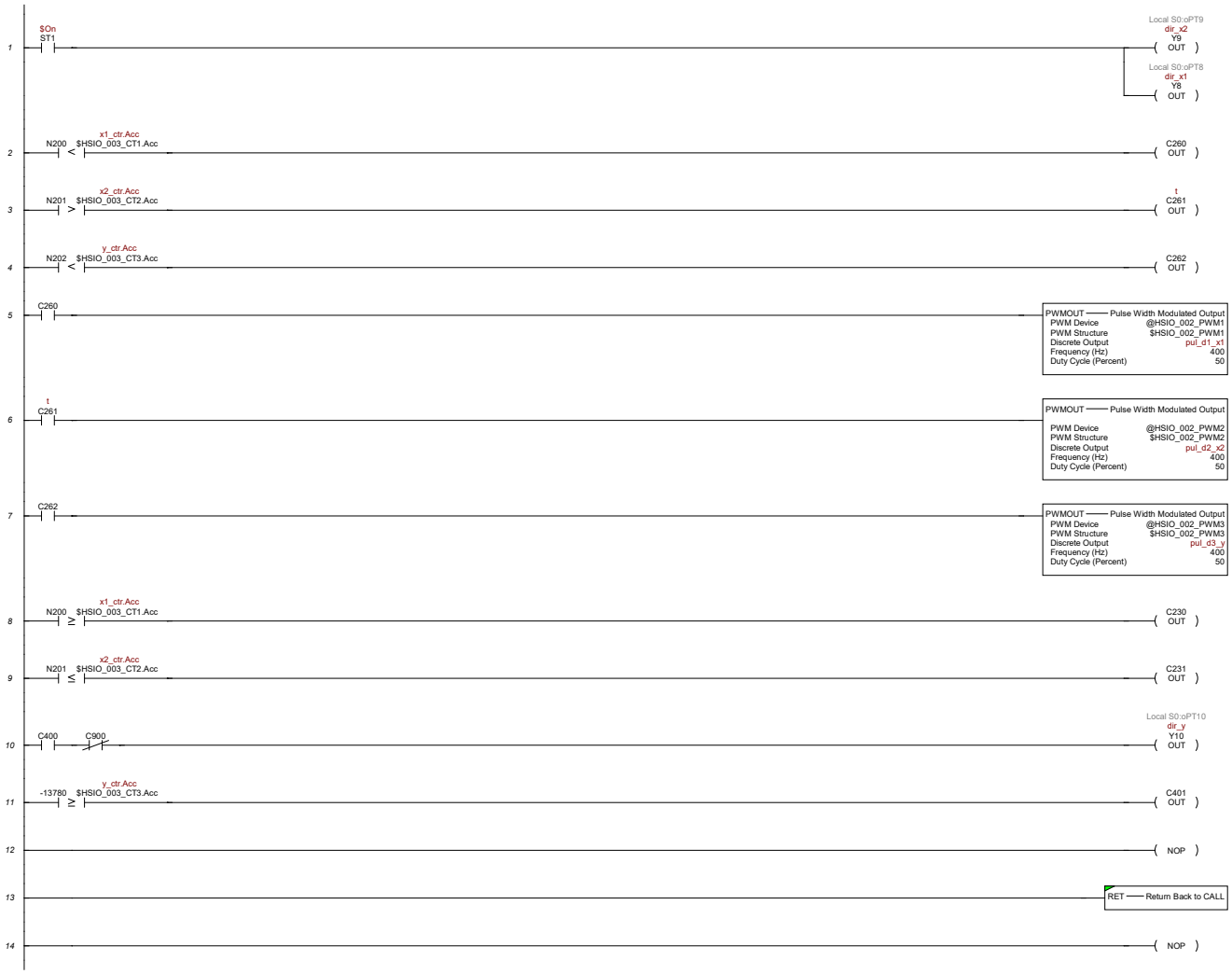

1
NOP
timer
C1
2
LocalPS2:oPT2
pul_d2_x2
Y18
OUT
3
RET
Return Back to CALL
4
NOP

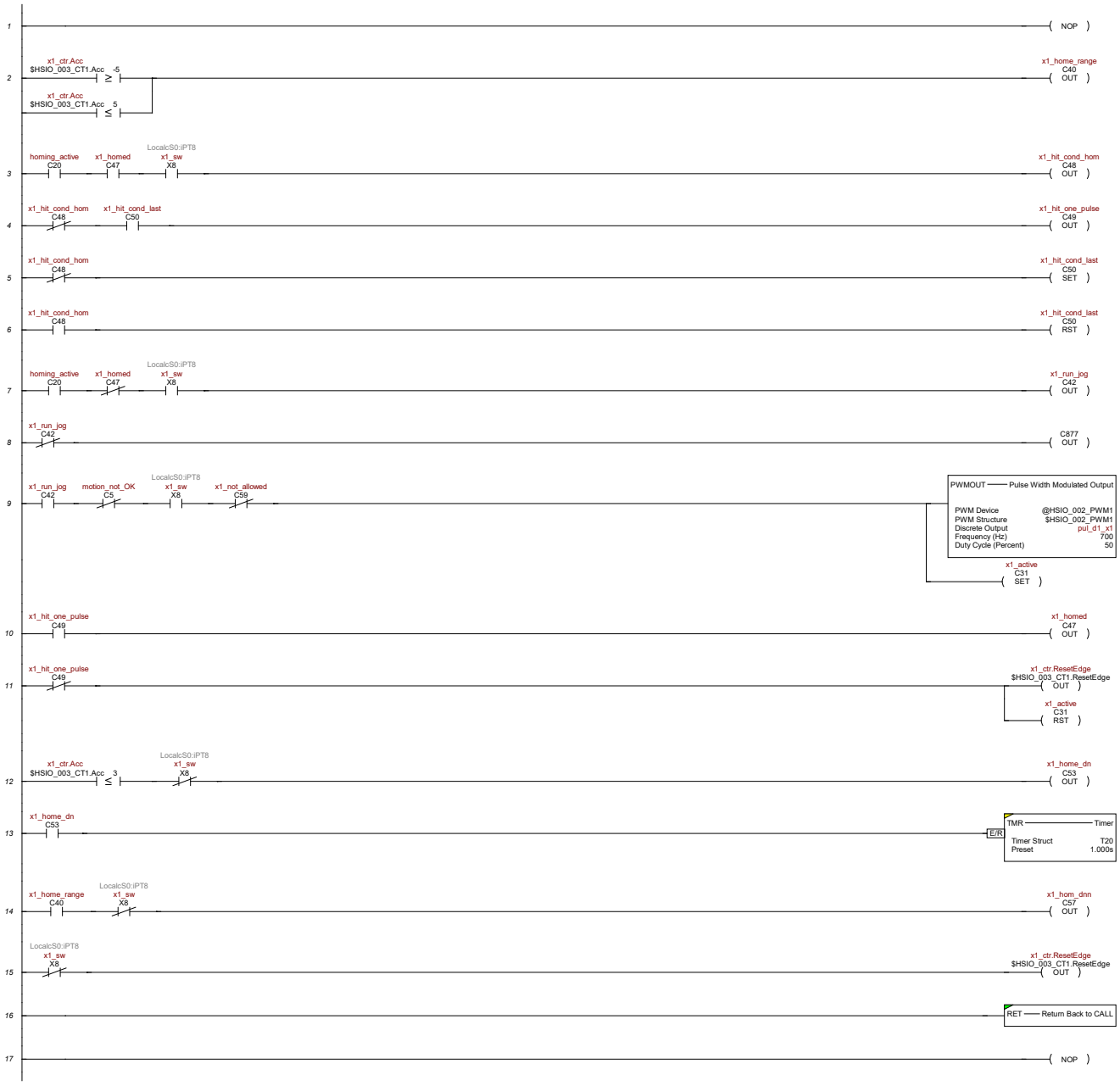

1
NOP
2
NOP
timer
C1
3
LocalPS2:oPT4
pul_d3_y
Y20
OUT
4
RET Return Back to CALL
5
NOP

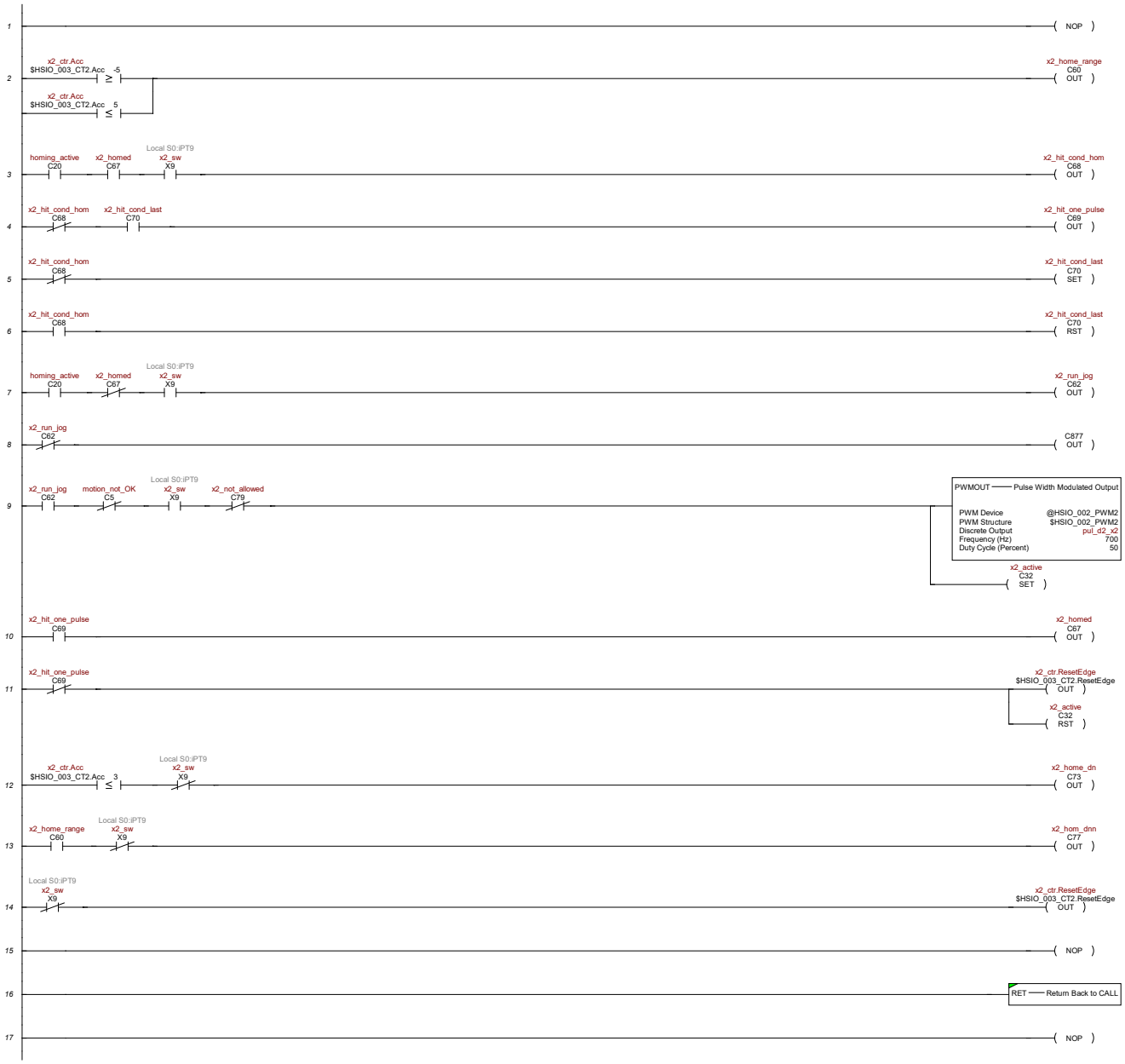

1
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT1
High Speed Input Structure x1_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X48 (Hs)
Count B Input X49 (Hs)
2
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT2
High Speed Input Structure x2_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X50 (Hs)
Count B Input X51 (Hs)
3
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT3
High Speed Input Structure y_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X52 (Hs)
Count B Input X53 (Hs)
4
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT4
High Speed Input Structure z_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X54 (Hs)
Count B Input X55 (Hs)
5
NOP
6
x_reset
C10
x1_ctr.ResetLevel
$HSIO_003_CT1.ResetLevel
OUT
x_reset
C10
x2_ctr.ResetLevel
$HSIO_003_CT2.ResetLevel
OUT
7
y_reset
C11
y_ctr.ResetLevel
$HSIO_003_CT3.ResetLevel
OUT
8
z_reset
C12
z_ctr.ResetLevel
$HSIO_003_CT4.ResetLevel
OUT
9
$On
ST1
COPY
Copy Data
Source Destination Count
1 x1_ctr.Acc N0 1
2 x2_ctr.Acc N1 1
3 y_ctr.Acc N2 1
4 z_ctr.Acc N3 1
10
$On
ST1
MATH
Calculate Expression
Result GantEnc0.X1_Enc_X1
Expression x1_ctr.Acc / 10
11
RET
Return Back to CALL
12
NOP

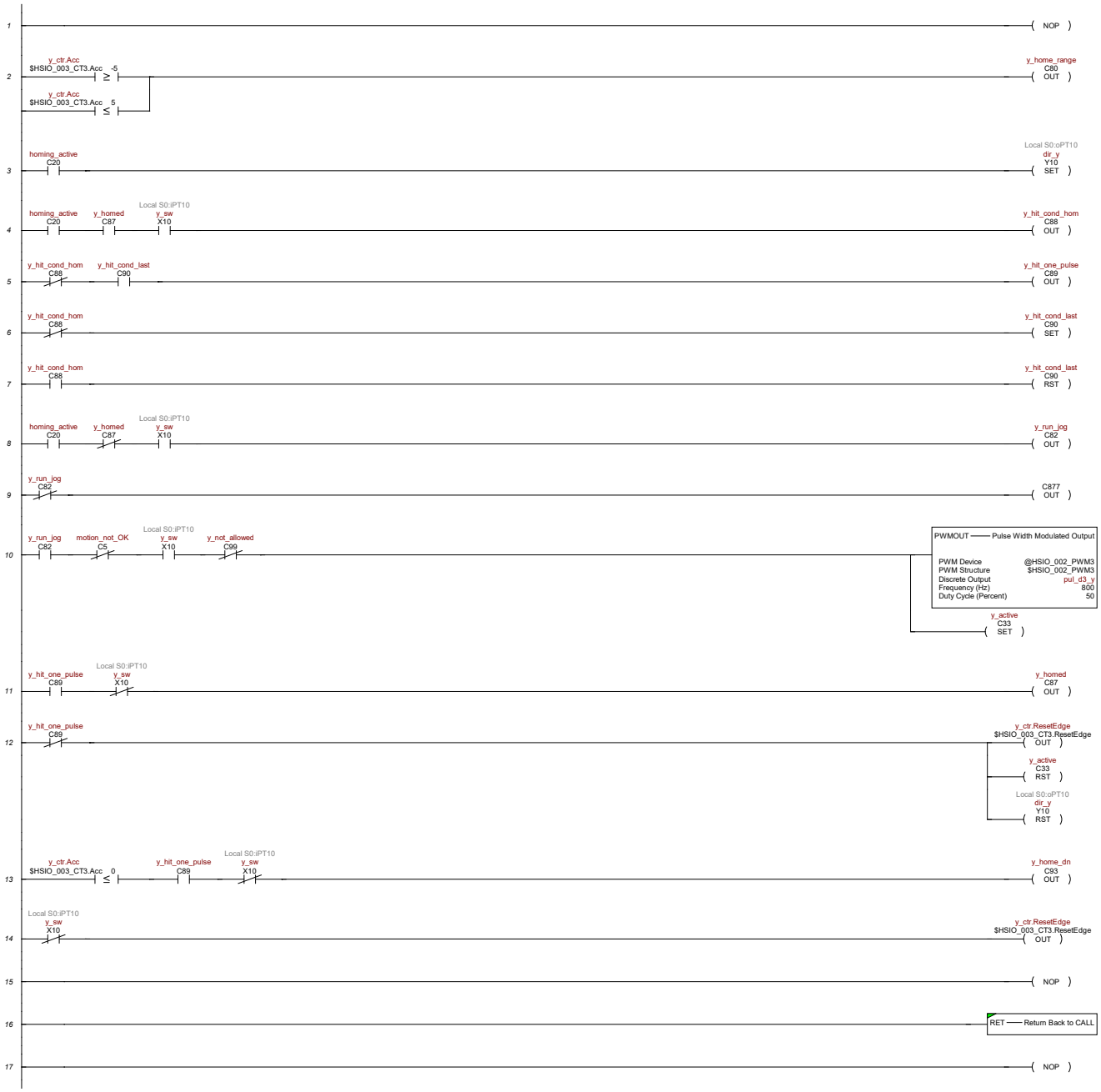

1
$Main.Running
Local S0:oPT3
green
Y3
OUT
2
system_ready
C16
pick_stg
C211
drop_stg
C139
Local S0:oPT2
white
Y2
OUT
Local S0:oPT3
green
Y3
RST
3
HOME_STG.Running
True_Calib_STG.Running
Local S0:oPT1
blue
Y1
OUT
Local S0:oPT3
green
Y3
RST
4
y_run_jog
C82
z_run_jog
C102
Local S0:oPT4
yellow
Y4
OUT
Local S0:oPT3
green
Y3
RST
5
Local S0:iPT0
ESTOP
X0
Local S0:iPT2
STOP
X2
Local S0:oPT0
red
Y0
OUT
Local S0:oPT3
green
Y3
RST
Local S0:oPT1
blue
Y1
RST
Local S0:oPT2
white
Y2
RST
Local S0:oPT4
yellow
Y4
RST
Local S0:oPT5
beep_sound
Y5
OUT
6
NOP
7
NOP
8
RET —— Return Back to CALL
9
NOP

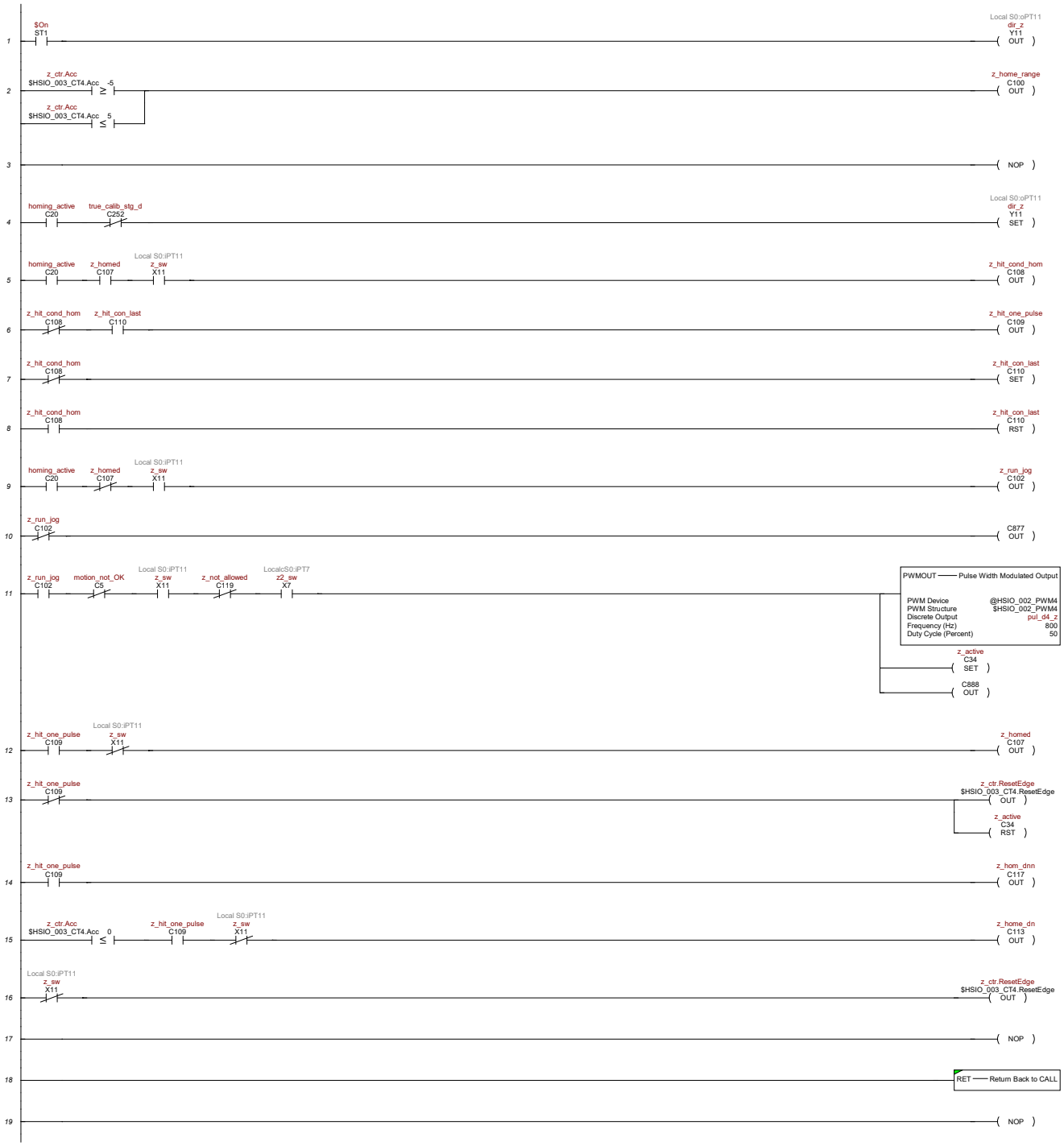

1
NOP
2
NOP
3
RET —— Return Back to CALL
4
NOP

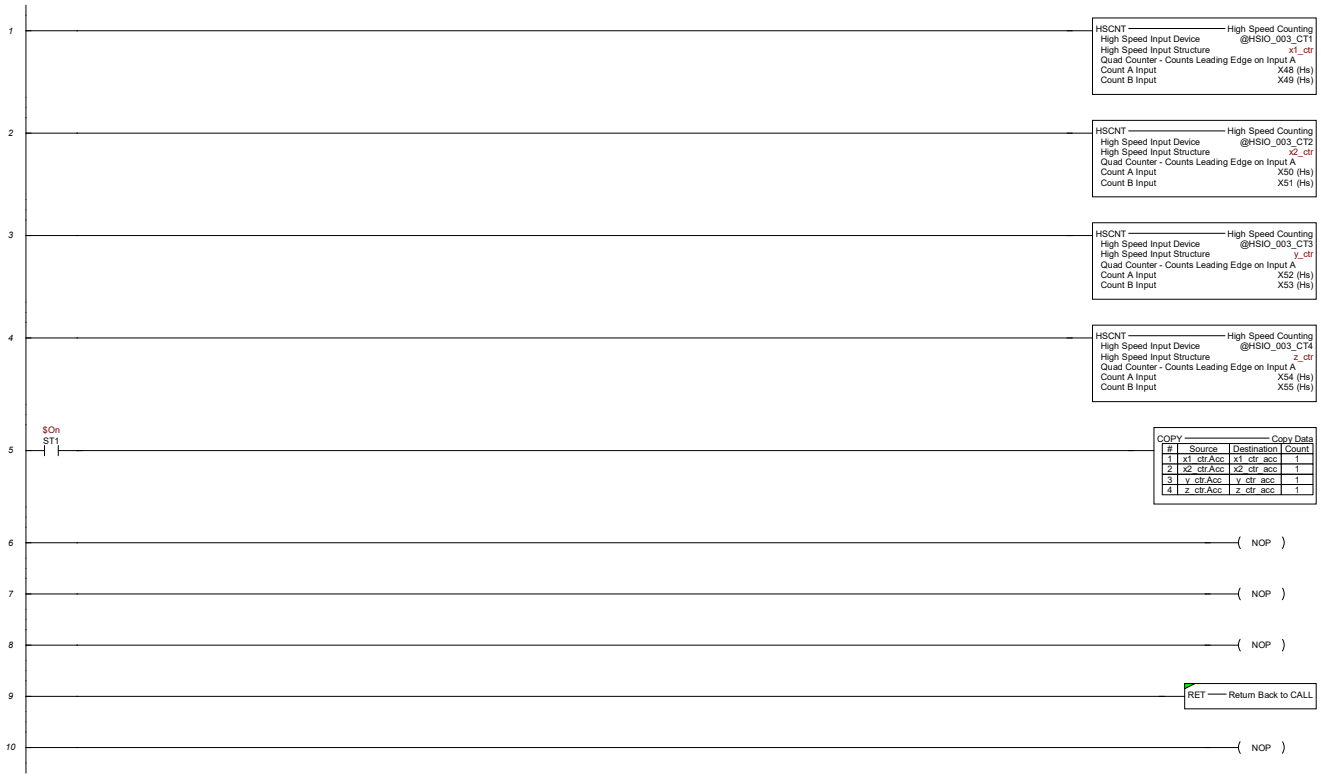

1
$On
ST1
Local S0:oPT9
dir_x2
Y9
OUT
Local S0:oPT8
dir_x1
Y8
OUT
2
x1_ctr.Acc
N200 < $HSIO_003_CT1.Acc
C260
OUT
3
x2_ctr.Acc
N201 > $HSIO_003_CT2.Acc
t
C261
OUT
4
y_ctr.Acc
N202 < $HSIO_003_CT3.Acc
C262
OUT
5
C260
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 400
Duty Cycle (Percent) 50
6
t
C261
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 400
Duty Cycle (Percent) 50
7
C262
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 400
Duty Cycle (Percent) 50
8
x1_ctr.Acc
N200 ≥ $HSIO_003_CT1.Acc
C230
OUT
9
x2_ctr.Acc
N201 ≤ $HSIO_003_CT2.Acc
C231
OUT
10
C400
C900
Local S0:oPT10
dir_y
Y10
OUT
11
y_ctr.Acc
-13780 ≥ $HSIO_003_CT3.Acc
C401
OUT
12
NOP
13
RET — Return Back to CALL
14
NOP


1
NOP
2
x1_ctr.Acc
$HSIO_003_CT1.Acc
≥
-5
x1_ctr.Acc
$HSIO_003_CT1.Acc
≤
5
x1_home_range
C40
OUT
3
homing_active
C20
x1_homed
C47
LocalcS0:iPT8
x1_sw
X8
x1_hit_cond_hom
C48
OUT
4
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
x1_hit_one_pulse
C49
OUT
5
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
SET
6
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
RST
7
homing_active
C20
x1_homed
C47
LocalcS0:iPT8
x1_sw
X8
x1_run_jog
C42
OUT
8
x1_run_jog
C42
C877
OUT
9
x1_run_jog
C42
motion_not_OK
C5
LocalcS0:iPT8
x1_sw
X8
x1_not_allowed
C59
PWMOUT
Pulse Width Modulated Output
PWM Device
@HSIO_002_PWM1
PWM Structure
$HSIO_002_PWM1
Discrete Output
pul_d1_x1
Frequency (Hz)
700
Duty Cycle (Percent)
50
x1_active
C31
SET
10
x1_hit_one_pulse
C49
x1_homed
C47
OUT
11
x1_hit_one_pulse
C49
x1_ctr.ResetEdge
$HSIO_003_CT1.ResetEdge
OUT
x1_active
C31
RST
12
x1_ctr.Acc
$HSIO_003_CT1.Acc
≤
3
LocalcS0:iPT8
x1_sw
X8
x1_home_dn
C53
OUT
13
x1_home_dn
C53
E/R
TMR
Timer
Timer Struct
T20
Preset
1.000s
14
x1_home_range
C40
LocalcS0:iPT8
x1_sw
X8
x1_hom_dnn
C57
OUT
15
LocalcS0:iPT8
x1_sw
X8
x1_ctr.ResetEdge
$HSIO_003_CT1.ResetEdge
OUT
16
RET
Return Back to CALL
17
NOP

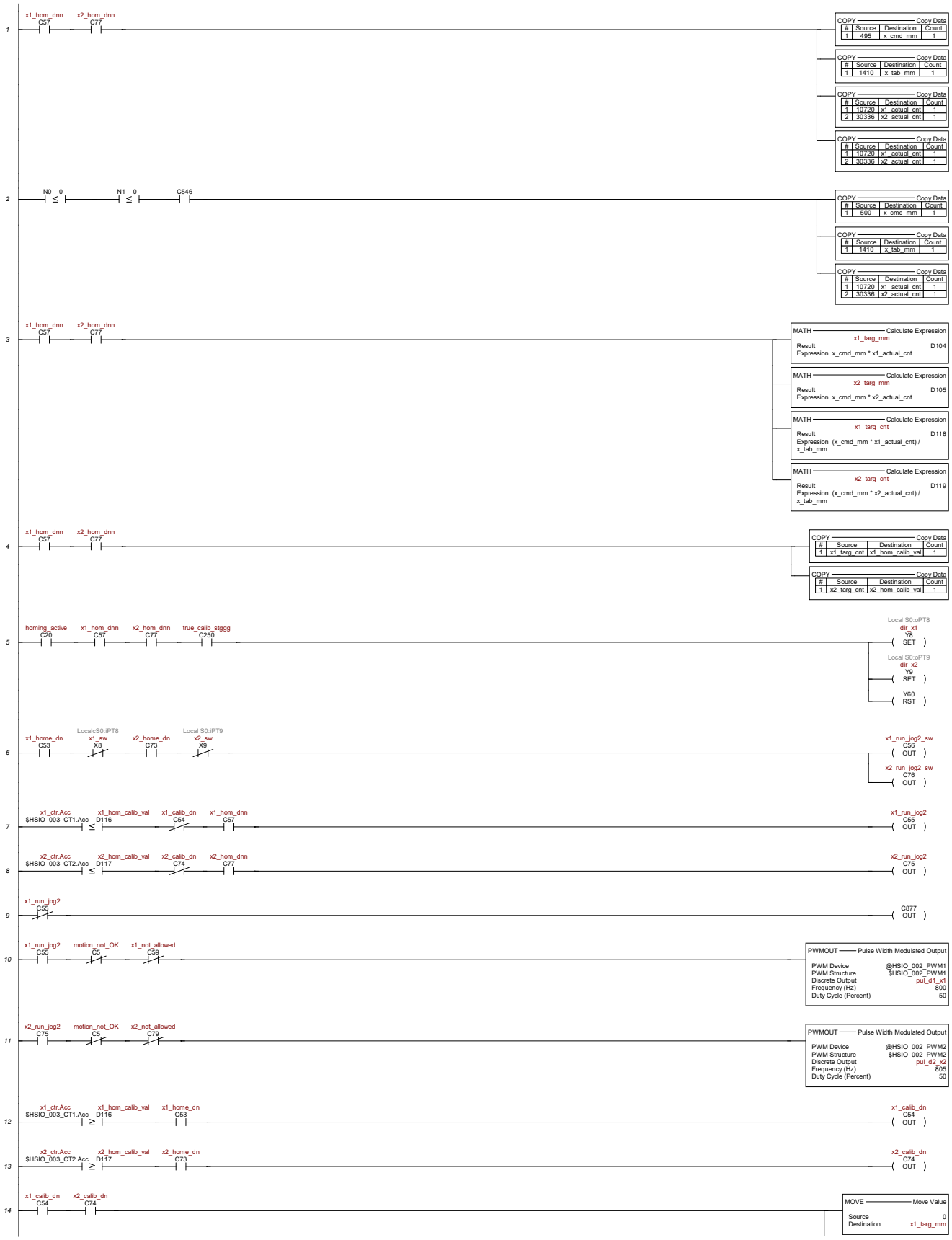

1
NOP
2
x2_ctr.Acc
$HSIO_003_CT2.Acc ≥ -5
x2_ctr.Acc
$HSIO_003_CT2.Acc ≤ 5
x2_home_range
C60
OUT
3
homing_active
C20
x2_homed
C67
Local S0:iPT9
x2_sw
X9
x2_hit_cond_hom
C68
OUT
4
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
x2_hit_one_pulse
C69
OUT
5
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
SET
6
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
RST
7
homing_active
C20
x2_homed
C67
Local S0:iPT9
x2_sw
X9
x2_run_jog
C62
OUT
8
x2_run_jog
C62
C877
OUT
9
x2_run_jog
C62
motion_not_OK
C5
Local S0:iPT9
x2_sw
X9
x2_not_allowed
C79
PWMOUT
Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output put_d2_x2
Frequency (Hz) 700
Duty Cycle (Percent) 50
x2_active
C32
SET
10
x2_hit_one_pulse
C69
x2_homed
C67
OUT
11
x2_hit_one_pulse
C69
x2_ctr.ResetEdge
$HSIO_003_CT2.ResetEdge
OUT
x2_active
C32
RST
12
x2_ctr.Acc
$HSIO_003_CT2.Acc ≤ 3
Local S0:iPT9
x2_sw
X9
x2_home_dn
C73
OUT
13
x2_home_range
C60
Local S0:iPT9
x2_sw
X9
x2_hom_dnn
C77
OUT
14
Local S0:iPT9
x2_sw
X9
x2_ctr.ResetEdge
$HSIO_003_CT2.ResetEdge
OUT
15
NOP
16
RET
Return Back to CALL
17
NOP


1
NOP
2
y_ctr.Acc
$HSIO_003_CT3.Acc -5
y_home_range
C80
OUT
y_ctr.Acc
$HSIO_003_CT3.Acc 5
3
homing_active
C20
Local S0:oPT10
dir_y
Y10
SET
4
homing_active
C20
y_homed
C87
Local S0:iPT10
y_sw
X10
y_hit_cond_hom
C88
OUT
5
y_hit_cond_hom
C88
y_hit_cond_last
C90
y_hit_one_pulse
C89
OUT
6
y_hit_cond_hom
C88
y_hit_cond_last
C90
SET
7
y_hit_cond_hom
C88
y_hit_cond_last
C90
RST
8
homing_active
C20
y_homed
C87
Local S0:iPT10
y_sw
X10
y_run_jog
C82
OUT
9
y_run_jog
C82
C877
OUT
10
y_run_jog
C82
motion_not_OK
C5
Local S0:iPT10
y_sw
X10
y_not_allowed
C99
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 800
Duty Cycle (Percent) 50
y_active
C33
SET
11
y_hit_one_pulse
C89
Local S0:iPT10
y_sw
X10
y_homed
C87
OUT
12
y_hit_one_pulse
C89
y_ctr.ResetEdge
$HSIO_003_CT3.ResetEdge
OUT
y_active
C33
RST
Local S0:oPT10
dir_y
Y10
RST
13
y_ctr.Acc
$HSIO_003_CT3.Acc 0
y_hit_one_pulse
C89
Local S0:iPT10
y_sw
X10
y_home_dn
C93
OUT
14
Local S0:iPT10
y_sw
X10
y_ctr.ResetEdge
$HSIO_003_CT3.ResetEdge
OUT
15
NOP
16
RET — Return Back to CALL
17
NOP


1
$On
ST1
Local S0:oPT11
dir_z
Y11
OUT
2
z_ctr.Acc
$HSIO_003_CT4.Acc
-5
≥
z_ctr.Acc
$HSIO_003_CT4.Acc
5
≤
z_home_range
C100
OUT
3
NOP
4
homing_active
C20
true_calib_stg_d
C252
Local S0:oPT11
dir_z
Y11
SET
5
homing_active
C20
z_homed
C107
Local S0:iPT11
z_sw
X11
z_hit_cond_hom
C108
OUT
6
z_hit_cond_hom
C108
z_hit_con_last
C110
z_hit_one_pulse
C109
OUT
7
z_hit_cond_hom
C108
z_hit_con_last
C110
SET
8
z_hit_cond_hom
C108
z_hit_con_last
C110
RST
9
homing_active
C20
z_homed
C107
Local S0:iPT11
z_sw
X11
z_run_jog
C102
OUT
10
z_run_jog
C102
C877
OUT
11
z_run_jog
C102
motion_not_OK
C6
Local S0:iPT11
z_sw
X11
z_not_allowed
C119
LocalcS0:iPT7
z2_sw
X7
PWMOUT
Pulse Width Modulated Output
PWM Device
@HSIO_002_PWM4
PWM Structure
$HSIO_002_PWM4
Discrete Output
pul_04_z
Frequency (Hz)
800
Duty Cycle (Percent)
50
z_active
C34
SET
C888
OUT
12
z_hit_one_pulse
C109
Local S0:iPT11
z_sw
X11
z_homed
C107
OUT
13
z_hit_one_pulse
C109
z_ctr.ResetEdge
$HSIO_003_CT4.ResetEdge
OUT
z_active
C34
RST
14
z_hit_one_pulse
C109
z_hom_dnn
C117
OUT
15
z_ctr.Acc
$HSIO_003_CT4.Acc
0
≤
z_hit_one_pulse
C109
Local S0:iPT11
z_sw
X11
z_home_dn
C113
OUT
16
Local S0:iPT11
z_sw
X11
z_ctr.ResetEdge
$HSIO_003_CT4.ResetEdge
OUT
17
NOP
18
RET
Return Back to CALL
19
NOP

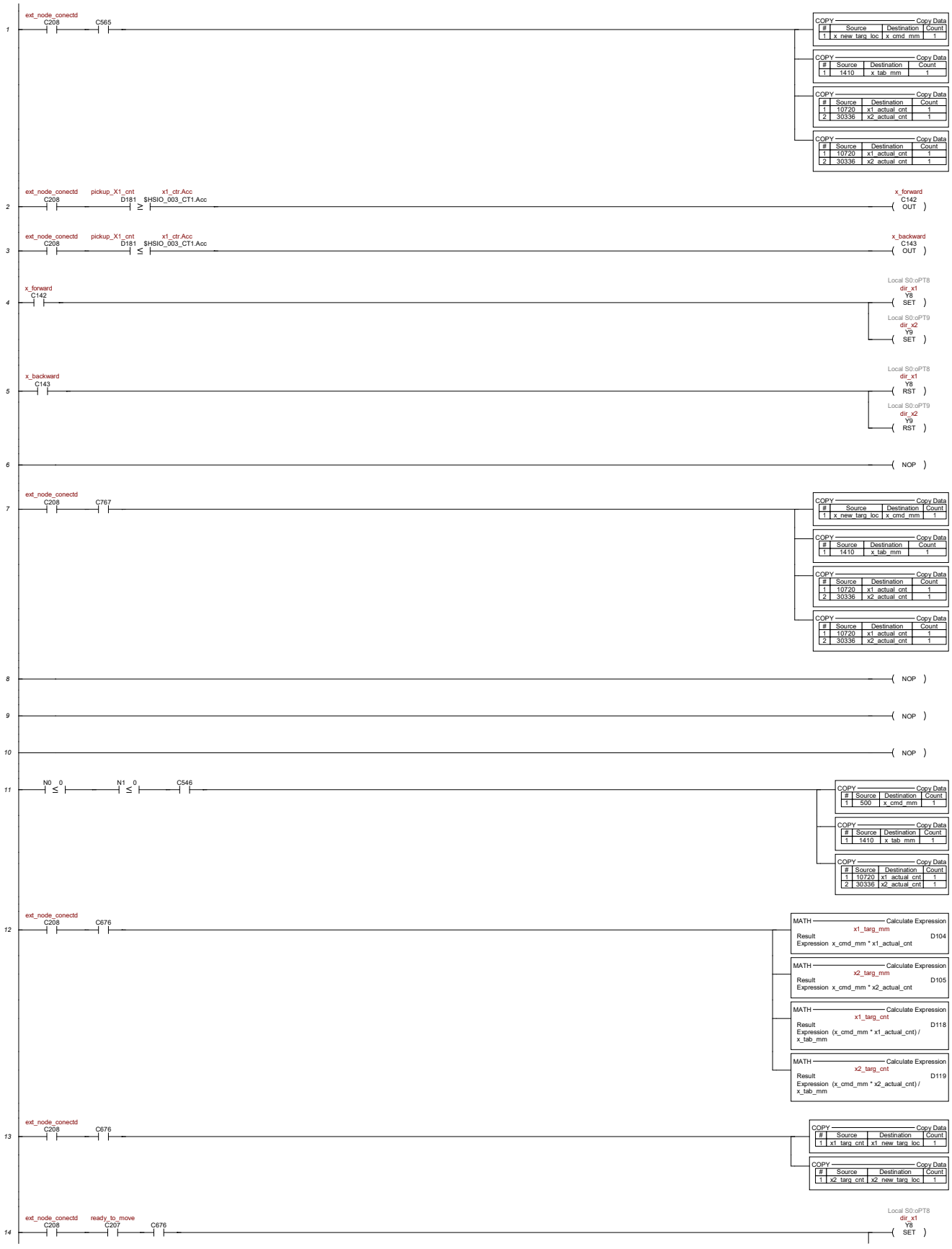

1
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT1
High Speed Input Structure x1_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X48 (Hs)
Count B Input X49 (Hs)
2
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT2
High Speed Input Structure x2_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X50 (Hs)
Count B Input X51 (Hs)
3
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT3
High Speed Input Structure y_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X52 (Hs)
Count B Input X53 (Hs)
4
HSCNT
High Speed Counting
High Speed Input Device @HSIO_003_CT4
High Speed Input Structure z_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input X54 (Hs)
Count B Input X55 (Hs)
5
$On
ST1
COPY
Copy Data
Source Destination Count
1 x1_ctr.Acc x1_ctr_acc 1
2 x2_ctr.Acc x2_ctr_acc 1
3 y_ctr.Acc y_ctr_acc 1
4 z_ctr.Acc z_ctr_acc 1
6
NOP
7
NOP
8
NOP
9
RET
Return Back to CALL
10
NOP

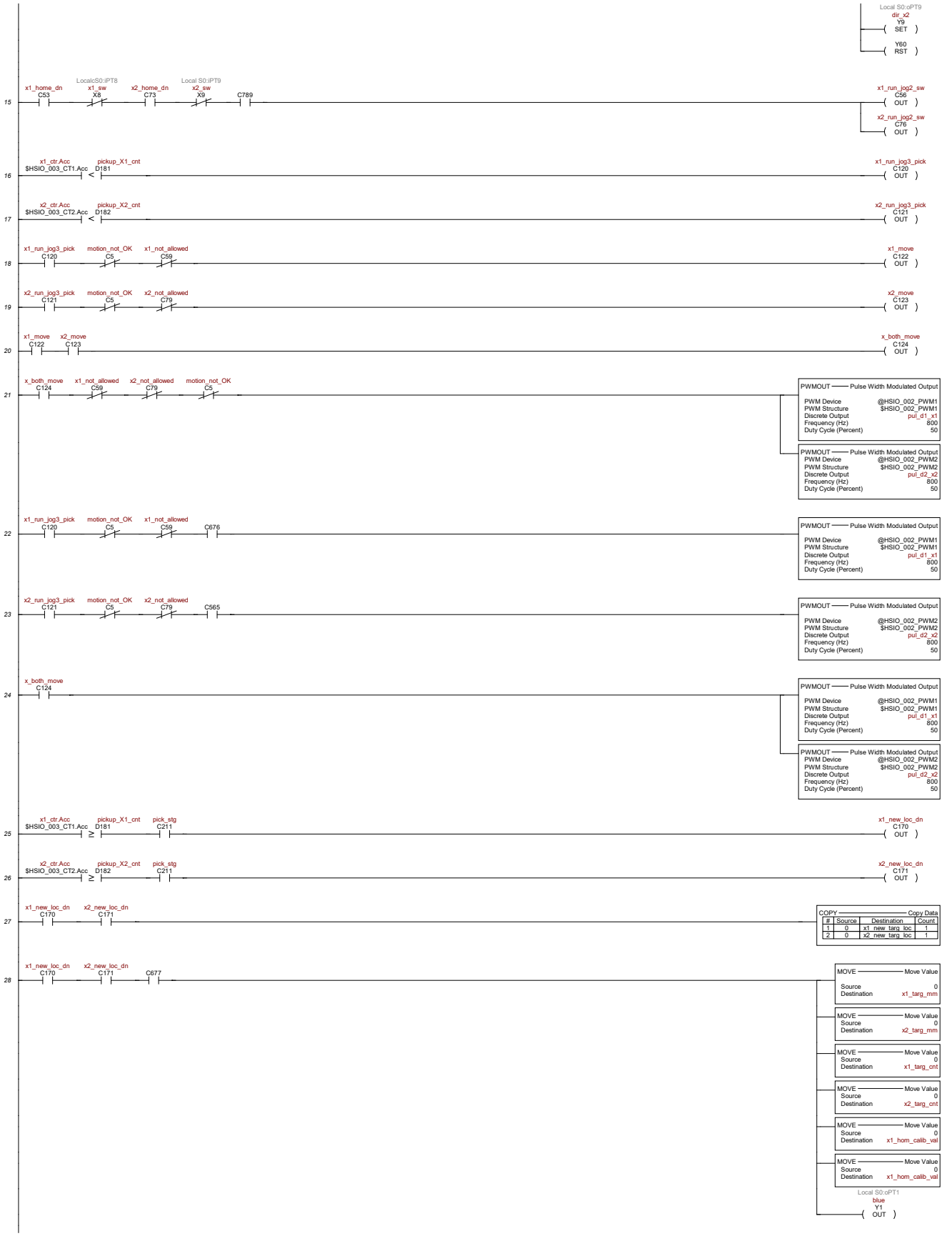

1
y_home_dn
C93
COPY — Copy Data
| Source | Destination | Count
1 | 15000 | y_hom_calib_val | 1
2
homing_active
C20
y_home_dn
C93
Local S0:oPT10
dir_y
Y10
RST
3
y_home_dn
C93
Local S0:iPT10
y_sw
X10
y_run_jog2_sw
C96
OUT
4
y_ctr.Acc
$HSIO_003_CT3.Acc
≤
y_hom_calib_val
D121
y_run_jog2
C95
OUT
5
y_run_jog2
C95
C877
OUT
6
y_run_jog2
C95
motion_not_OK
C5
y_not_allowed
C99
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 800
Duty Cycle (Percent) 50
7
y_ctr.Acc
$HSIO_003_CT3.Acc
≥
y_hom_calib_val
D121
y_home_dn
C93
y_calib_dn
C94
OUT
8
RET — Return Back to CALL
9
NOP

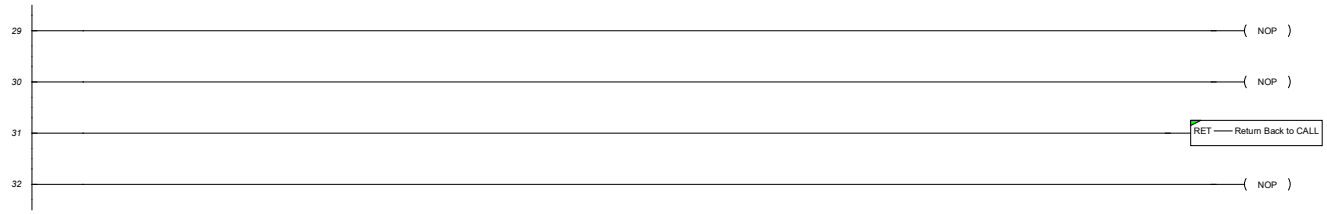

1
x1_hom_dnn C57
x2_hom_dnn C77
COPY Copy Data
Source Destination Count
1 495 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
2
N0 0
N1 0
C546
COPY Copy Data
Source Destination Count
1 500 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
3
x1_hom_dnn C57
x2_hom_dnn C77
MATH Calculate Expression
x1_targ_mm
Result D104
Expression x_cmd_mm * x1_actual_cnt
MATH Calculate Expression
x2_targ_mm
Result D105
Expression x_cmd_mm * x2_actual_cnt
MATH Calculate Expression
x1_targ_cnt
Result D118
Expression (x_cmd_mm * x1_actual_cnt) / x_tab_mm
MATH Calculate Expression
x2_targ_cnt
Result D119
Expression (x_cmd_mm * x2_actual_cnt) / x_tab_mm
4
x1_hom_dnn C57
x2_hom_dnn C77
COPY Copy Data
Source Destination Count
1 x1_targ_cnt x1_hom_calib_val 1
COPY Copy Data
Source Destination Count
1 x2_targ_cnt x2_hom_calib_val 1
5
homing_active C20
x1_hom_dnn C57
x2_hom_dnn C77
true_calib_stggg C250
Local S0:oPT8
dir_x1
Y8
SET
Local S0:oPT9
dir_x2
Y9
SET
Y60
RST
6
x1_home_dn C53
LocalcS0:iPT8
x1_sw X8
x2_home_dn C73
Local S0:iPT9
x2_sw X9
x1_run_jog2_sw C56
OUT
x2_run_jog2_sw C76
OUT
7
x1_ctr.Acc $HSIO_003_CT1.Acc
x1_hom_calib_val D116
x1_calib_dn C54
x1_hom_dnn C57
x1_run_jog2 C55
OUT
8
x2_ctr.Acc $HSIO_003_CT2.Acc
x2_hom_calib_val D117
x2_calib_dn C74
x2_hom_dnn C77
x2_run_jog2 C75
OUT
9
x1_run_jog2 C55
C877
OUT
10
x1_run_jog2 C55
motion_not_OK C5
x1_not_allowed C59
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
11
x2_run_jog2 C75
motion_not_OK C5
x2_not_allowed C79
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 805
Duty Cycle (Percent) 50
12
x1_ctr.Acc $HSIO_003_CT1.Acc
x1_hom_calib_val D116
x1_home_dn C53
x1_calib_dn C54
OUT
13
x2_ctr.Acc $HSIO_003_CT2.Acc
x2_hom_calib_val D117
x2_home_dn C73
x2_calib_dn C74
OUT
14
x1_calib_dn C54
x2_calib_dn C74
MOVE Move Value
Source 0
Destination x1_targ_mm

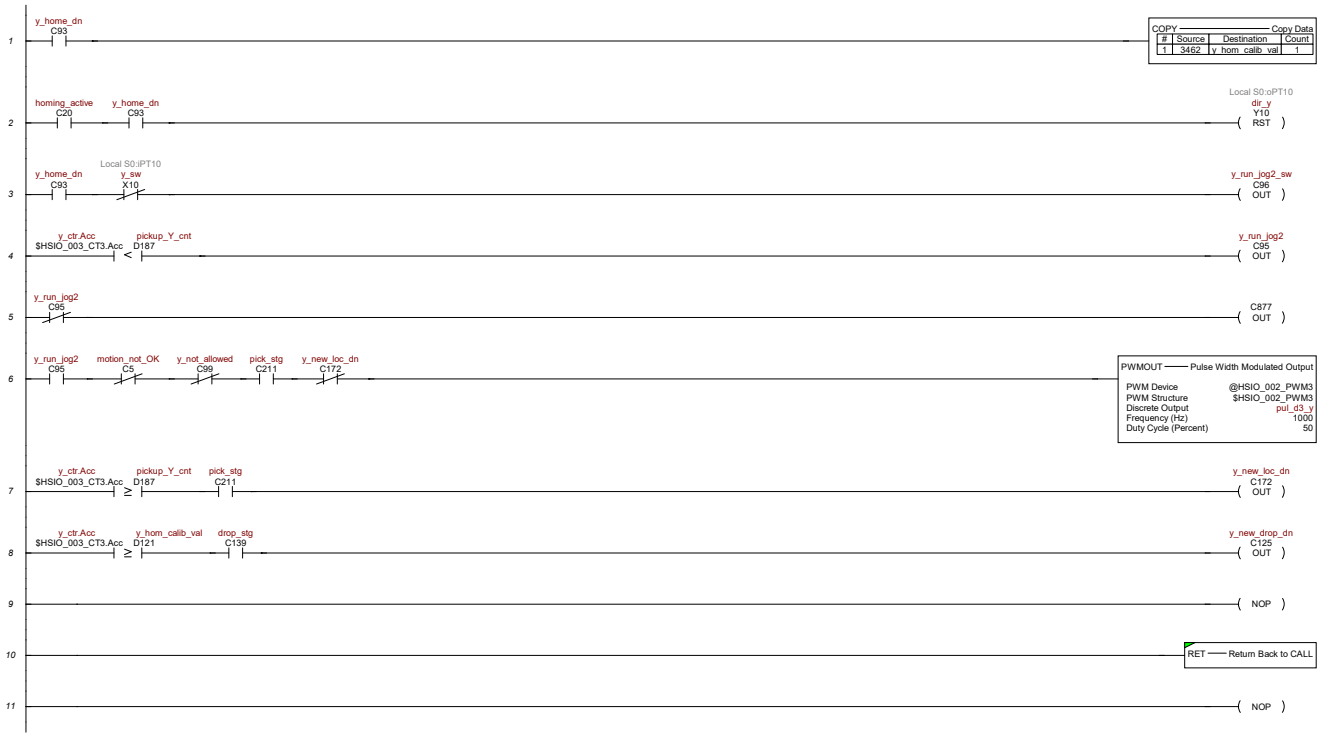

MOVE
Source
Destination
Move Value
0
x2_targ_mm
MOVE
Source
Destination
Move Value
0
x1_targ_cnt
MOVE
Source
Destination
Move Value
0
x2_targ_cnt
MOVE
Source
Destination
Move Value
0
x1_hom_calib_val
MOVE
Source
Destination
Move Value
0
x1_hom_calib_val
15
RET
Return Back to CALL
16
NOP

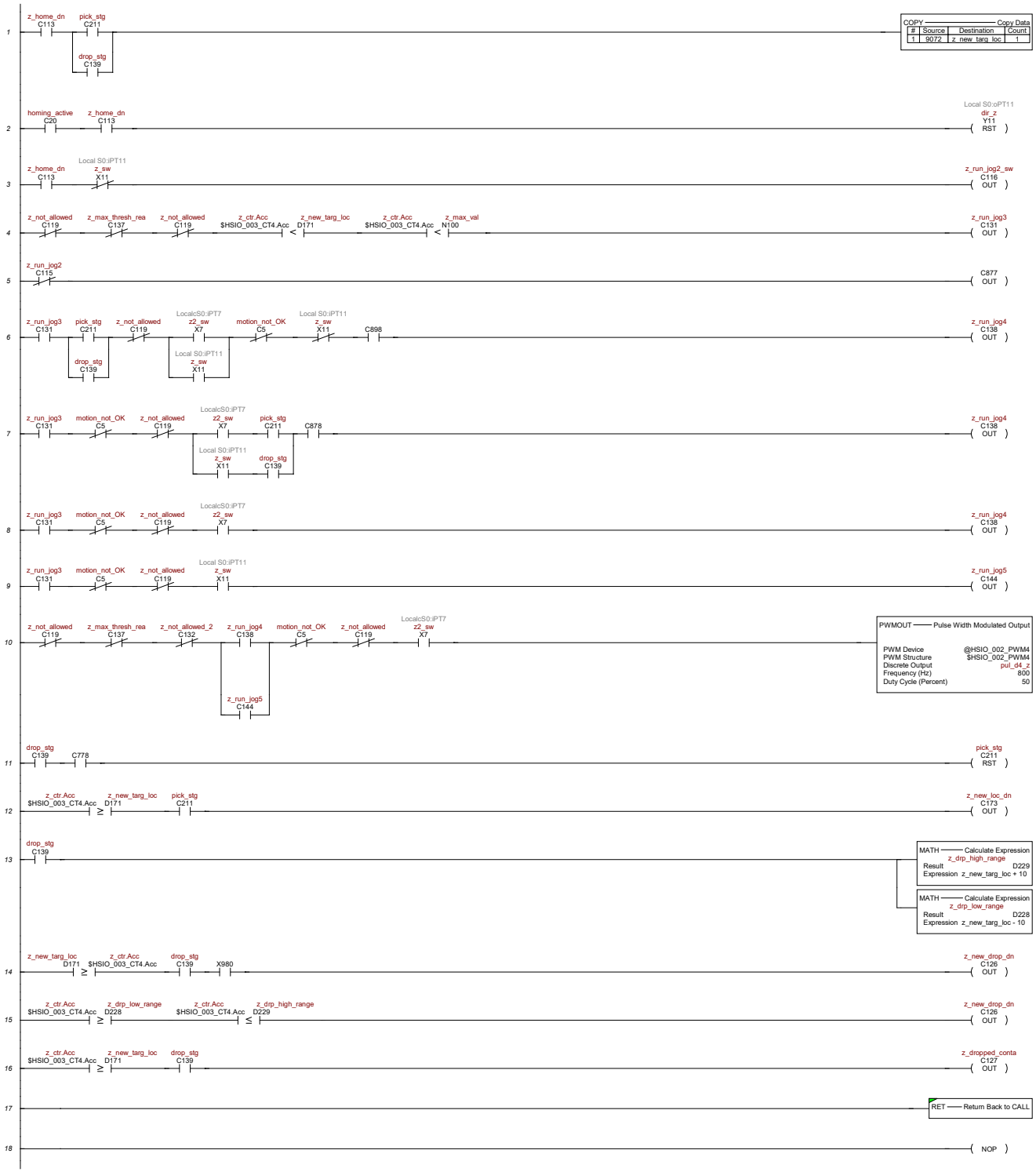

1
z_home_dn
C113
COPY
Copy Data
| Source | Destination | Count
1 | 1500 | z_hom_calib_val | 1
2
homing_active
C20
z_home_dn
C113
Local S0:oPT11
dir_z
Y11
RST
3
z_home_dn
C113
Local S0:iPT11
z_sw
X11
z_run_jog2_sw
C116
OUT
4
z_ctr.Acc
$HSIO_003_CT4.Acc
≤
z_hom_calib_val
D120
z_calib_dn
C114
z_run_jog2
C115
OUT
5
z_run_jog2
C115
C877
OUT
6
z_run_jog2
C115
motion_not_OK
C5
z_not_allowed
C119
PWMOUT
Pulse Width Modulated Output
PWM Device @HSIO_002_PWM4
PWM Structure $HSIO_002_PWM4
Discrete Output pul_d4_z
Frequency (Hz) 1000
Duty Cycle (Percent) 50
7
z_ctr.Acc
$HSIO_003_CT4.Acc
≥
z_hom_calib_val
D120
z_hom_dnn
C117
true_calib_stggg
C250
z_calib_dn
C114
OUT
8
z_calib_dn
C114
C565
MOVE
Move Value
Source 0
Destination z_hom_calib_val
9
RET
Return Back to CALL
10
NOP


1
ext_node_conectd
C208
C565
COPY Copy Data
Source Destination Count
1 x_new_targ_loc x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
2
ext_node_conectd
C208
pickup_X1_cnt
D181
≥
x1_ctr.Acc
$HSIO_003_CT1.Acc
x_forward
C142
OUT
3
ext_node_conectd
C208
pickup_X1_cnt
D181
≤
x1_ctr.Acc
$HSIO_003_CT1.Acc
x_backward
C143
OUT
4
x_forward
C142
Local S0:oPT8
dir_x1
Y8
SET
Local S0:oPT9
dir_x2
Y9
SET
5
x_backward
C143
Local S0:oPT8
dir_x1
Y8
RST
Local S0:oPT9
dir_x2
Y9
RST
6
NOP
7
ext_node_conectd
C208
C767
COPY Copy Data
Source Destination Count
1 x_new_targ_loc x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
8
NOP
9
NOP
10
NOP
11
N0 0
≤
N1 0
≤
C546
COPY Copy Data
Source Destination Count
1 500 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
12
ext_node_conectd
C208
C676
MATH Calculate Expression
x1_targ_mm
Result D104
Expression x_cmd_mm * x1_actual_cnt
MATH Calculate Expression
x2_targ_mm
Result D105
Expression x_cmd_mm * x2_actual_cnt
MATH Calculate Expression
x1_targ_cnt
Result D118
Expression (x_cmd_mm * x1_actual_cnt) / x_tab_mm
MATH Calculate Expression
x2_targ_cnt
Result D119
Expression (x_cmd_mm * x2_actual_cnt) / x_tab_mm
13
ext_node_conectd
C208
C676
COPY Copy Data
Source Destination Count
1 x1_targ_cnt x1_new_targ_loc 1
COPY Copy Data
Source Destination Count
1 x2_targ_cnt x2_new_targ_loc 1
14
ext_node_conectd
C208
ready_to_move
C207
C676
Local S0:oPT8
dir_x1
Y8
SET

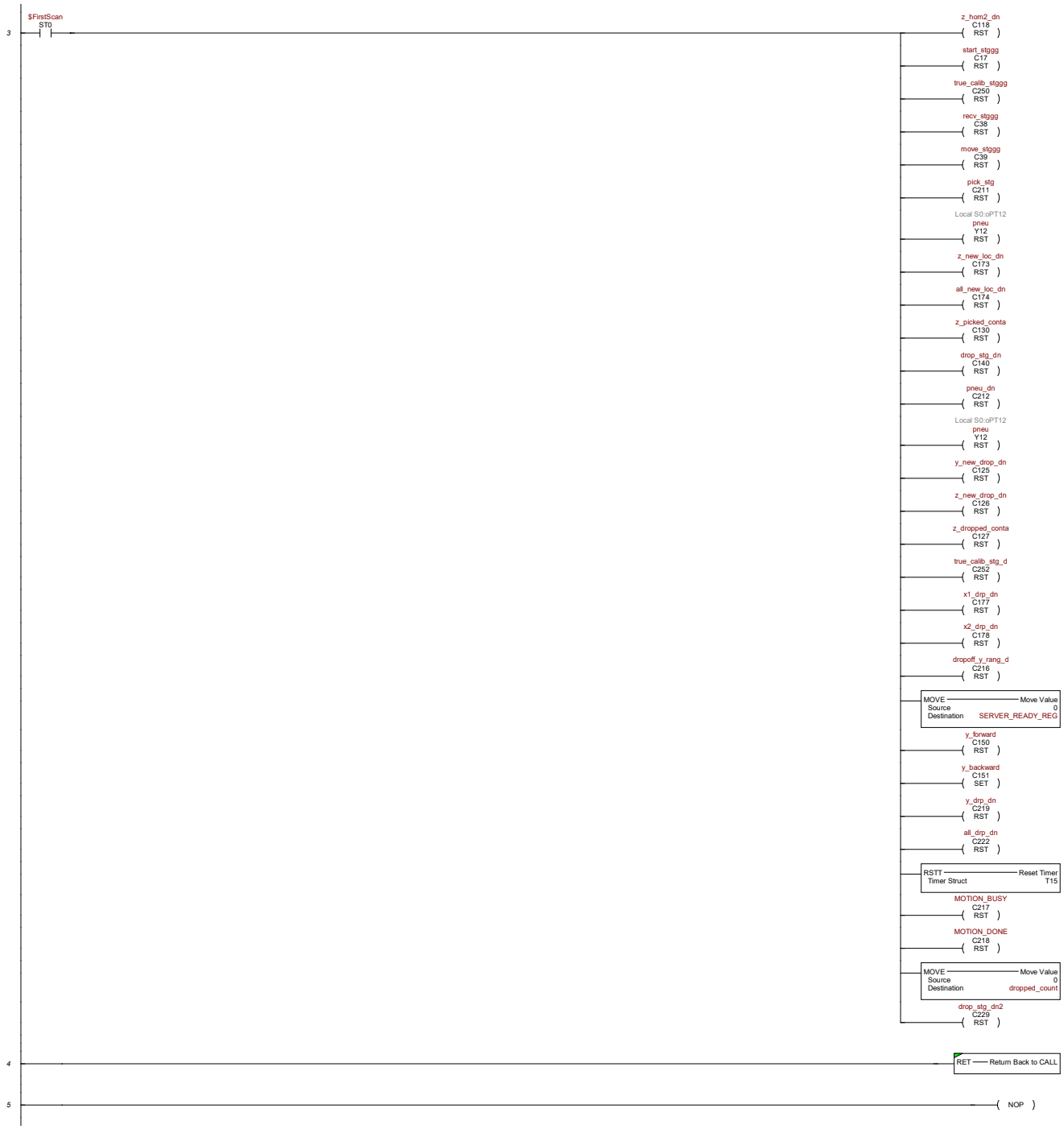

Local S0:oPT9
dir_x2
Y9
SET
Y60
RST
15
x1_home_dn
C53
LocalcS0:iPT8
x1_sw
X8
x2_home_dn
C73
Local S0:iPT9
x2_sw
X9
C789
x1_run_jog2_sw
C56
OUT
x2_run_jog2_sw
C76
OUT
16
x1_ctr.Acc
$HSIO_003_CT1.Acc
pickup_X1_cnt
D181
x1_run_jog3_pick
C120
OUT
17
x2_ctr.Acc
$HSIO_003_CT2.Acc
pickup_X2_cnt
D182
x2_run_jog3_pick
C121
OUT
18
x1_run_jog3_pick
C120
motion_not_OK
C5
x1_not_allowed
C59
x1_move
C122
OUT
19
x2_run_jog3_pick
C121
motion_not_OK
C5
x2_not_allowed
C79
x2_move
C123
OUT
20
x1_move
C122
x2_move
C123
x_both_move
C124
OUT
21
x_both_move
C124
x1_not_allowed
C59
x2_not_allowed
C79
motion_not_OK
C5
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
22
x1_run_jog3_pick
C120
motion_not_OK
C5
x1_not_allowed
C59
C676
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
23
x2_run_jog3_pick
C121
motion_not_OK
C5
x2_not_allowed
C79
C565
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
24
x_both_move
C124
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
25
x1_ctr.Acc
$HSIO_003_CT1.Acc
pickup_X1_cnt
D181
pick_stg
C211
x1_new_loc_dn
C170
OUT
26
x2_ctr.Acc
$HSIO_003_CT2.Acc
pickup_X2_cnt
D182
pick_stg
C211
x2_new_loc_dn
C171
OUT
27
x1_new_loc_dn
C170
x2_new_loc_dn
C171
COPY Copy Data
Source Destination Count
1 0 x1_new_targ_loc 1
2 0 x2_new_targ_loc 1
28
x1_new_loc_dn
C170
x2_new_loc_dn
C171
C677
MOVE Move Value
Source 0
Destination x1_targ_mm
MOVE Move Value
Source 0
Destination x2_targ_mm
MOVE Move Value
Source 0
Destination x1_targ_cnt
MOVE Move Value
Source 0
Destination x2_targ_cnt
MOVE Move Value
Source 0
Destination x1_hom_calib_val
MOVE Move Value
Source 0
Destination x1_hom_calib_val
Local S0:oPT1
blue
Y1
OUT

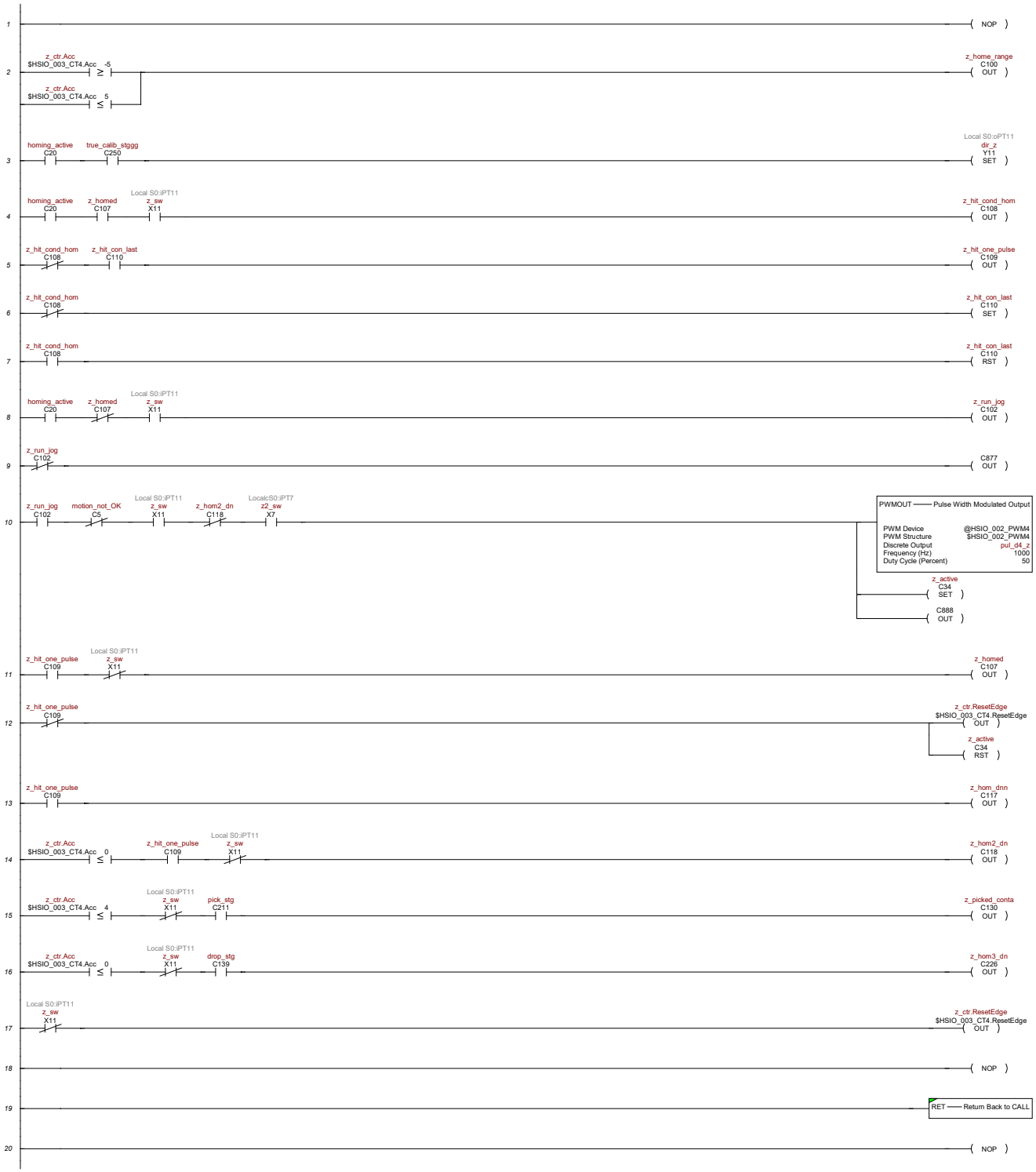

29
NOP
30
NOP
31
RET —— Return Back to CALL
32
NOP


1
y_home_dn
C93
COPY Copy Data
Source Destination Count
1 3462 y_hom_calib_val 1
2
homing_active
C20
y_home_dn
C93
Local S0:oPT10
dir_y
Y10
RST
3
y_home_dn
C93
Local S0:iPT10
y_sw
X10
y_run_jog2_sw
C96
OUT
4
y_ctr.Acc
$HSIO_003_CT3.Acc
<
pickup_Y_cnt
D187
y_run_jog2
C95
OUT
5
y_run_jog2
C95
C877
OUT
6
y_run_jog2
C95
motion_not_OK
C5
y_not_allowed
C99
pick_stg
C211
y_new_loc_dn
C172
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 1000
Duty Cycle (Percent) 50
7
y_ctr.Acc
$HSIO_003_CT3.Acc
≥
pickup_Y_cnt
D187
pick_stg
C211
y_new_loc_dn
C172
OUT
8
y_ctr.Acc
$HSIO_003_CT3.Acc
≥
y_hom_calib_val
D121
drop_stg
C139
y_new_drop_dn
C125
OUT
9
NOP
10
RET Return Back to CALL
11
NOP

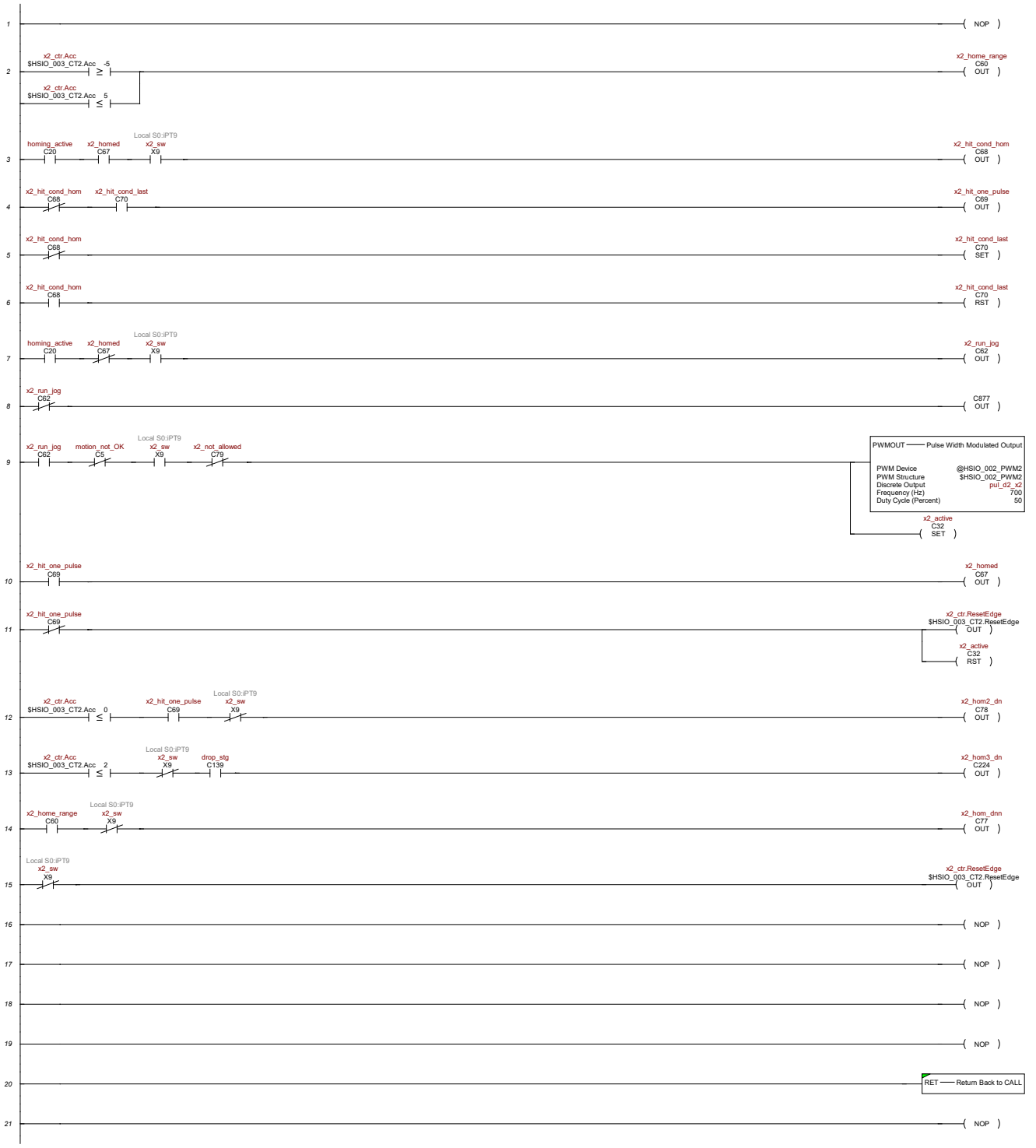

1
z_home_dn C113
pick_stg C211
drop_stg C139
COPY Copy Data
Source Destination Count
1 9072 z_new_targ_loc 1
2
homing_active C20
z_home_dn C113
Local S0:oPT11
dir_z Y11
RST
3
z_home_dn C113
Local S0:iPT11
z_sw X11
z_run_jog2_sw C116
OUT
4
z_not_allowed C119
z_max_thresh_rea C137
z_not_allowed C119
z_ctr.Acc $HSIO_003_CT4.Acc
<
z_new_targ_loc D171
z_ctr.Acc $HSIO_003_CT4.Acc
<
z_max_val N100
z_run_jog3 C131
OUT
5
z_run_jog2 C115
C877
OUT
6
z_run_jog3 C131
pick_stg C211
drop_stg C139
z_not_allowed C119
LocalcS0:iPT7
z2_sw X7
Local S0:iPT11
z_sw X11
motion_not_OK C5
Local S0:iPT11
z_sw X11
C898
z_run_jog4 C138
OUT
7
z_run_jog3 C131
motion_not_OK C5
z_not_allowed C119
LocalcS0:iPT7
z2_sw X7
pick_stg C211
C878
Local S0:iPT11
z_sw X11
drop_stg C139
z_run_jog4 C138
OUT
8
z_run_jog3 C131
motion_not_OK C5
z_not_allowed C119
LocalcS0:iPT7
z2_sw X7
z_run_jog4 C138
OUT
9
z_run_jog3 C131
motion_not_OK C5
z_not_allowed C119
Local S0:iPT11
z_sw X11
z_run_jog5 C144
OUT
10
z_not_allowed C119
z_max_thresh_rea C137
z_not_allowed_2 C132
z_run_jog4 C138
motion_not_OK C5
z_not_allowed C119
LocalcS0:iPT7
z2_sw X7
z_run_jog5 C144
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM4
PWM Structure $HSIO_002_PWM4
Discrete Output pul_d4_z
Frequency (Hz) 800
Duty Cycle (Percent) 50
11
drop_stg C139
C778
pick_stg C211
RST
12
z_ctr.Acc $HSIO_003_CT4.Acc
≥
z_new_targ_loc D171
pick_stg C211
z_new_loc_dn C173
OUT
13
drop_stg C139
MATH Calculate Expression
z_drp_high_range
Result D229
Expression z_new_targ_loc + 10
MATH Calculate Expression
z_drp_low_range
Result D228
Expression z_new_targ_loc - 10
14
z_new_targ_loc D171
z_ctr.Acc $HSIO_003_CT4.Acc
≥
drop_stg C139
X980
z_new_drop_dn C126
OUT
15
z_ctr.Acc $HSIO_003_CT4.Acc
≥
z_drp_low_range D228
z_ctr.Acc $HSIO_003_CT4.Acc
≤
z_drp_high_range D229
z_new_drop_dn C126
OUT
16
z_ctr.Acc $HSIO_003_CT4.Acc
≥
z_new_targ_loc D171
drop_stg C139
z_dropped_conta C127
OUT
17
RET Return Back to CALL
18
NOP

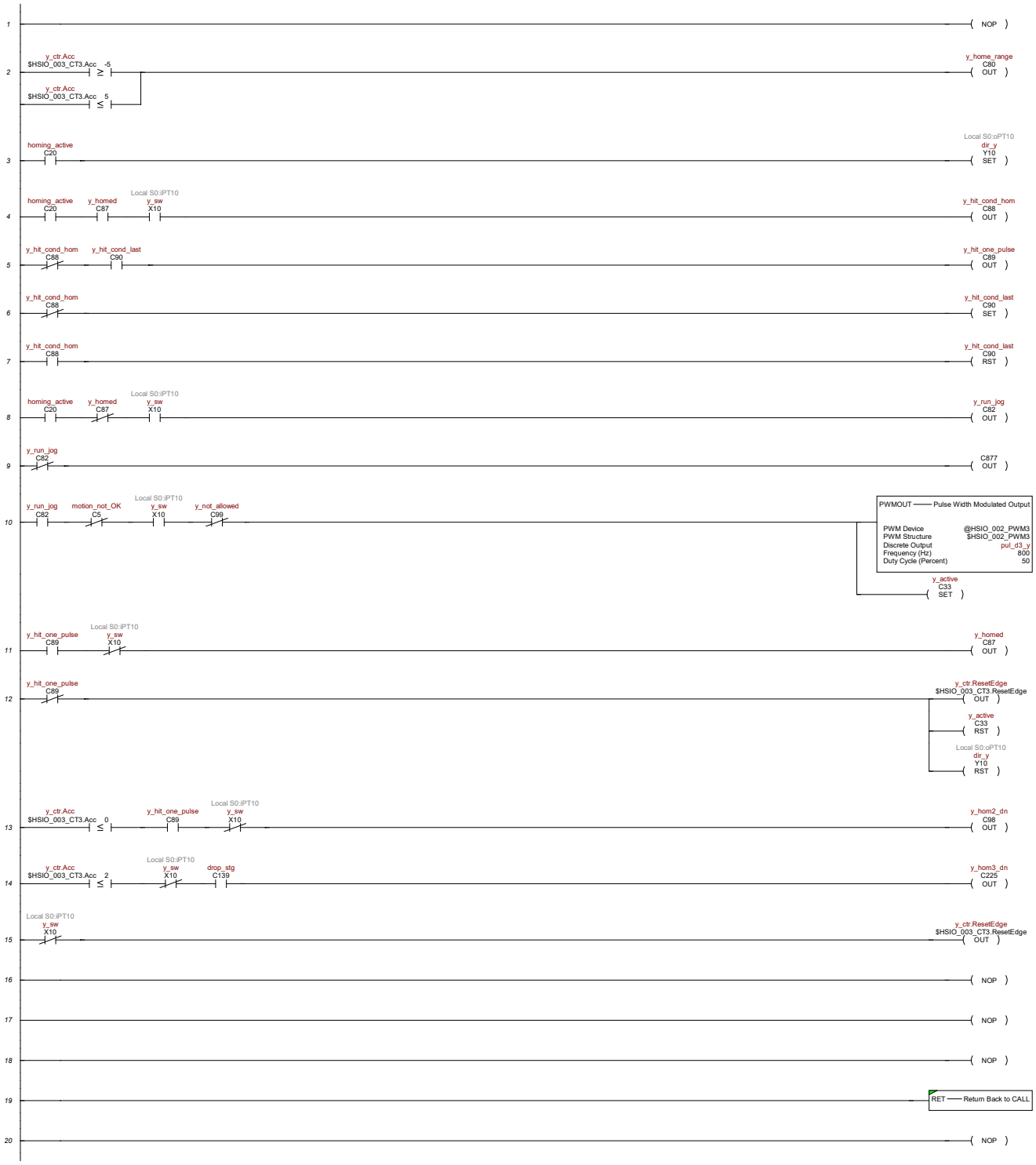

1
$FirstScan
ST0
system_homed
C21
RST
axis_pose_valid
C22
RST
x1_homed
C47
OUT
Local S0:oPT5
beep_sound
Y5
OUT
x2_homed
C67
OUT
y_homed
C87
OUT
z_homed
C107
OUT
x1_ctr.ResetEdge
$HSIO_003_CT1.ResetEdge
OUT
x2_ctr.ResetEdge
$HSIO_003_CT2.ResetEdge
OUT
y_ctr.ResetEdge
$HSIO_003_CT3.ResetEdge
OUT
z_ctr.ResetEdge
$HSIO_003_CT4.ResetEdge
OUT
Local S0:oPT1
blue
Y1
RST
true_calib_stggg
C250
RST
x1_home_dn
C53
RST
x2_home_dn
C73
RST
y_home_dn
C93
RST
z_home_dn
C113
RST
start_stggg
C17
RST
x1_run_jog
C42
RST
x2_run_jog
C62
RST
y_run_jog
C82
RST
z_run_jog
C102
RST
server_snd_move
C206
RST
z_new_drop_dn
C126
RST
x1_hom3_dn
C223
RST
x2_hom3_dn
C224
RST
y_hom3_dn
C225
RST
z_hom3_dn
C226
RST
readyyy
C228
RST
2
$FirstScan
ST0
x1_calib_dn
C54
RST
x2_calib_dn
C74
RST
y_calib_dn
C94
RST
z_calib_dn
C114
RST
x1_new_loc_dn
C170
RST
x2_new_loc_dn
C171
RST
y_new_loc_dn
C172
RST
z_new_loc_dn
C173
RST
server_ready
C203
RST
demo_ready
C204
RST
dav_ready
C205
RST

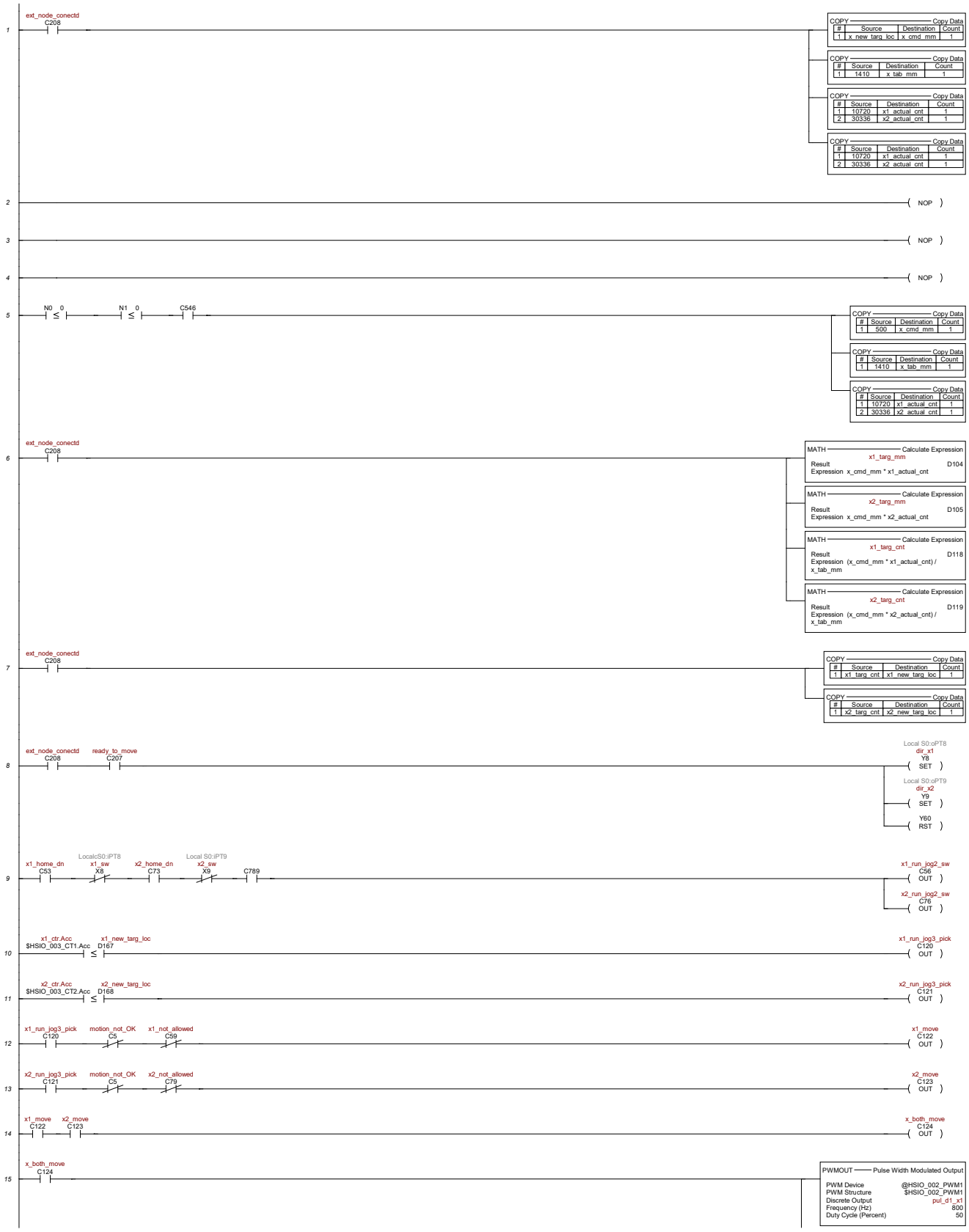

3
$FirstScan
ST0
z_hom2_dn
C118
RST
start_stggg
C17
RST
true_calib_stggg
C250
RST
recv_stggg
C38
RST
move_stggg
C39
RST
pick_stg
C211
RST
Local S0:oPT12
pneu
Y12
RST
z_new_loc_dn
C173
RST
all_new_loc_dn
C174
RST
z_picked_conta
C130
RST
drop_stg_dn
C140
RST
pneu_dn
C212
RST
Local S0:oPT12
pneu
Y12
RST
y_new_drop_dn
C125
RST
z_new_drop_dn
C126
RST
z_dropped_conta
C127
RST
true_calib_stg_d
C252
RST
x1_drp_dn
C177
RST
x2_drp_dn
C178
RST
dropoff_y_rang_d
C216
RST
MOVE
Source 0
Destination SERVER_READY_REG
Move Value
y_forward
C150
RST
y_backward
C151
SET
y_drp_dn
C219
RST
all_drp_dn
C222
RST
RSTT
Timer Struct
Reset Timer
T15
MOTION_BUSY
C217
RST
MOTION_DONE
C218
RST
MOVE
Source 0
Destination dropped_count
Move Value
drop_stg_dn2
C229
RST
4
RET
Return Back to CALL
5
NOP

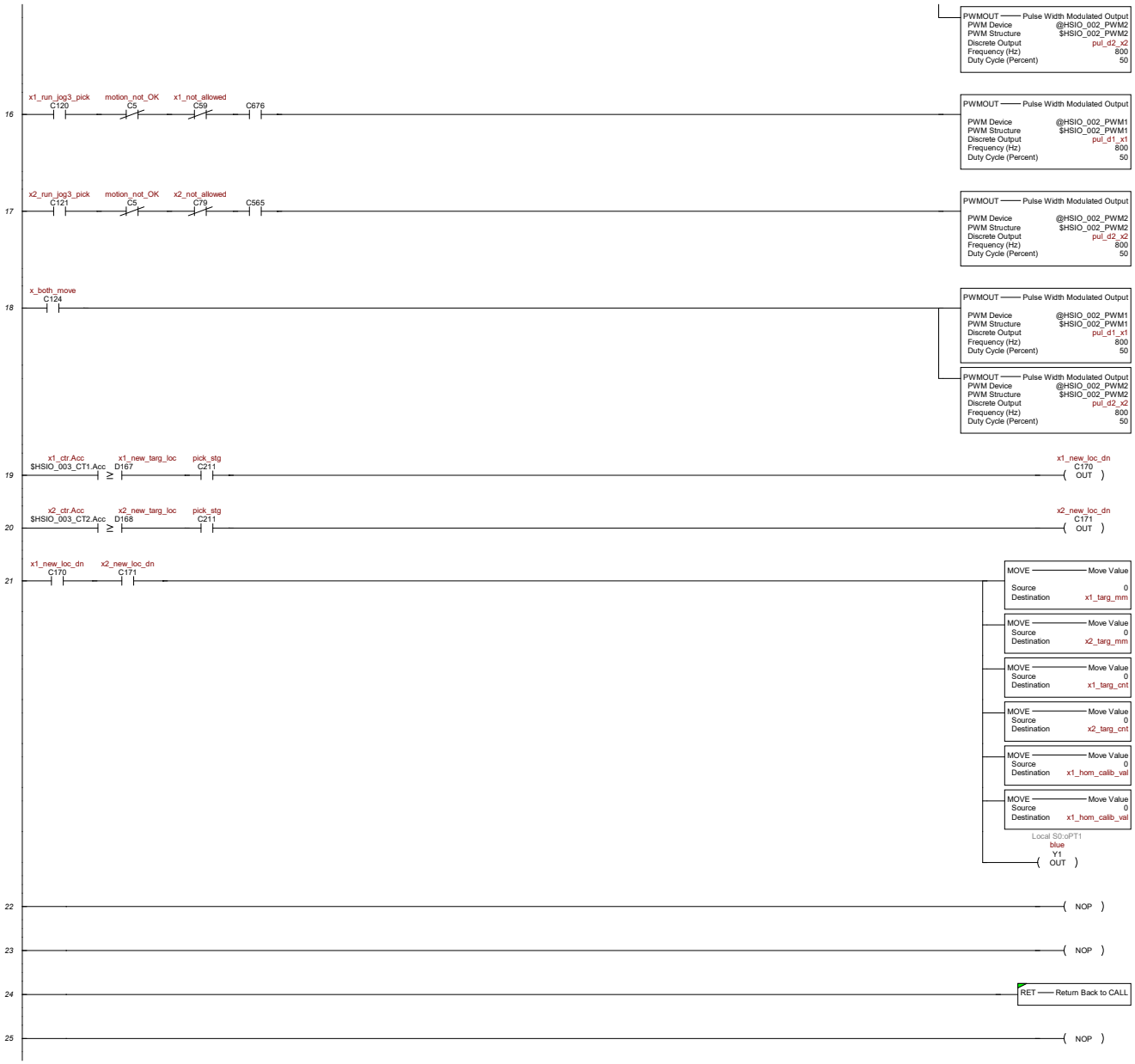

1
NOP
2
z_ctr.Acc
$HSIO_003_CT4.Acc
-5
z_ctr.Acc
$HSIO_003_CT4.Acc
5
z_home_range
C100
OUT
3
homing_active
C20
true_calib_stggg
C250
Local S0:oPT11
dir_z
Y11
SET
4
homing_active
C20
z_homed
C107
Local S0:iPT11
z_sw
X11
z_hit_cond_hom
C108
OUT
5
z_hit_cond_hom
C108
z_hit_con_last
C110
z_hit_one_pulse
C109
OUT
6
z_hit_cond_hom
C108
z_hit_con_last
C110
SET
7
z_hit_cond_hom
C108
z_hit_con_last
C110
RST
8
homing_active
C20
z_homed
C107
Local S0:iPT11
z_sw
X11
z_run_jog
C102
OUT
9
z_run_jog
C102
C877
OUT
10
z_run_jog
C102
motion_not_OK
C5
Local S0:iPT11
z_sw
X11
z_hom2_dn
C118
LocalcS0:iPT7
z2_sw
X7
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM4
PWM Structure $HSIO_002_PWM4
Discrete Output pul_d4_z
Frequency (Hz) 1000
Duty Cycle (Percent) 50
z_active
C34
SET
C888
OUT
11
z_hit_one_pulse
C109
Local S0:iPT11
z_sw
X11
z_homed
C107
OUT
12
z_hit_one_pulse
C109
z_ctr.ResetEdge
$HSIO_003_CT4.ResetEdge
OUT
z_active
C34
RST
13
z_hit_one_pulse
C109
z_hom_dnn
C117
OUT
14
z_ctr.Acc
$HSIO_003_CT4.Acc
0
z_hit_one_pulse
C109
Local S0:iPT11
z_sw
X11
z_hom2_dn
C118
OUT
15
z_ctr.Acc
$HSIO_003_CT4.Acc
4
Local S0:iPT11
z_sw
X11
pick_stg
C211
z_picked_conta
C130
OUT
16
z_ctr.Acc
$HSIO_003_CT4.Acc
0
Local S0:iPT11
z_sw
X11
drop_stg
C139
z_hom3_dn
C226
OUT
17
Local S0:iPT11
z_sw
X11
z_ctr.ResetEdge
$HSIO_003_CT4.ResetEdge
OUT
18
NOP
19
RET — Return Back to CALL
20
NOP

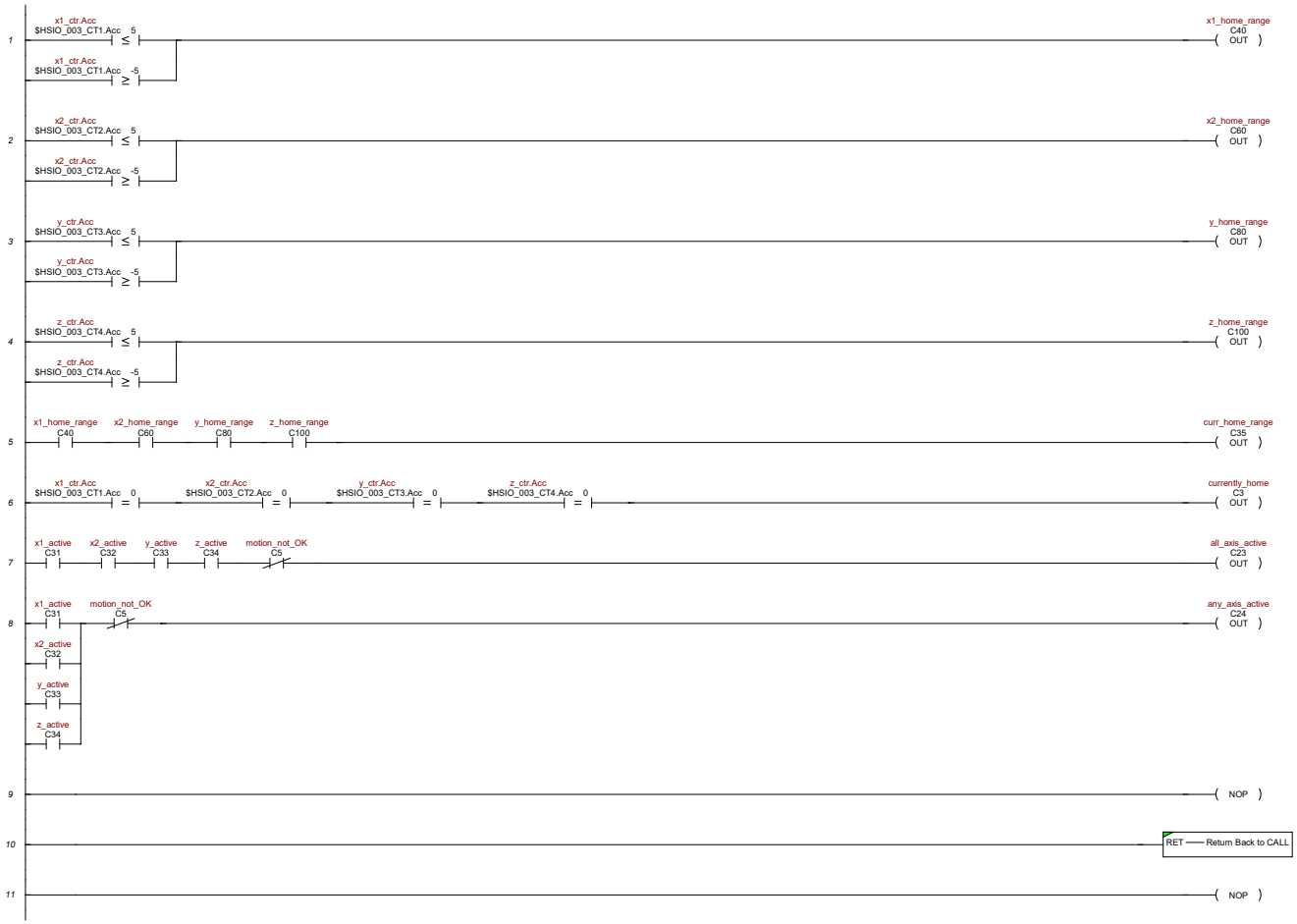

1
NOP
2
x1_ctr.Acc
$HSIO_003_CT1.Acc
-5
x1_ctr.Acc
$HSIO_003_CT1.Acc
5
x1_home_range
C40
OUT
3
homing_active
C20
x1_homed
C47
LocalcS0:iPT8
x1_sw
X8
x1_hit_cond_hom
C48
OUT
4
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
x1_hit_one_pulse
C49
OUT
5
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
SET
6
x1_hit_cond_hom
C48
x1_hit_cond_last
C50
RST
7
homing_active
C20
x1_homed
C47
LocalcS0:iPT8
x1_sw
X8
x1_run_jog
C42
OUT
8
x1_run_jog
C42
C877
OUT
9
x1_run_jog
C42
motion_not_OK
C5
LocalcS0:iPT8
x1_sw
X8
x1_not_allowed
C59
PWMOUT
Pulse Width Modulated Output
PWM Device
@HSIO_002_PWM1
PWM Structure
$HSIO_002_PWM1
Discrete Output
pul_d1_x1
Frequency (Hz)
700
Duty Cycle (Percent)
50
x1_active
C31
SET
10
x1_hit_one_pulse
C49
x1_homed
C47
OUT
11
x1_hit_one_pulse
C49
x1_ctr.ResetEdge
$HSIO_003_CT1.ResetEdge
OUT
x1_active
C31
RST
12
x1_ctr.Acc
$HSIO_003_CT1.Acc
0
x1_hit_one_pulse
C49
LocalcS0:iPT8
x1_sw
X8
x1_hom2_dn
C58
OUT
13
x1_ctr.Acc
$HSIO_003_CT1.Acc
2
LocalcS0:iPT8
x1_sw
X8
drop_stg
C139
x1_hom3_dn
C223
OUT
14
x1_home_range
C40
LocalcS0:iPT8
x1_sw
X8
x1_hom_dnn
C57
OUT
15
LocalcS0:iPT8
x1_sw
X8
x1_ctr.ResetEdge
$HSIO_003_CT1.ResetEdge
OUT
16
NOP
17
NOP
18
NOP
19
RET
Return Back to CALL
20
NOP

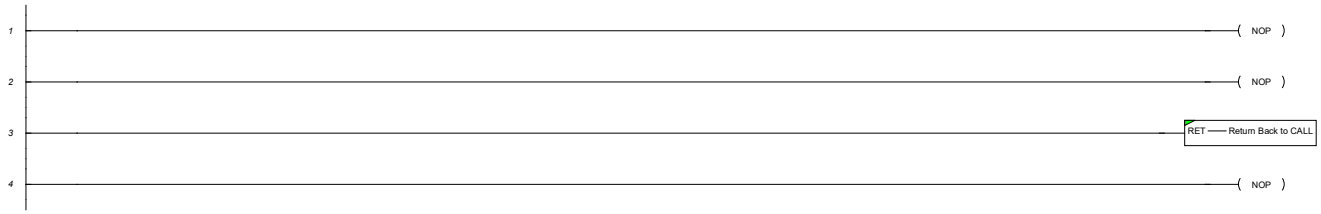

1
NOP
2
x2_ctr.Acc
$HSIO_003_CT2.Acc
≥
-5
x2_home_range
C60
OUT
x2_ctr.Acc
$HSIO_003_CT2.Acc
≤
5
3
homing_active
C20
x2_homed
C67
Local S0:iPT9
x2_sw
X9
x2_hit_cond_hom
C68
OUT
4
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
x2_hit_one_pulse
C69
OUT
5
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
SET
6
x2_hit_cond_hom
C68
x2_hit_cond_last
C70
RST
7
homing_active
C20
x2_homed
C67
Local S0:iPT9
x2_sw
X9
x2_run_jog
C62
OUT
8
x2_run_jog
C62
C877
OUT
9
x2_run_jog
C62
motion_not_OK
C5
Local S0:iPT9
x2_sw
X9
x2_not_allowed
C79
PWMOUT
Pulse Width Modulated Output
PWM Device
@HSIO_002_PWM2
PWM Structure
$HSIO_002_PWM2
Discrete Output
put_d2_x2
Frequency (Hz)
700
Duty Cycle (Percent)
50
x2_active
C32
SET
10
x2_hit_one_pulse
C69
x2_homed
C67
OUT
11
x2_hit_one_pulse
C69
x2_ctr.ResetEdge
$HSIO_003_CT2.ResetEdge
OUT
x2_active
C32
RST
12
x2_ctr.Acc
$HSIO_003_CT2.Acc
≤
0
x2_hit_one_pulse
C69
Local S0:iPT9
x2_sw
X9
x2_hom2_dn
C78
OUT
13
x2_ctr.Acc
$HSIO_003_CT2.Acc
≤
2
Local S0:iPT9
x2_sw
X9
drop_stg
C139
x2_hom3_dn
C224
OUT
14
x2_home_range
C60
Local S0:iPT9
x2_sw
X9
x2_hom_dnn
C77
OUT
15
Local S0:iPT9
x2_sw
X9
x2_ctr.ResetEdge
$HSIO_003_CT2.ResetEdge
OUT
16
NOP
17
NOP
18
NOP
19
NOP
20
RET
Return Back to CALL
21
NOP

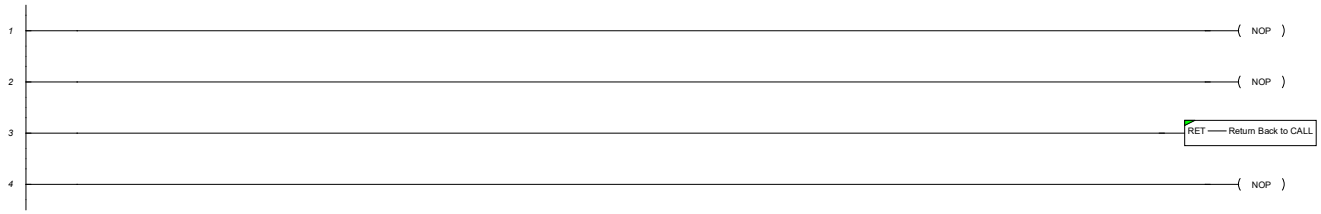

1
NOP
2
y_ctr.Acc
$HSIO_003_CT3.Acc
≥ -5
≤ 5
y_home_range
C80
OUT
3
homing_active
C20
Local S0:oPT10
dir_y
Y10
SET
4
homing_active
C20
y_homed
C87
Local S0:iPT10
y_sw
X10
y_hit_cond_hom
C88
OUT
5
y_hit_cond_hom
C88
y_hit_cond_last
C90
y_hit_one_pulse
C89
OUT
6
y_hit_cond_hom
C88
y_hit_cond_last
C90
SET
7
y_hit_cond_hom
C88
y_hit_cond_last
C90
RST
8
homing_active
C20
y_homed
C87
Local S0:iPT10
y_sw
X10
y_run_jog
C82
OUT
9
y_run_jog
C82
C877
OUT
10
y_run_jog
C82
motion_not_OK
C5
Local S0:iPT10
y_sw
X10
y_not_allowed
C99
PWMOUT — Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 800
Duty Cycle (Percent) 50
y_active
C33
SET
11
y_hit_one_pulse
C89
Local S0:iPT10
y_sw
X10
y_homed
C87
OUT
12
y_hit_one_pulse
C89
y_ctr.ResetEdge
$HSIO_003_CT3.ResetEdge
OUT
y_active
C33
RST
Local S0:oPT10
dir_y
Y10
RST
13
y_ctr.Acc
$HSIO_003_CT3.Acc
≤ 0
y_hit_one_pulse
C89
Local S0:iPT10
y_sw
X10
y_hom2_dn
C98
OUT
14
y_ctr.Acc
$HSIO_003_CT3.Acc
≤ 2
Local S0:iPT10
y_sw
X10
drop_stg
C139
y_hom3_dn
C225
OUT
15
Local S0:iPT10
y_sw
X10
y_ctr.ResetEdge
$HSIO_003_CT3.ResetEdge
OUT
16
NOP
17
NOP
18
NOP
19
RET — Return Back to CALL
20
NOP

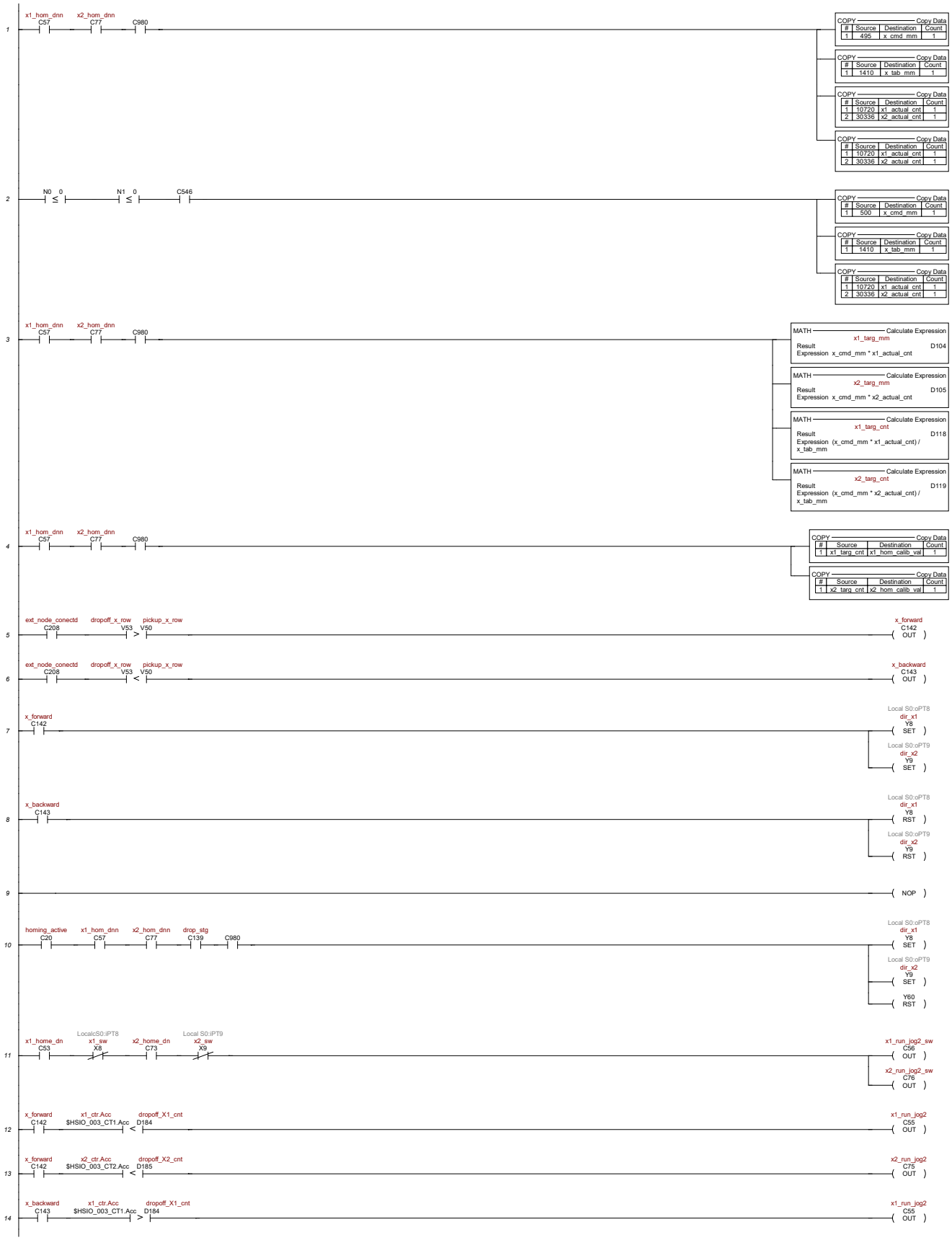

1
ext_node_conectd
C208
COPY Copy Data
Source Destination Count
1 x_new_targ_loc x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
2
NOP
3
NOP
4
NOP
5
N0 0
N1 0
C546
COPY Copy Data
Source Destination Count
1 500 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
6
ext_node_conectd
C208
MATH Calculate Expression
x1_targ_mm
Result D104
Expression x_cmd_mm * x1_actual_cnt
MATH Calculate Expression
x2_targ_mm
Result D105
Expression x_cmd_mm * x2_actual_cnt
MATH Calculate Expression
x1_targ_cnt
Result D118
Expression (x_cmd_mm * x1_actual_cnt) / x_tab_mm
MATH Calculate Expression
x2_targ_cnt
Result D119
Expression (x_cmd_mm * x2_actual_cnt) / x_tab_mm
7
ext_node_conectd
C208
COPY Copy Data
Source Destination Count
1 x1_targ_cnt x1_new_targ_loc 1
COPY Copy Data
Source Destination Count
1 x2_targ_cnt x2_new_targ_loc 1
8
ext_node_conectd
C208
ready_to_move
C207
Local S0:oPT8
dir_x1
Y8
SET
Local S0:oPT9
dir_x2
Y9
SET
Y60
RST
9
x1_home_dn
C53
LocalcS0:iPT8
x1_sw
X8
x2_home_dn
C73
Local S0:iPT9
x2_sw
X9
C789
x1_run_jog2_sw
C56
OUT
x2_run_jog2_sw
C76
OUT
10
x1_ctr.Acc
$HSIO_003_CT1.Acc
x1_new_targ_loc
D167
x1_run_jog3_pick
C120
OUT
11
x2_ctr.Acc
$HSIO_003_CT2.Acc
x2_new_targ_loc
D168
x2_run_jog3_pick
C121
OUT
12
x1_run_jog3_pick
C120
motion_not_OK
C5
x1_not_allowed
C59
x1_move
C122
OUT
13
x2_run_jog3_pick
C121
motion_not_OK
C5
x2_not_allowed
C79
x2_move
C123
OUT
14
x1_move
C122
x2_move
C123
x_both_move
C124
OUT
15
x_both_move
C124
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50

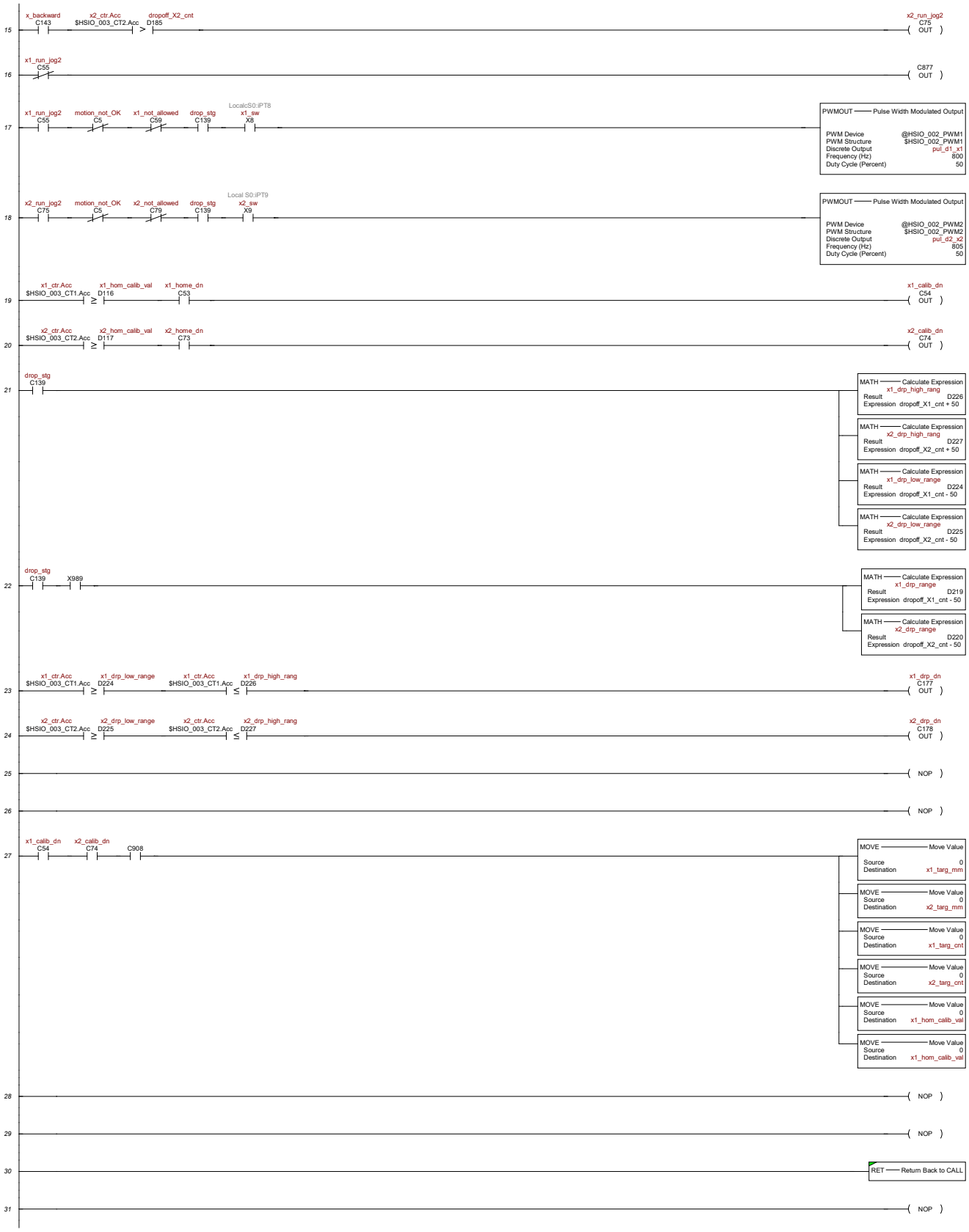

PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
16
x1_run_jog3_pick C120
motion_not_OK C5
x1_not_allowed C59
C676
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
17
x2_run_jog3_pick C121
motion_not_OK C5
x2_not_allowed C79
C565
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
18
x_both_move C124
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 800
Duty Cycle (Percent) 50
19
x1_ctr.Acc $HSIO_003_CT1.Acc ≥ x1_new_targ_loc D167
pick_stg C211
x1_new_loc_dn C170 OUT
20
x2_ctr.Acc $HSIO_003_CT2.Acc ≥ x2_new_targ_loc D168
pick_stg C211
x2_new_loc_dn C171 OUT
21
x1_new_loc_dn C170
x2_new_loc_dn C171
MOVE Move Value
Source 0
Destination x1_targ_mm
MOVE Move Value
Source 0
Destination x2_targ_mm
MOVE Move Value
Source 0
Destination x1_targ_cnt
MOVE Move Value
Source 0
Destination x2_targ_cnt
MOVE Move Value
Source 0
Destination x1_hom_calib_val
MOVE Move Value
Source 0
Destination x1_hom_calib_val
Local S0:oPT1
blue
Y1
OUT
22
NOP
23
NOP
24
RET Return Back to CALL
25
NOP

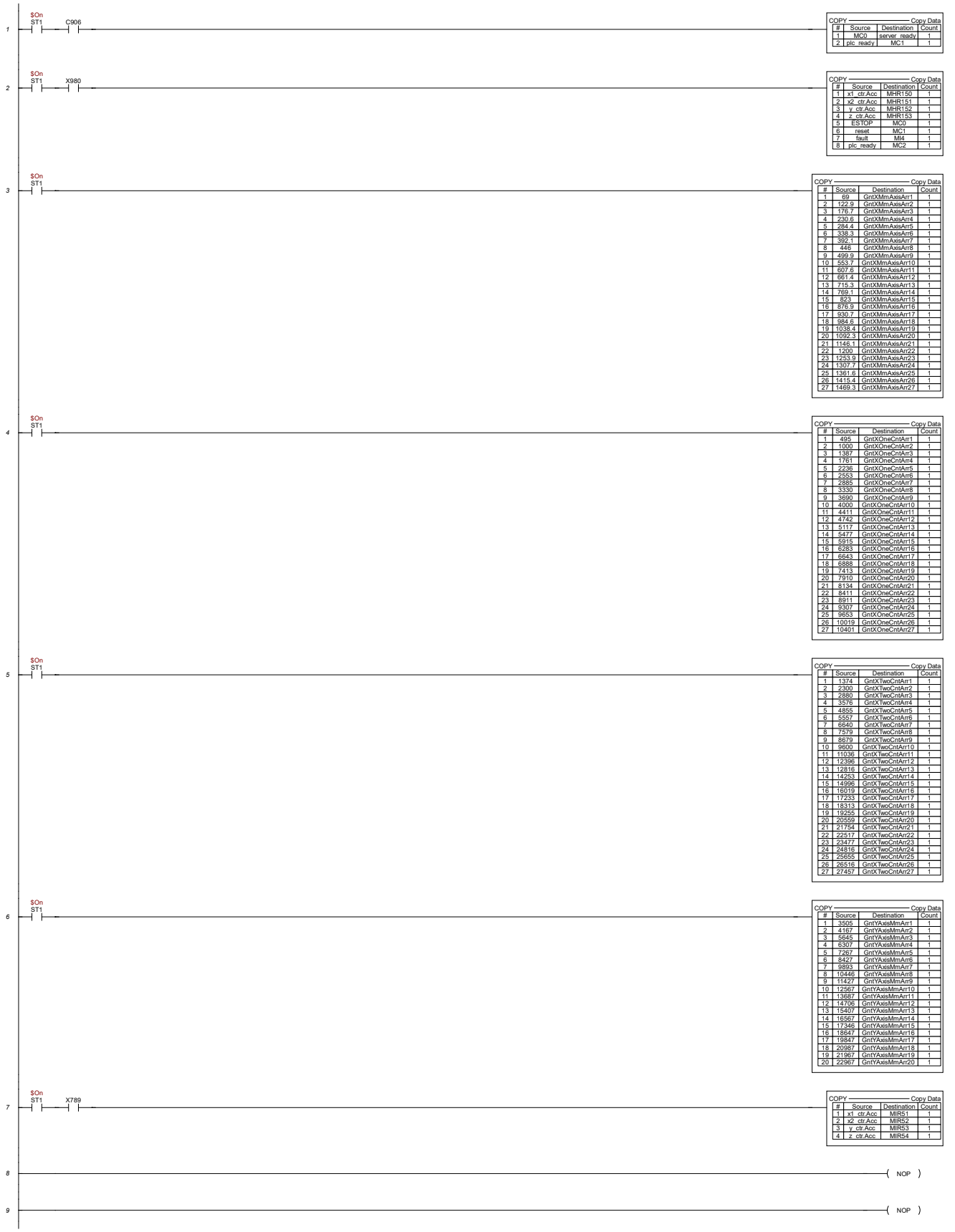

1
x1_ctr.Acc
$HSIO_003_CT1.Acc 5
x1_ctr.Acc
$HSIO_003_CT1.Acc -5
x1_home_range
C40
OUT
2
x2_ctr.Acc
$HSIO_003_CT2.Acc 5
x2_ctr.Acc
$HSIO_003_CT2.Acc -5
x2_home_range
C60
OUT
3
y_ctr.Acc
$HSIO_003_CT3.Acc 5
y_ctr.Acc
$HSIO_003_CT3.Acc -5
y_home_range
C80
OUT
4
z_ctr.Acc
$HSIO_003_CT4.Acc 5
z_ctr.Acc
$HSIO_003_CT4.Acc -5
z_home_range
C100
OUT
5
x1_home_range C40
x2_home_range C60
y_home_range C80
z_home_range C100
curr_home_range
C35
OUT
6
x1_ctr.Acc $HSIO_003_CT1.Acc 0
x2_ctr.Acc $HSIO_003_CT2.Acc 0
y_ctr.Acc $HSIO_003_CT3.Acc 0
z_ctr.Acc $HSIO_003_CT4.Acc 0
currently_home
C3
OUT
7
x1_active C31
x2_active C32
y_active C33
z_active C34
motion_not_OK C5
all_axis_active
C23
OUT
8
x1_active C31
motion_not_OK C5
x2_active C32
y_active C33
z_active C34
any_axis_active
C24
OUT
9
NOP
10
RET —— Return Back to CALL
11
NOP


1
NOP
2
NOP
3
RET —— Return Back to CALL
4
NOP


1
NOP
2
NOP
3
RET — Return Back to CALL
4
NOP


1
x1_hom_dnn C57
x2_hom_dnn C77
C980
COPY Copy Data
Source Destination Count
1 495 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
2
N0 0
N1 0
C546
COPY Copy Data
Source Destination Count
1 500 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
3
x1_hom_dnn C57
x2_hom_dnn C77
C980
MATH Calculate Expression
x1_targ_mm
Result D104
Expression x_cmd_mm * x1_actual_cnt
MATH Calculate Expression
x2_targ_mm
Result D105
Expression x_cmd_mm * x2_actual_cnt
MATH Calculate Expression
x1_targ_cnt
Result D118
Expression (x_cmd_mm * x1_actual_cnt) / x_tab_mm
MATH Calculate Expression
x2_targ_cnt
Result D119
Expression (x_cmd_mm * x2_actual_cnt) / x_tab_mm
4
x1_hom_dnn C57
x2_hom_dnn C77
C980
COPY Copy Data
Source Destination Count
1 x1_targ_cnt x1_hom_calib_val 1
COPY Copy Data
Source Destination Count
1 x2_targ_cnt x2_hom_calib_val 1
5
ext_node_conectd C208
dropoff_x_row V53
pickup_x_row V50
x_forward C142 OUT
6
ext_node_conectd C208
dropoff_x_row V53
pickup_x_row V50
x_backward C143 OUT
7
x_forward C142
Local S0:oPT8 dir_x1 Y8 SET
Local S0:oPT9 dir_x2 Y9 SET
8
x_backward C143
Local S0:oPT8 dir_x1 Y8 RST
Local S0:oPT9 dir_x2 Y9 RST
9
NOP
10
homing_active C20
x1_hom_dnn C57
x2_hom_dnn C77
drop_stg C139
C980
Local S0:oPT8 dir_x1 Y8 SET
Local S0:oPT9 dir_x2 Y9 SET
Y60 RST
11
x1_home_dn C53
LocalcS0:iPT8 x1_sw X8
x2_home_dn C73
Local S0:iPT9 x2_sw X9
x1_run_jog2_sw C56 OUT
x2_run_jog2_sw C76 OUT
12
x_forward C142
x1_ctr.Acc $HSIO_003_CT1.Acc
dropoff_X1_cnt D184
x1_run_jog2 C55 OUT
13
x_forward C142
x2_ctr.Acc $HSIO_003_CT2.Acc
dropoff_X2_cnt D185
x2_run_jog2 C75 OUT
14
x_backward C143
x1_ctr.Acc $HSIO_003_CT1.Acc
dropoff_X1_cnt D184
x1_run_jog2 C55 OUT


15
x_backward C143
x2_ctr.Acc $HSIO_003_CT2.Acc > dropoff_X2_cnt D185
x2_run_jog2 C75 OUT
16
x1_run_jog2 C55
C877 OUT
17
LocalcS0:iPT8
x1_run_jog2 C55
motion_not_OK C5
x1_not_allowed C59
drop_stg C139
x1_sw X8
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 800
Duty Cycle (Percent) 50
18
Local:S0:iPT9
x2_run_jog2 C75
motion_not_OK C5
x2_not_allowed C79
drop_stg C139
x2_sw X9
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 805
Duty Cycle (Percent) 50
19
x1_ctr.Acc $HSIO_003_CT1.Acc ≥ x1_hom_calib_val D116
x1_home_dn C53
x1_calib_dn C54 OUT
20
x2_ctr.Acc $HSIO_003_CT2.Acc ≥ x2_hom_calib_val D117
x2_home_dn C73
x2_calib_dn C74 OUT
21
drop_stg C139
MATH Calculate Expression x1_drp_high_rang Result D226 Expression dropoff_X1_cnt + 50
MATH Calculate Expression x2_drp_high_rang Result D227 Expression dropoff_X2_cnt + 50
MATH Calculate Expression x1_drp_low_range Result D224 Expression dropoff_X1_cnt - 50
MATH Calculate Expression x2_drp_low_range Result D225 Expression dropoff_X2_cnt - 50
22
drop_stg C139
X989
MATH Calculate Expression x1_drp_range Result D219 Expression dropoff_X1_cnt - 50
MATH Calculate Expression x2_drp_range Result D220 Expression dropoff_X2_cnt - 50
23
x1_ctr.Acc $HSIO_003_CT1.Acc ≥ x1_drp_low_range D224
x1_ctr.Acc $HSIO_003_CT1.Acc ≤ x1_drp_high_rang D226
x1_drp_dn C177 OUT
24
x2_ctr.Acc $HSIO_003_CT2.Acc ≥ x2_drp_low_range D225
x2_ctr.Acc $HSIO_003_CT2.Acc ≤ x2_drp_high_rang D227
x2_drp_dn C178 OUT
25
NOP
26
NOP
27
x1_calib_dn C54
x2_calib_dn C74
C908
MOVE Move Value Source 0 Destination x1_targ_mm
MOVE Move Value Source 0 Destination x2_targ_mm
MOVE Move Value Source 0 Destination x1_targ_cnt
MOVE Move Value Source 0 Destination x2_targ_cnt
MOVE Move Value Source 0 Destination x1_hom_calib_val
MOVE Move Value Source 0 Destination x1_hom_calib_val
28
NOP
29
NOP
30
RET Return Back to CALL
31
NOP



COPY — Copy Data

#	Source	Destination	Count
1	MC0	server_ready	1
2	plc_ready	MC1	1

COPY — Copy Data

#	Source	Destination	Count
1	x1_ctr.Acc	MHR150	1
2	x2_ctr.Acc	MHR151	1
3	y_ctr.Acc	MHR152	1
4	z_ctr.Acc	MHR153	1
5	ESTOP	MC0	1
6	reset	MC1	1
7	fault	MI4	1
8	plc_ready	MC2	1

COPY — Copy Data

#	Source	Destination	Count
1	69	GntXMmAxisArr1	1
2	122.9	GntXMmAxisArr2	1
3	176.7	GntXMmAxisArr3	1
4	230.6	GntXMmAxisArr4	1
5	284.4	GntXMmAxisArr5	1
6	338.3	GntXMmAxisArr6	1
7	392.1	GntXMmAxisArr7	1
8	446	GntXMmAxisArr8	1
9	499.9	GntXMmAxisArr9	1
10	553.7	GntXMmAxisArr10	1
11	607.6	GntXMmAxisArr11	1
12	661.4	GntXMmAxisArr12	1
13	715.3	GntXMmAxisArr13	1
14	769.1	GntXMmAxisArr14	1
15	823	GntXMmAxisArr15	1
16	876.9	GntXMmAxisArr16	1
17	930.7	GntXMmAxisArr17	1
18	984.6	GntXMmAxisArr18	1
19	1038.4	GntXMmAxisArr19	1
20	1092.3	GntXMmAxisArr20	1
21	1146.1	GntXMmAxisArr21	1
22	1200	GntXMmAxisArr22	1
23	1253.9	GntXMmAxisArr23	1
24	1307.7	GntXMmAxisArr24	1
25	1361.6	GntXMmAxisArr25	1
26	1415.4	GntXMmAxisArr26	1
27	1469.3	GntXMmAxisArr27	1

COPY — Copy Data

#	Source	Destination	Count
1	495	GntXOneCntArr1	1
2	1000	GntXOneCntArr2	1
3	1387	GntXOneCntArr3	1
4	1761	GntXOneCntArr4	1
5	2236	GntXOneCntArr5	1
6	2553	GntXOneCntArr6	1
7	2885	GntXOneCntArr7	1
8	3330	GntXOneCntArr8	1
9	3690	GntXOneCntArr9	1
10	4000	GntXOneCntArr10	1
11	4411	GntXOneCntArr11	1
12	4742	GntXOneCntArr12	1
13	5117	GntXOneCntArr13	1
14	5477	GntXOneCntArr14	1
15	5915	GntXOneCntArr15	1
16	6283	GntXOneCntArr16	1
17	6643	GntXOneCntArr17	1
18	6888	GntXOneCntArr18	1
19	7413	GntXOneCntArr19	1
20	7910	GntXOneCntArr20	1
21	8134	GntXOneCntArr21	1
22	8411	GntXOneCntArr22	1
23	8911	GntXOneCntArr23	1
24	9307	GntXOneCntArr24	1
25	9653	GntXOneCntArr25	1
26	10019	GntXOneCntArr26	1
27	10401	GntXOneCntArr27	1

COPY — Copy Data

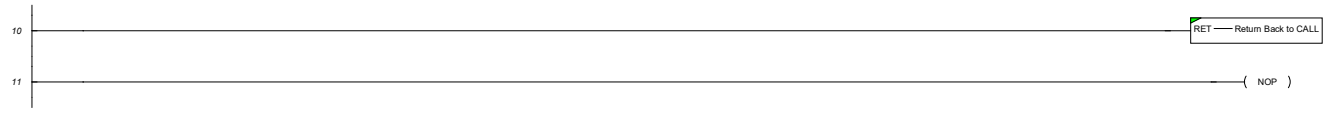
#	Source	Destination	Count
1	1374	GntXTwoCntArr1	1
2	2300	GntXTwoCntArr2	1
3	2880	GntXTwoCntArr3	1
4	3576	GntXTwoCntArr4	1
5	4855	GntXTwoCntArr5	1
6	5557	GntXTwoCntArr6	1
7	6640	GntXTwoCntArr7	1
8	7579	GntXTwoCntArr8	1
9	8679	GntXTwoCntArr9	1
10	9600	GntXTwoCntArr10	1
11	11036	GntXTwoCntArr11	1
12	12396	GntXTwoCntArr12	1
13	12816	GntXTwoCntArr13	1
14	14253	GntXTwoCntArr14	1
15	14996	GntXTwoCntArr15	1
16	16019	GntXTwoCntArr16	1
17	17233	GntXTwoCntArr17	1
18	18313	GntXTwoCntArr18	1
19	19255	GntXTwoCntArr19	1
20	20559	GntXTwoCntArr20	1
21	21754	GntXTwoCntArr21	1
22	22517	GntXTwoCntArr22	1
23	23477	GntXTwoCntArr23	1
24	24816	GntXTwoCntArr24	1
25	25655	GntXTwoCntArr25	1
26	26516	GntXTwoCntArr26	1
27	27457	GntXTwoCntArr27	1

COPY — Copy Data

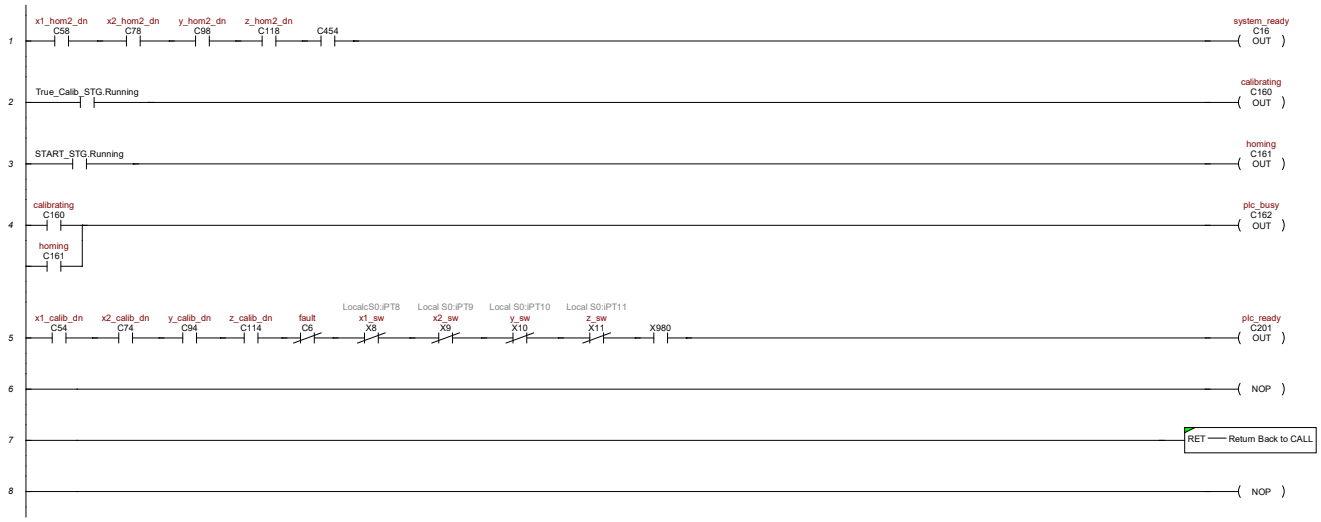
#	Source	Destination	Count
1	3505	GntYAxisMmArr1	1
2	4167	GntYAxisMmArr2	1
3	5645	GntYAxisMmArr3	1
4	6307	GntYAxisMmArr4	1
5	7267	GntYAxisMmArr5	1
6	8427	GntYAxisMmArr6	1
7	9893	GntYAxisMmArr7	1
8	10446	GntYAxisMmArr8	1
9	11427	GntYAxisMmArr9	1
10	12567	GntYAxisMmArr10	1
11	13687	GntYAxisMmArr11	1
12	14706	GntYAxisMmArr12	1
13	15407	GntYAxisMmArr13	1
14	16567	GntYAxisMmArr14	1
15	17346	GntYAxisMmArr15	1
16	18647	GntYAxisMmArr16	1
17	19847	GntYAxisMmArr17	1
18	20987	GntYAxisMmArr18	1
19	21967	GntYAxisMmArr19	1
20	22967	GntYAxisMmArr20	1

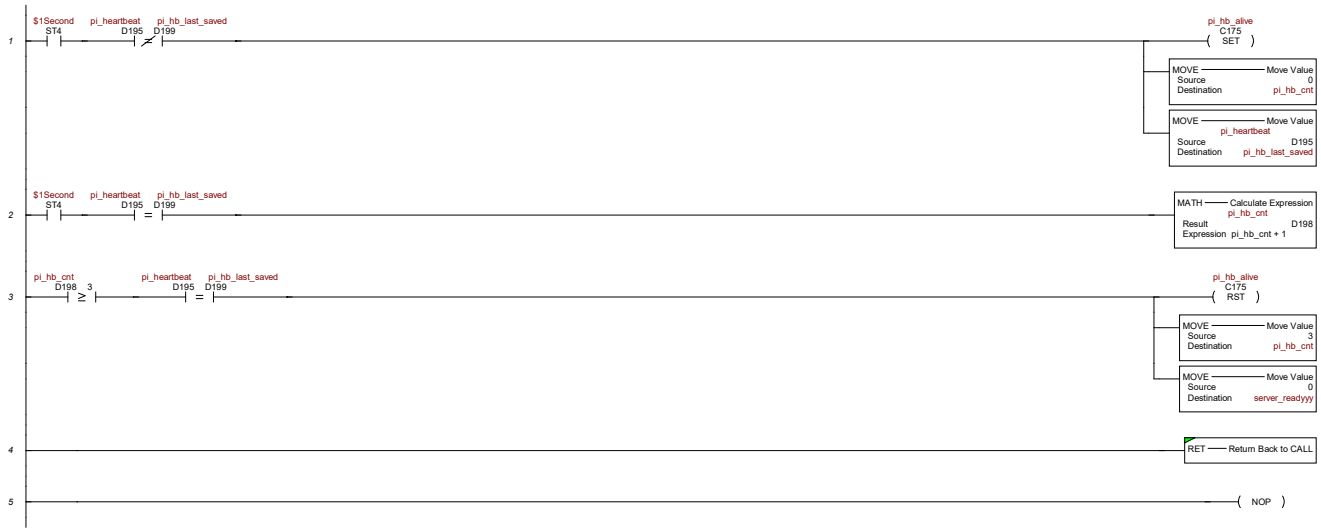
COPY — Copy Data

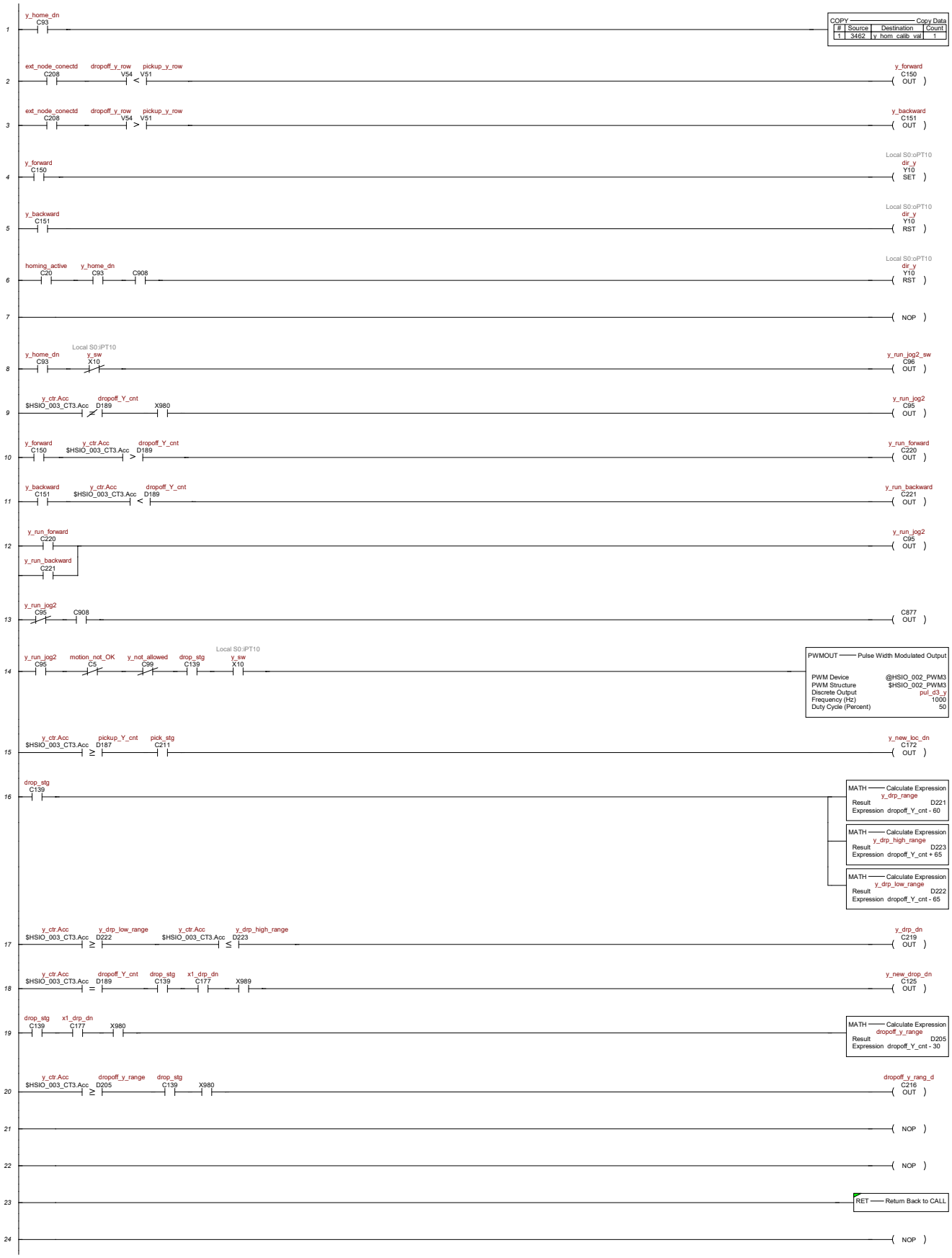
#	Source	Destination	Count
1	x1_ctr.Acc	MIR51	1
2	x2_ctr.Acc	MIR52	1
3	y_ctr.Acc	MIR53	1
4	z_ctr.Acc	MIR54	1

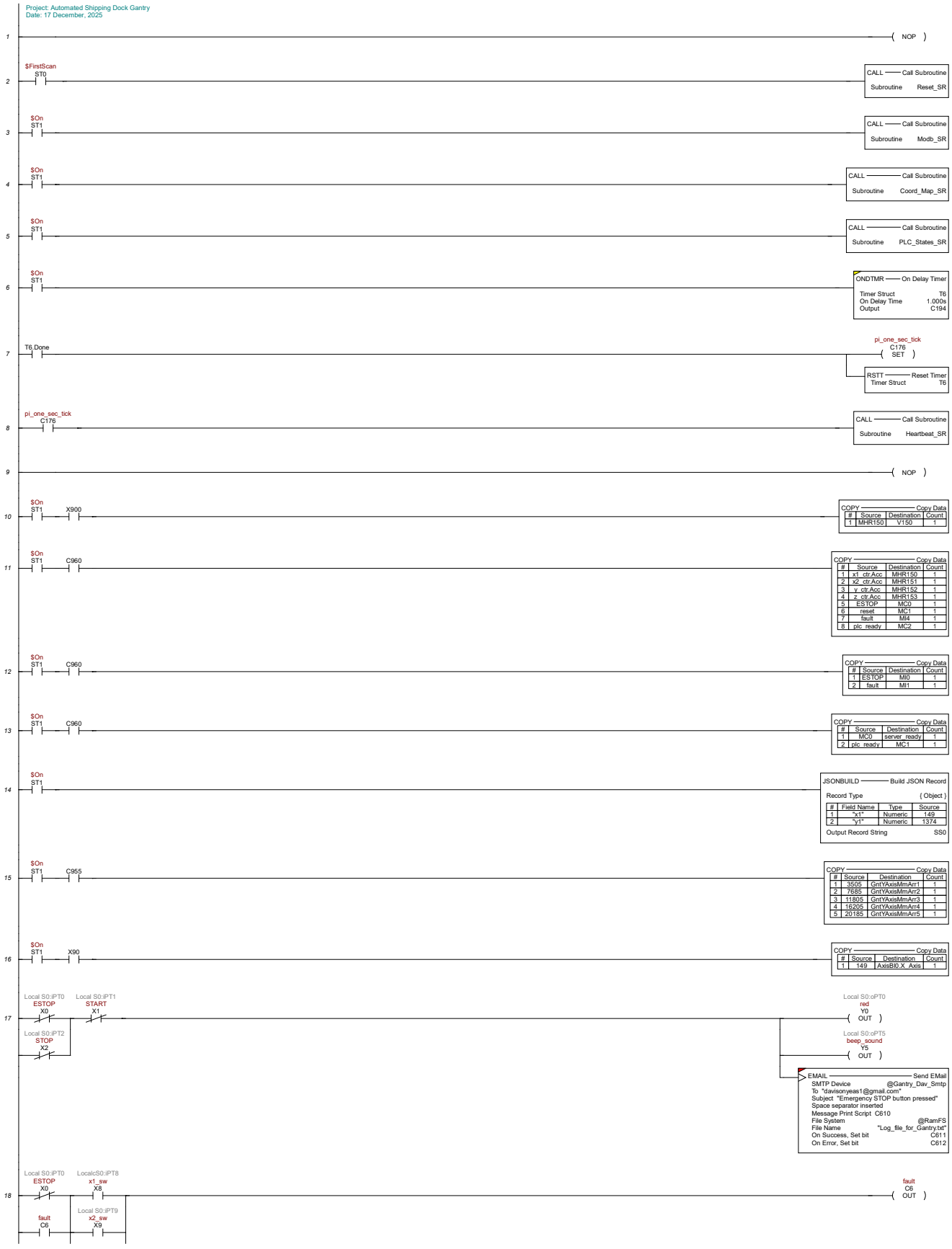

10
RET —— Return Back to CALL
11
(NOP)

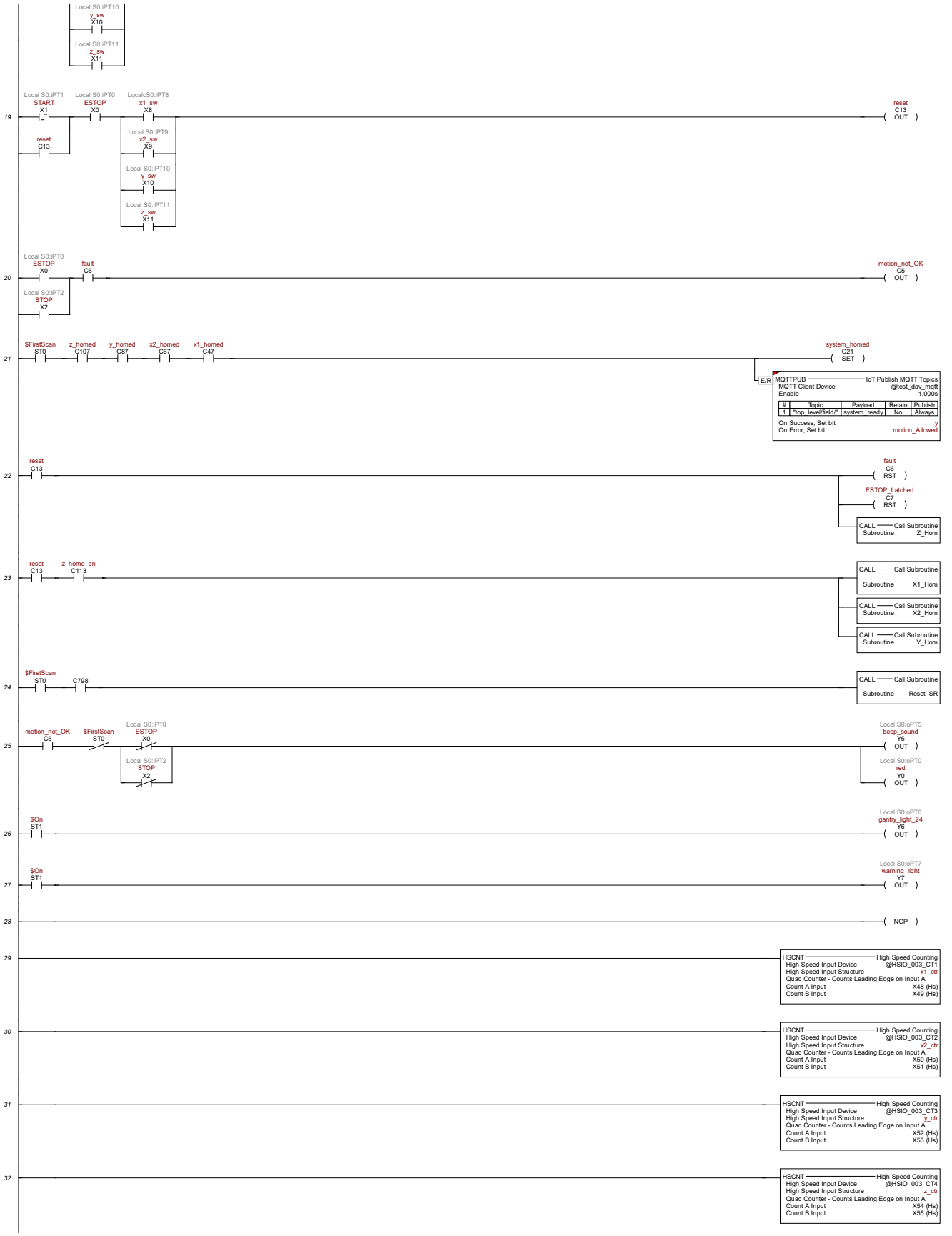

Local S0:iPT0
ESTOP
X0
estop_rev
C145
SET
Local S0:iPT0
ESTOP
X0
estop_rev
C145
RST
LocalcS0:iPT8
x1_sw
X8
Local S0:iPT9
x2_sw
X9
C905
COPY
Copy Data
Source Destination Count
1 0 x2_ctr_acc 1
$On
ST1
COPY
Copy Data
Source Destination Count
1 ESTOP MC1 1
2 estop_rev MC2 1
3 fault MC3 1
4 plc_busy MC4 1
5 plc_ready MC5 1
6 server_ready MC6 1
7 all_sys_homed MC7 1
$On
ST1
COPY
Copy Data
Source Destination Count
1 x1_ctr.Acc MHR151 1
2 x2_ctr.Acc MHR152 1
3 y_ctr.Acc MHR153 1
4 z_ctr.Acc MHR154 1
5 MHR197 pl_heartbeat 1
$On
ST1
COPY
Copy Data
Source Destination Count
1 MHR50 pickup_x_row 1
2 MHR51 pickup_y_row 1
3 MHR52 pickup_z_row 1
4 MHR53 dropoff_x_row 1
5 MHR54 dropoff_y_row 1
6 MHR55 dropoff_z_row 1
7 MHR211 CMD_ID_REG 1
8 MHR212 START_CMD_REG 1
9 MHR213 ACK_ID_REG 1
10 MHR214 SERVER_READY_REG 1
11 MHR215 PLC_READY_REG 1
RET Return Back to CALL
NOP


1
x1_hom2_dn C58
x2_hom2_dn C78
y_hom2_dn C98
z_hom2_dn C118
C454
system_ready C16 OUT
2
True_Calib_STG.Running
calibrating C160 OUT
3
START_STG.Running
homing C161 OUT
4
calibrating C160
homing C161
plc_busy C162 OUT
5
x1_calib_dn C54
x2_calib_dn C74
y_calib_dn C94
z_calib_dn C114
fault C6
Local:S0:iPT8
x1_sw X8
Local:S0:iPT9
x2_sw X9
Local:S0:iPT10
y_sw X10
Local:S0:iPT11
z_sw X11
X980
plc_ready C201 OUT
6
NOP
7
RET Return Back to CALL
8
NOP


1
$1Second
ST4
pi_heartbeat
D195
pi_hb_last_saved
D199
pi_hb_alive
C175
SET
MOVE
Move Value
Source
0
Destination
pi_hb_cnt
MOVE
Move Value
pi_heartbeat
Source
D195
Destination
pi_hb_last_saved
2
$1Second
ST4
pi_heartbeat
D195
=
pi_hb_last_saved
D199
MATH
Calculate Expression
pi_hb_cnt
Result
D198
Expression
pi_hb_cnt + 1
3
pi_hb_cnt
D198
3
≥
pi_heartbeat
D195
=
pi_hb_last_saved
D199
pi_hb_alive
C175
RST
MOVE
Move Value
Source
3
Destination
pi_hb_cnt
MOVE
Move Value
Source
0
Destination
server_readyyy
4
RET
Return Back to CALL
5
NOP

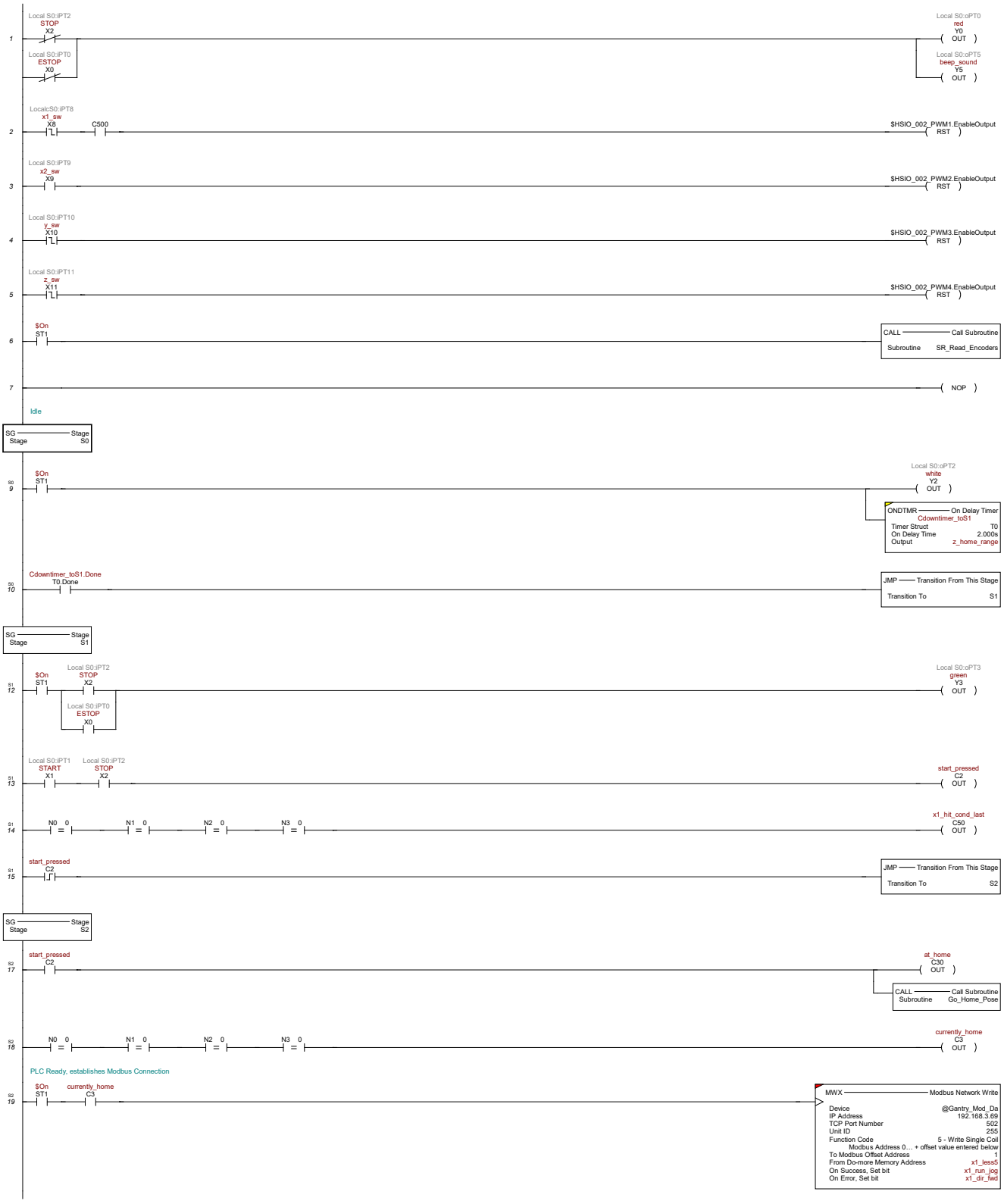

1 y_home_dn C93 COPY Copy Data F Source Destination Count 1 3462 y_hom_calib_val 1
2 ext_node_conectd C208 dropoff_y_row V54 < pickup_y_row V51 y_forward C150 OUT
3 ext_node_conectd C208 dropoff_y_row V54 > pickup_y_row V51 y_backward C151 OUT
4 y_forward C150 Local S0:oPT10 dir_y Y10 SET
5 y_backward C151 Local S0:oPT10 dir_y Y10 RST
6 homing_active C20 y_home_dn C93 C908 Local S0:oPT10 dir_y Y10 RST
7 NOP
8 y_home_dn C93 Local S0:iPT10 y_sw X10 y_run_jog2_sw C96 OUT
9 y_ctr.Acc $HSIO_003_CT3.Acc < dropoff_Y_cnt D189 X980 y_run_jog2 C95 OUT
10 y_forward C150 y_ctr.Acc $HSIO_003_CT3.Acc > dropoff_Y_cnt D189 y_run_forward C220 OUT
11 y_backward C151 y_ctr.Acc $HSIO_003_CT3.Acc < dropoff_Y_cnt D189 y_run_backward C221 OUT
12 y_run_forward C220 y_run_backward C221 y_run_jog2 C95 OUT
13 y_run_jog2 C95 C908 C877 OUT
14 y_run_jog2 C95 motion_not_OK C5 y_not_allowed C99 drop_stg C139 Local S0:iPT10 y_sw X10 PWMOUT Pulse Width Modulated Output PWM Device @HSIO_002_PWM3 PWM Structure $HSIO_002_PWM3 Discrete Output pul_d3_y Frequency (Hz) 1000 Duty Cycle (Percent) 50
15 y_ctr.Acc $HSIO_003_CT3.Acc ≥ pickup_Y_cnt D187 pick_stg C211 y_new_loc_dn C172 OUT
16 drop_stg C139 MATH Calculate Expression y_drp_range Result D221 Expression dropoff_Y_cnt - 60 MATH Calculate Expression y_drp_high_range Result D223 Expression dropoff_Y_cnt + 65 MATH Calculate Expression y_drp_low_range Result D222 Expression dropoff_Y_cnt - 65
17 y_ctr.Acc $HSIO_003_CT3.Acc ≥ y_drp_low_range D222 y_ctr.Acc $HSIO_003_CT3.Acc ≤ y_drp_high_range D223 y_drp_dn C219 OUT
18 y_ctr.Acc $HSIO_003_CT3.Acc = dropoff_Y_cnt D189 drop_stg C139 x1_drp_dn C177 X989 y_new_drop_dn C125 OUT
19 drop_stg C139 x1_drp_dn C177 X980 MATH Calculate Expression dropoff_y_range Result D205 Expression dropoff_Y_cnt - 30
20 y_ctr.Acc $HSIO_003_CT3.Acc ≥ dropoff_y_range D205 drop_stg C139 X980 dropoff_y_rang_d C216 OUT
21 NOP
22 NOP
23 RET Return Back to CALL
24 NOP

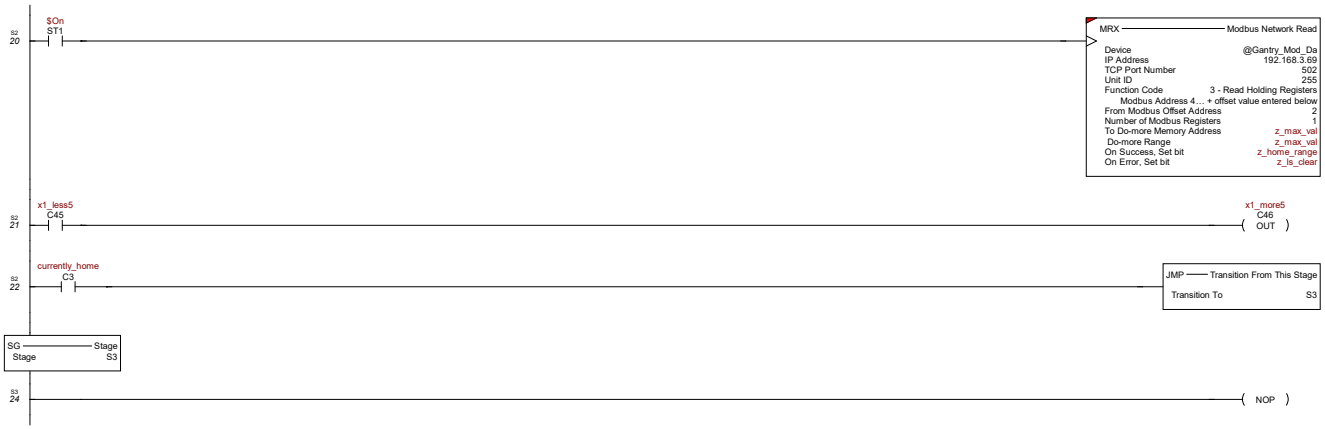

Project: Automated Shipping Dock Gantry
Date: 17 December, 2025
1 NOP
2 $FirstScan ST0 CALL Call Subroutine Subroutine Reset_SR
3 $On ST1 CALL Call Subroutine Subroutine Modb_SR
4 $On ST1 CALL Call Subroutine Subroutine Coord_Map_SR
5 $On ST1 CALL Call Subroutine Subroutine PLC_States_SR
6 $On ST1 ONDTMR On Delay Timer Timer Struct T6 On Delay Time 1.000s Output C194
7 T6.Done pi_one_sec_tick C176 SET RSTT Reset Timer Timer Struct T6
8 pi_one_sec_tick C176 CALL Call Subroutine Subroutine Heartbeat_SR
9 NOP
10 $On ST1 X900 COPY Copy Data # Source Destination Count 1 MHR150 V150 1
11 $On ST1 C960 COPY Copy Data # Source Destination Count 1 x1_ctr.Acc MHR150 1 2 x2_ctr.Acc MHR151 1 3 y_ctr.Acc MHR152 1 4 z_ctr.Acc MHR153 1 5 ESTOP MC0 1 6 reset MC1 1 7 fault MI4 1 8 plc_ready MC2 1
12 $On ST1 C960 COPY Copy Data # Source Destination Count 1 ESTOP MI0 1 2 fault MI1 1
13 $On ST1 C960 COPY Copy Data # Source Destination Count 1 MC0 server_ready 1 2 plc_ready MC1 1
14 $On ST1 JSONBUILD Build JSON Record Record Type { Object } # Field Name Type Source 1 "x1" Numeric 149 2 "y1" Numeric 1374 Output Record String SS0
15 $On ST1 C955 COPY Copy Data # Source Destination Count 1 3505 GntYAxisMmArr1 1 2 7685 GntYAxisMmArr2 1 3 11805 GntYAxisMmArr3 1 4 16205 GntYAxisMmArr4 1 5 20185 GntYAxisMmArr5 1
16 $On ST1 X90 COPY Copy Data # Source Destination Count 1 149 AxisB0.X_Axis 1
17 Local S0:iPT0 ESTOP X0 Local S0:iPT1 START X1 Local S0:iPT2 STOP X2 Local S0:oPT0 red Y0 OUT Local S0:oPT5 beep_sound Y5 OUT EMAIL Send EMail SMTP Device @Gantry_Dav_Smtp To "davisonyeas1@gmail.com" Subject "Emergency STOP button pressed" Space separator inserted Message Print Script C610 File System @RamFS File Name "Log_file_for_Gantry.txt" On Success, Set bit C611 On Error, Set bit C612
18 Local S0:iPT0 ESTOP X0 LocalcS0:iPT8 x1_sw X8 fault C6 Local S0:iPT9 x2_sw X9 fault C6 OUT

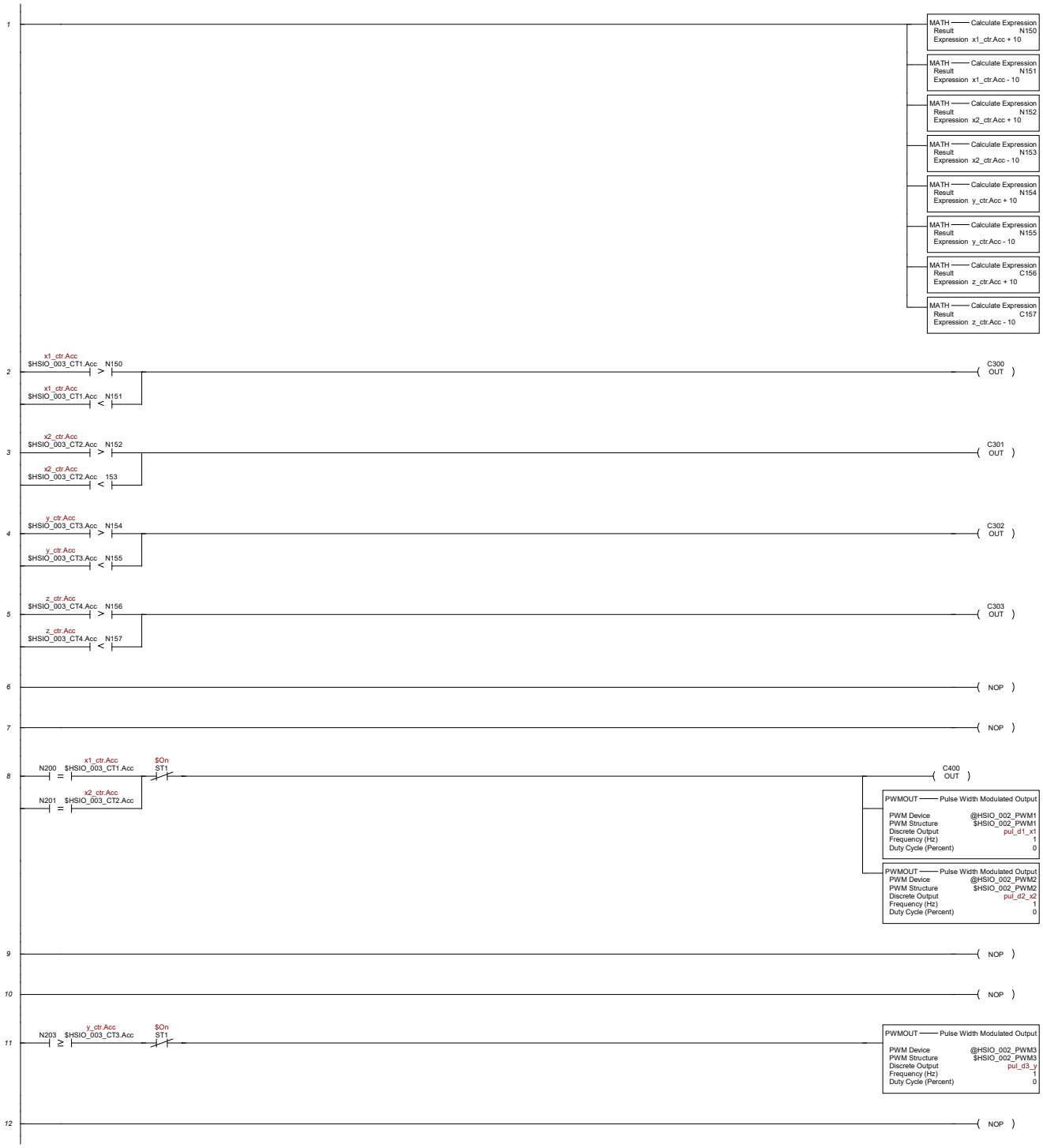

Local S0:iPT10
y_sw
X10
Local S0:iPT11
z_sw
X11
19
Local S0:iPT1
START
X1
Local S0:iPT0
ESTOP
X0
LocalcS0:iPT8
x1_sw
X8
reset
C13
Local S0:iPT9
x2_sw
X9
Local S0:iPT10
y_sw
X10
Local S0:iPT11
z_sw
X11
reset
C13
OUT
20
Local S0:iPT0
ESTOP
X0
fault
C6
Local S0:iPT2
STOP
X2
motion_not_OK
C5
OUT
21
$FirstScan
ST0
z_homed
C107
y_homed
C87
x2_homed
C67
x1_homed
C47
system_homed
C21
SET
E/R
MQTTPUB
IoT Publish MQTT Topics
MQTT Client Device
@test_dav_mqtt
Enable
1.000s
Topic Payload Retain Publish
1 "top_level/field/" system_ready No Always
On Success, Set bit
y
On Error, Set bit
motion_Allowed
22
reset
C13
fault
C6
RST
ESTOP_Latched
C7
RST
CALL
Call Subroutine
Subroutine
Z_Hom
23
reset
C13
z_home_dn
C113
CALL
Call Subroutine
Subroutine
X1_Hom
CALL
Call Subroutine
Subroutine
X2_Hom
CALL
Call Subroutine
Subroutine
Y_Hom
24
$FirstScan
ST0
C798
CALL
Call Subroutine
Subroutine
Reset_SR
25
motion_not_OK
C5
$FirstScan
ST0
Local S0:iPT0
ESTOP
X0
Local S0:iPT2
STOP
X2
Local S0:oPT5
beep_sound
Y5
OUT
Local S0:oPT0
red
Y0
OUT
26
$On
ST1
Local S0:oPT6
gantry_light_24
Y6
OUT
27
$On
ST1
Local S0:oPT7
warning_light
Y7
OUT
28
NOP
29
HSCNT
High Speed Counting
High Speed Input Device
@HSIO_003_CT1
High Speed Input Structure
x1_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input
X48 (Hs)
Count B Input
X49 (Hs)
30
HSCNT
High Speed Counting
High Speed Input Device
@HSIO_003_CT2
High Speed Input Structure
x2_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input
X50 (Hs)
Count B Input
X51 (Hs)
31
HSCNT
High Speed Counting
High Speed Input Device
@HSIO_003_CT3
High Speed Input Structure
y_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input
X52 (Hs)
Count B Input
X53 (Hs)
32
HSCNT
High Speed Counting
High Speed Input Device
@HSIO_003_CT4
High Speed Input Structure
z_ctr
Quad Counter - Counts Leading Edge on Input A
Count A Input
X54 (Hs)
Count B Input
X55 (Hs)

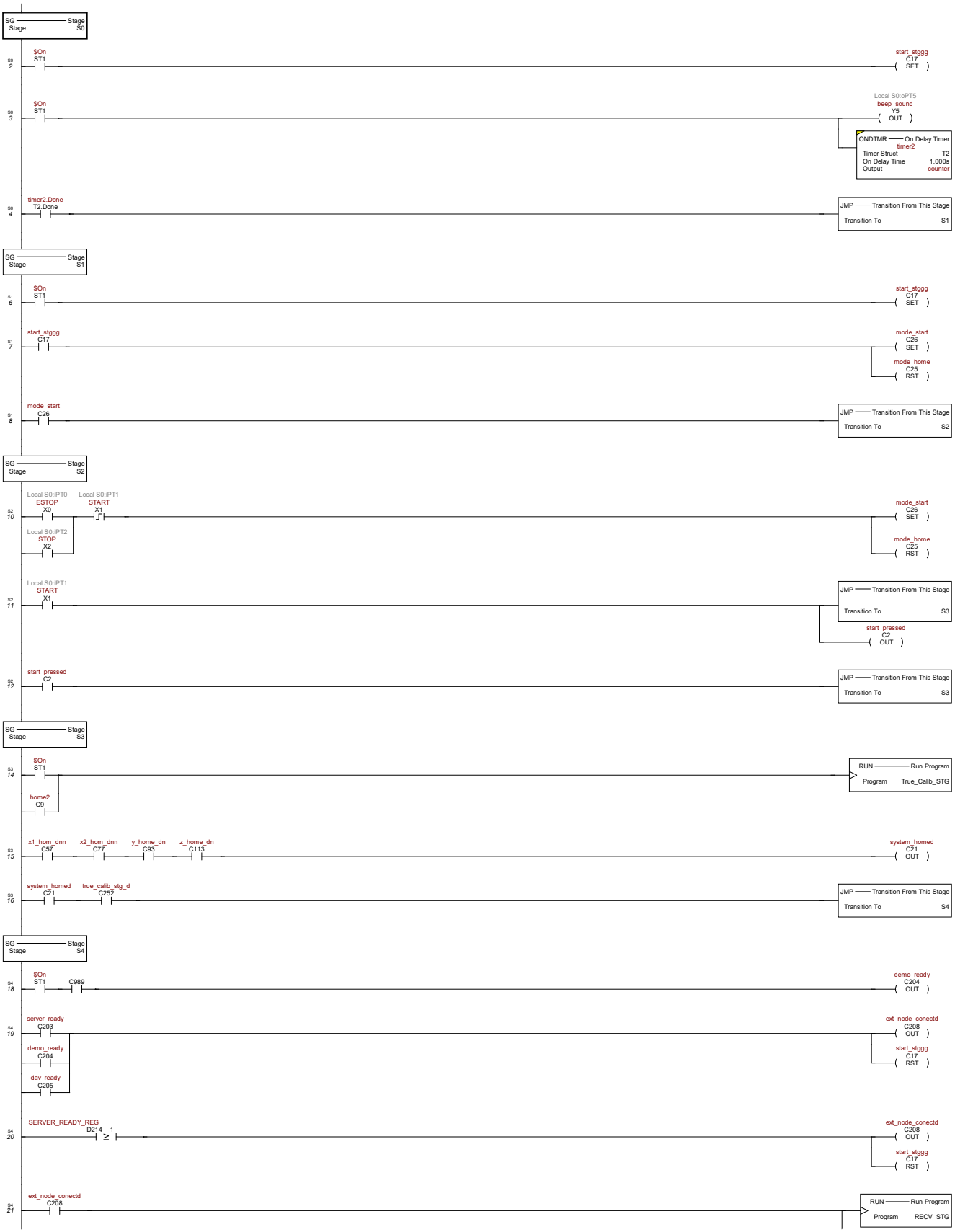

33
$On
ST1
COPY Copy Data
Source Destination Count
1 x1_ctr.Acc x1_ctr_acc 1
2 x2_ctr.Acc x2_ctr_acc 1
3 y_ctr.Acc y_ctr_acc 1
4 z_ctr.Acc z_ctr_acc 1
34
$On
ST1
COPY Copy Data
Source Destination Count
1 9120 z_max_vall 1
COPY Copy Data
Source Destination Count
1 25000 y_max_vall 1
COPY Copy Data
Source Destination Count
1 30010 x2_max_vall 1
COPY Copy Data
Source Destination Count
1 10720 x1_max_vall 1
35
$On
ST1
MATH Calculate Expression
z_not_allowed
Result C119
Expression z_ctr.Acc >= z_max_vall
MATH Calculate Expression
y_not_allowed
Result C99
Expression y_ctr.Acc >= y_max_vall
MATH Calculate Expression
x2_not_allowed
Result C79
Expression x2_ctr.Acc >= x2_max_vall
MATH Calculate Expression
x1_not_allowed
Result C59
Expression x1_ctr.Acc >= x1_max_vall
MATH Calculate Expression
z_not_allowed_2
Result C132
Expression z_ctr.Acc >= z_max_vall
36
$On
ST1
C960
MATH Calculate Expression
z_not_allowed
Result C119
Expression z_ctr.Acc >= z_max_vall
37
z_ctr.Acc
$HSIO_003_CT4.Acc 9100
C687
C789
z_max_thresh_rea
C137
OUT
INTCONFIG Configure Interrupt
HSIO Module HSIO_003
ISR Code-Block Z_Max_Thr
Trigger Resource Match Register 1
Fire ISR when "@HSIO_003_CT4 (High speed Ctr/Tmr 4 Accumulator)"
is Greater Than or Equal to (>=) z_max_vall
38
z_ctr.Acc
$HSIO_003_CT4.Acc 915
C676
C789
C960
LocalPS2:oPT6
pul_d4_z
Y22
RSTI
39
$On
ST1
CALL Call Subroutine
Subroutine Axis_Ranges_SR
40
$On
ST1
Counter1.DnDone
CT0.DnDone
FREQCNT Frequency Counter
Counter1
Counter Struct CT0
Sample Time 0.100s
Frequency Units Counts per Minute
Output R0
RST
41
LocalcS0:iPT8
x1_sw
X8
Local S0:iPT9
x2_sw
X9
N0 0
N1 0
C454
COPY Copy Data
Source Destination Count
1 500 x_cmd_mm 1
COPY Copy Data
Source Destination Count
1 1410 x_tab_mm 1
COPY Copy Data
Source Destination Count
1 10720 x1_actual_cnt 1
2 30336 x2_actual_cnt 1
42
$On
ST1
MOVE Move Value
Source 7.6028
Destination x1_scale_const
MOVE Move Value
Source 21.5156
Destination x2_scale_const
43
x1_homed
C47
x2_homed
C67
C454
MATH Calculate Expression
x1_targ_mm
Result D104
Expression x_cmd_mm * x1_actual_cnt
MATH Calculate Expression
x2_targ_mm
Result D105
Expression x_cmd_mm * x2_actual_cnt
44
$On
ST1
C565
C676
OUT
45
LocalcS0:iPT8
x1_sw
X8
Local S0:iPT9
x2_sw
X9
Local S0:iPT10
y_sw
X10
Local S0:iPT11
z_sw
X11
all_sys_homed
C37
OUT

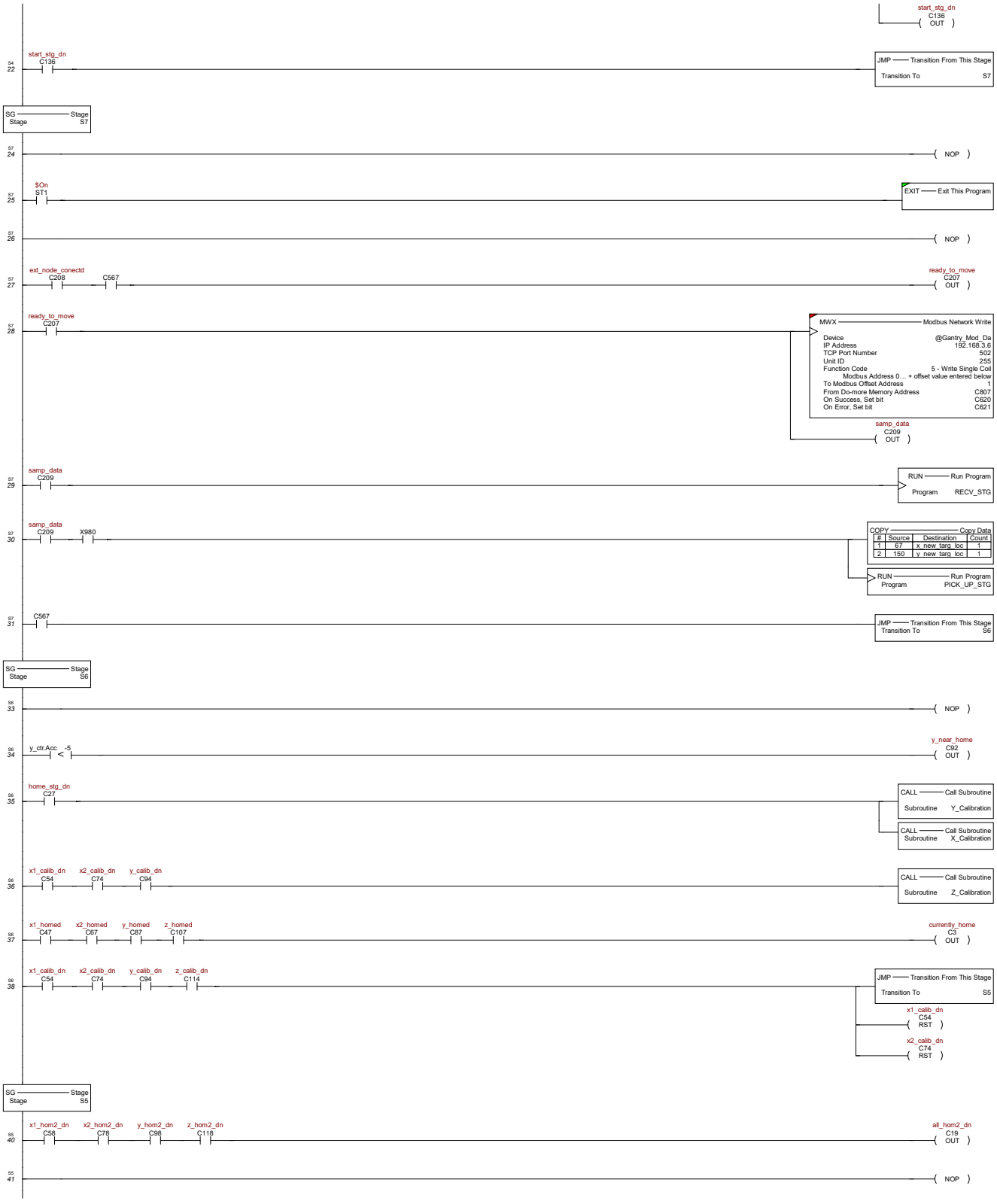

46
pick_stg C211
z_not_allowed C119
C960
z_run_jog3 C131 RST
z_run_jog4 C138 RST
47
pick_stg C211
y_not_allowed C99
y_run_jog2 C95 RST
48
$On ST1
RUN Run Program
Program START_STG
49
$On ST1
COPY Copy Data
Source Destination Count
1 -7769 MHR2:D 1
2 20960 MHR4:D 1
3 -13780 MHR6:D 1
4 4500 MHR8:D 1
50
$On ST1
COPY Copy Data
Source Destination Count
1 MHR2:D N200 1
2 MHR4:D N201 1
3 MHR6:D N202 1
4 MHR8:D N203 1
51
$50ms ST6
C676
COPY Copy Data
Source Destination Count
1 2000 z_up 1
2 9000 z_down 1
52
z_home_dn C113
pick_stg C211
y_new_loc_dn C172
C878
COPY Copy Data
Source Destination Count
1 9092 z_hom_calib_val 1
53
$On ST1
CALL Call Subroutine
Subroutine Lights
54
NOP

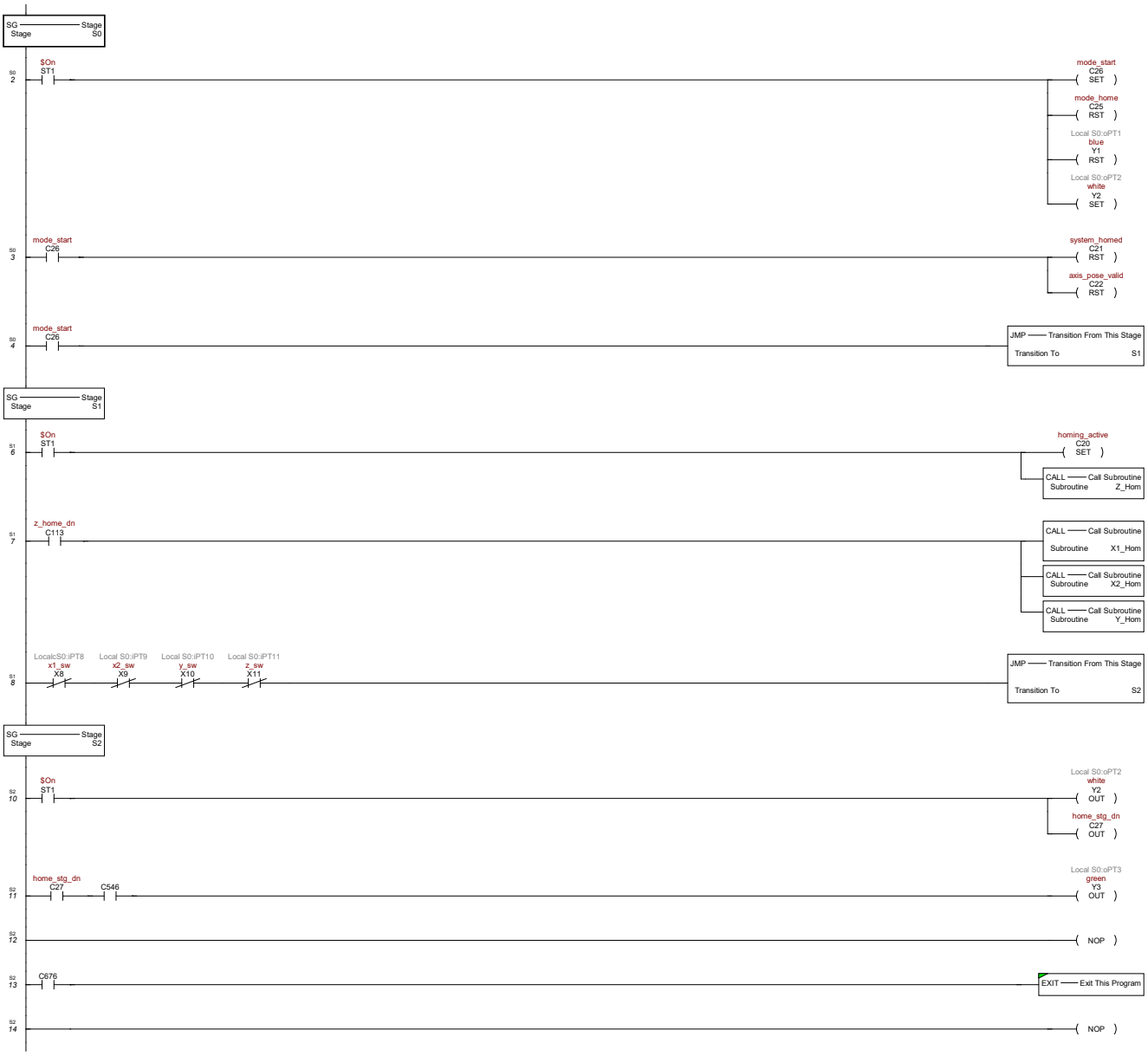

Local S0:iPT2
STOP
X2
Local S0:iPT0
ESTOP
X0
Local S0:oPT0
red
Y0
OUT
Local S0:oPT5
beep_sound
Y5
OUT
LocalcS0:iPT8
x1_sw
X8
C500
$HSIO_002_PWM1.EnableOutput
RST
Local S0:iPT9
x2_sw
X9
$HSIO_002_PWM2.EnableOutput
RST
Local S0:iPT10
y_sw
X10
$HSIO_002_PWM3.EnableOutput
RST
Local S0:iPT11
z_sw
X11
$HSIO_002_PWM4.EnableOutput
RST
$On
ST1
CALL
Call Subroutine
Subroutine
SR_Read_Encoders
NOP
Idle
SG
Stage
Stage
S0
Local S0:oPT2
white
Y2
OUT
ONDTMR
On Delay Timer
Cdowntimer_toS1
Timer Struct
T0
On Delay Time
2.000s
Output
z_home_range
Cdowntimer_toS1.Done
T0.Done
JMP
Transition From This Stage
Transition To
S1
Local S0:iPT3
green
Y3
N0 0
N1 0
N2 0
N3 0
x1_hit_cond_last
C50
start_pressed
C2
S2
at_home
C50
Go_Home_Pose
currently_home
C3
PLC Ready, establishes Modbus Connection
MWX
Modbus Network Write
Device
@Gantry_Mod_Da
IP Address
192.168.3.69
TCP Port Number
502
Unit ID
255
Function Code
5 - Write Single Coil
Modbus Address 0... + offset value entered below
To Modbus Offset Address
1
From Do-more Memory Address
x1_less5
On Success, Set bit
x1_run_jog
On Error, Set bit
x1_dir_fwd

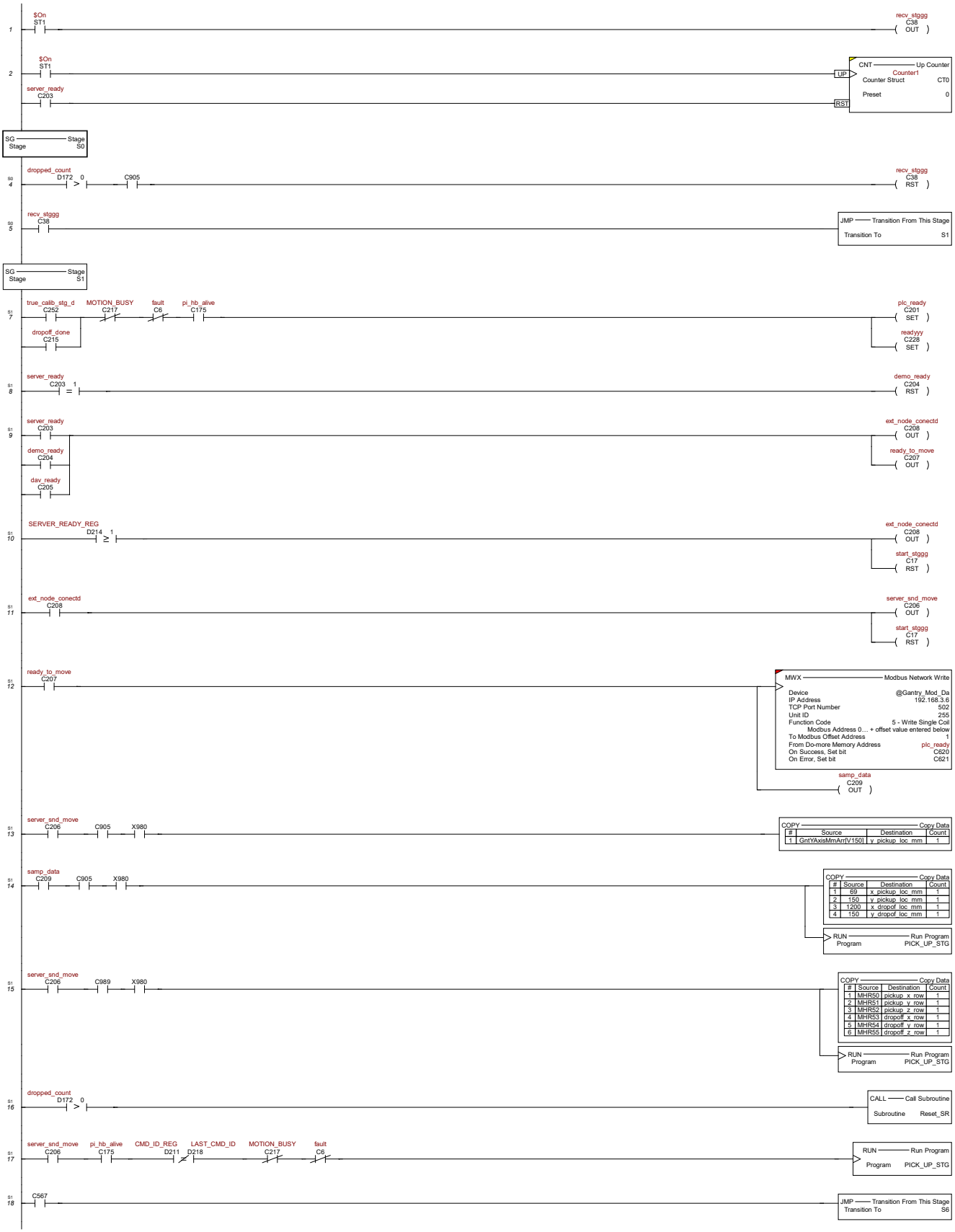

$On
ST1
MRX — Modbus Network Read
Device @Gantry_Mod_Da
IP Address 192.168.3.69
TCP Port Number 502
Unit ID 255
Function Code 3 - Read Holding Registers
Modbus Address 4... + offset value entered below
From Modbus Offset Address 2
Number of Modbus Registers 1
To Do-more Memory Address z_max_val
Do-more Range z_max_val
On Success, Set bit z_home_range
On Error, Set bit z_ls_clear
x1_less5
C45
x1_more5
C46
OUT
currently_home
C3
JMP — Transition From This Stage
Transition To S3
SG — Stage
Stage S3
NOP

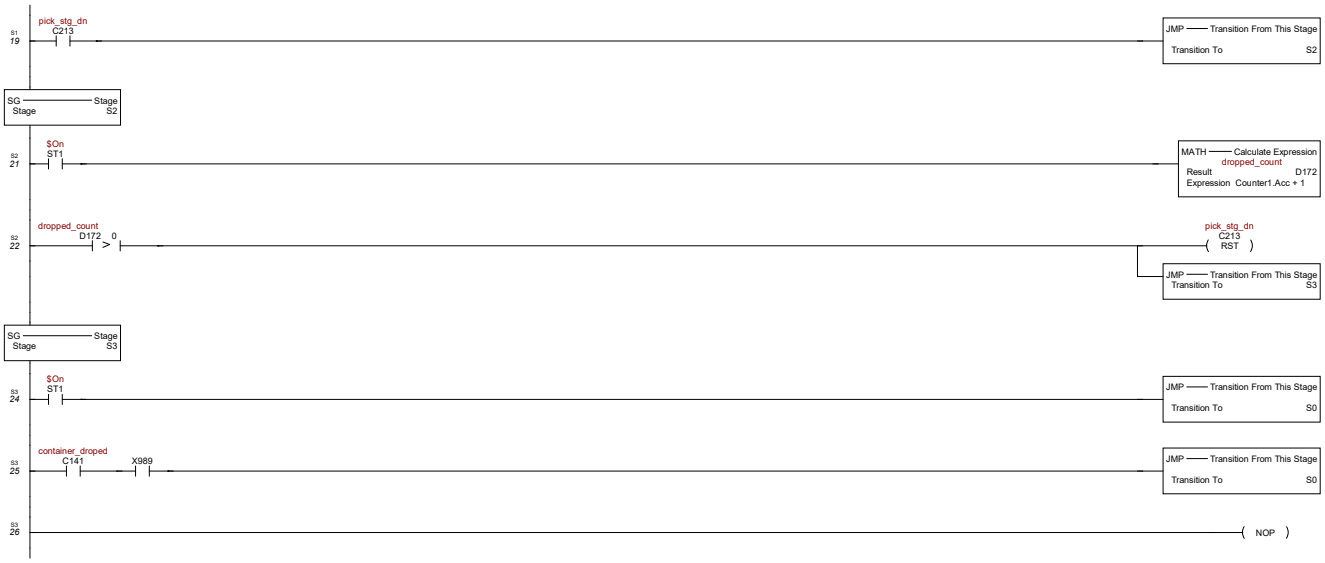

1
MATH Calculate Expression
Result N150
Expression x1_ctr.Acc + 10
MATH Calculate Expression
Result N151
Expression x1_ctr.Acc - 10
MATH Calculate Expression
Result N152
Expression x2_ctr.Acc + 10
MATH Calculate Expression
Result N153
Expression x2_ctr.Acc - 10
MATH Calculate Expression
Result N154
Expression y_ctr.Acc + 10
MATH Calculate Expression
Result N155
Expression y_ctr.Acc - 10
MATH Calculate Expression
Result C156
Expression z_ctr.Acc + 10
MATH Calculate Expression
Result C157
Expression z_ctr.Acc - 10
2
x1_ctr.Acc
$HSIO_003_CT1.Acc > N150
x1_ctr.Acc
$HSIO_003_CT1.Acc < N151
C300 OUT
3
x2_ctr.Acc
$HSIO_003_CT2.Acc > N152
x2_ctr.Acc
$HSIO_003_CT2.Acc < 153
C301 OUT
4
y_ctr.Acc
$HSIO_003_CT3.Acc > N154
y_ctr.Acc
$HSIO_003_CT3.Acc < N155
C302 OUT
5
z_ctr.Acc
$HSIO_003_CT4.Acc > N156
z_ctr.Acc
$HSIO_003_CT4.Acc < N157
C303 OUT
6
NOP
7
NOP
8
N200 = x1_ctr.Acc $HSIO_003_CT1.Acc $On ST1
N201 = x2_ctr.Acc $HSIO_003_CT2.Acc
C400 OUT
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM1
PWM Structure $HSIO_002_PWM1
Discrete Output pul_d1_x1
Frequency (Hz) 1
Duty Cycle (Percent) 0
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM2
PWM Structure $HSIO_002_PWM2
Discrete Output pul_d2_x2
Frequency (Hz) 1
Duty Cycle (Percent) 0
9
NOP
10
NOP
11
N203 ≥ y_ctr.Acc $HSIO_003_CT3.Acc $On ST1
PWMOUT Pulse Width Modulated Output
PWM Device @HSIO_002_PWM3
PWM Structure $HSIO_002_PWM3
Discrete Output pul_d3_y
Frequency (Hz) 1
Duty Cycle (Percent) 0
12
NOP

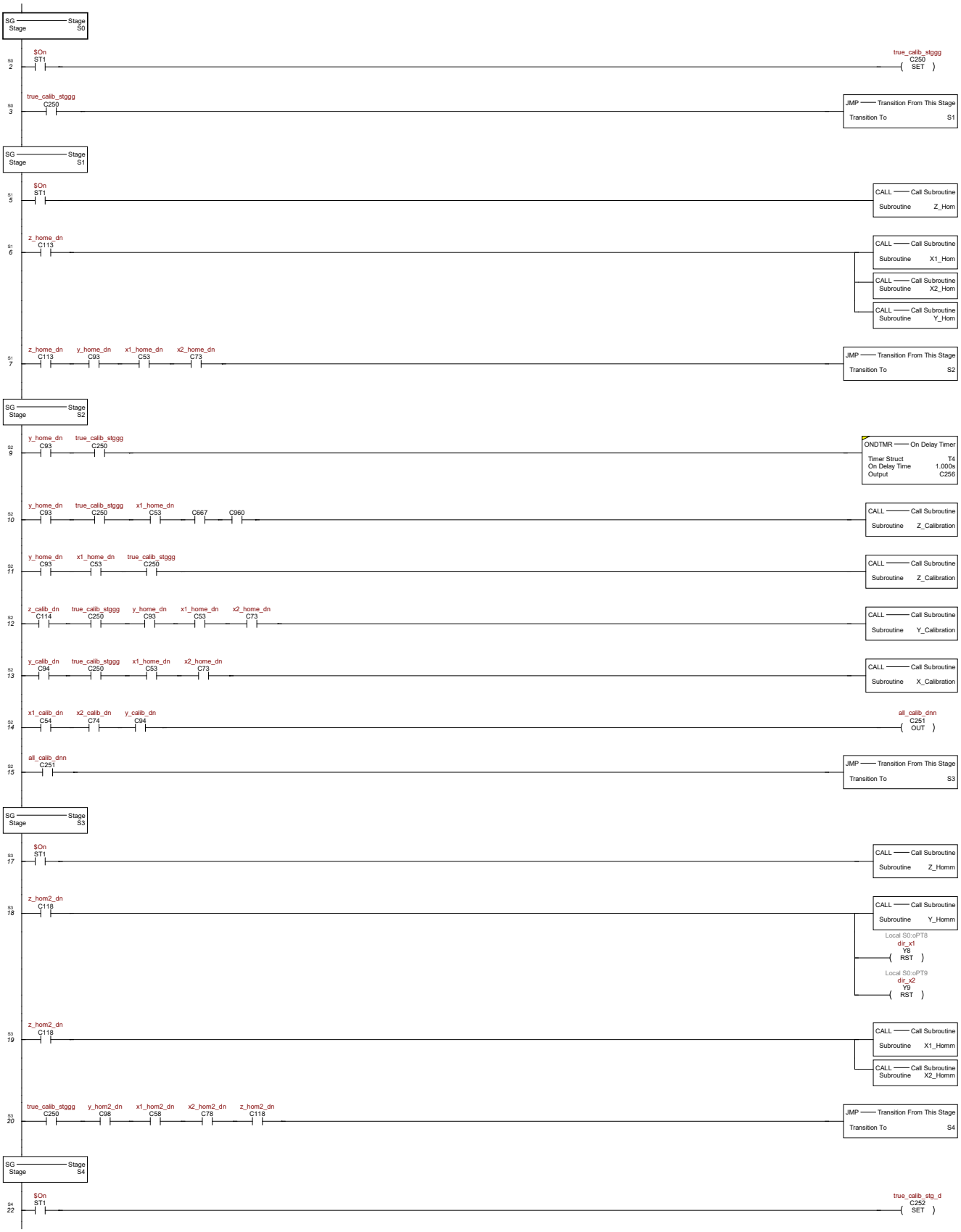

SG Stage
Stage S0
$On
ST1
start_stggg
C17
SET
Local S0:oPT5
beep_sound
Y5
OUT
ONDTMR On Delay Timer
timer2
Timer Struct T2
On Delay Time 1.000s
Output counter
timer2.Done
T2.Done
JMP Transition From This Stage
Transition To S1
SG Stage
Stage S1
mode_start
C26
SET
mode_home
C25
RST
JMP Transition From This Stage
Transition To S2
SG Stage
Stage S2
Local S0:iPT0
ESTOP
X0
Local S0:iPT1
START
X1
Local S0:iPT2
STOP
X2
JMP Transition From This Stage
Transition To S3
start_pressed
C2
OUT
SG Stage
Stage S3
RUN Run Program
Program True_Calib_STG
home2
C9
x1_hom_dnn
C57
x2_hom_dnn
C77
y_home_dn
C93
z_home_dn
C113
system_homed
C21
true_calib_stg_d
C252
SG Stage
Stage S4
C989
demo_ready
C204
server_ready
C203
ext_node_conectd
C208
dav_ready
C205
SERVER_READY_REG
D214
≥ 1
RUN Run Program
Program RECV_STG

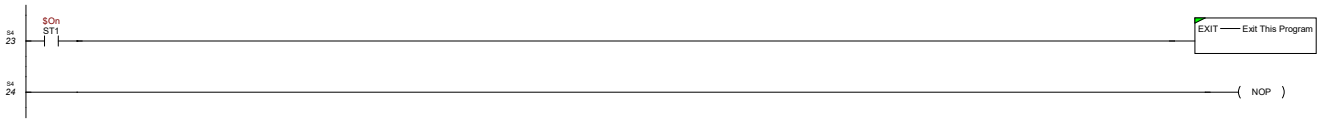

start_stg_dn
C136
OUT
start_stg_dn
C136
JMP Transition From This Stage
Transition To S7
SG Stage
Stage S7
NOP
$On
ST1
EXIT Exit This Program
NOP
ext_node_conectd
C208
C567
ready_to_move
C207
OUT
ready_to_move
C207
MWX Modbus Network Write
Device @Gantry_Mod_Da
IP Address 192.168.3.6
TCP Port Number 502
Unit ID 255
Function Code 5 - Write Single Coil
Modbus Address 0... + offset value entered below
To Modbus Offset Address 1
From Do-more Memory Address C807
On Success, Set bit C620
On Error, Set bit C621
samp_data
C209
OUT
samp_data
C209
RUN Run Program
Program RECV_STG
samp_data
C209
X980
COPY Copy Data
Source Destination Count
1 67 x_new_targ_loc 1
2 150 y_new_targ_loc 1
RUN Run Program
Program PICK_UP_STG
C567
JMP Transition From This Stage
Transition To S6
SG Stage
Stage S6
NOP
y_ctr.Acc
-5
y_near_home
C92
OUT
home_stg_dn
C27
CALL Call Subroutine
Subroutine Y_Calibration
CALL Call Subroutine
Subroutine X_Calibration
x1_calib_dn
C54
x2_calib_dn
C74
y_calib_dn
C94
CALL Call Subroutine
Subroutine Z_Calibration
x1_homed
C47
x2_homed
C67
y_homed
C87
z_homed
C107
currently_home
C3
OUT
x1_calib_dn
C54
x2_calib_dn
C74
y_calib_dn
C94
z_calib_dn
C114
JMP Transition From This Stage
Transition To S5
x1_calib_dn
C54
RST
x2_calib_dn
C74
RST
SG Stage
Stage S5
x1_hom2_dn
C58
x2_hom2_dn
C78
y_hom2_dn
C98
z_hom2_dn
C118
all_hom2_dn
C19
OUT
NOP

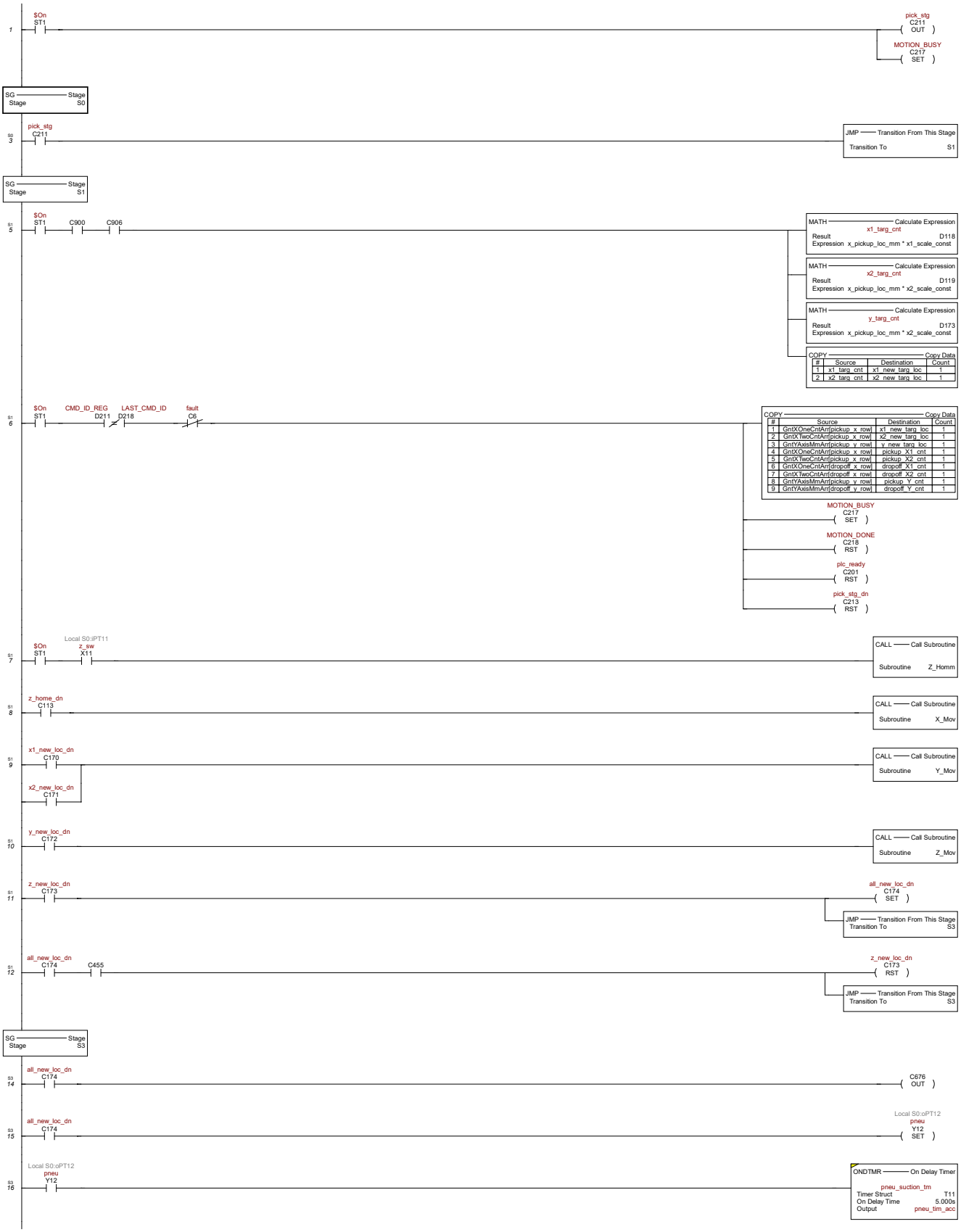

SG Stage
Stage S0
$On
ST1
mode_start
C26
SET
mode_home
C25
RST
Local S0:oPT1
blue
Y1
RST
Local S0:oPT2
white
Y2
SET
mode_start
C26
system_homed
C21
RST
axis_pose_valid
C22
RST
mode_start
C26
JMP Transition From This Stage
Transition To S1
SG Stage
Stage S1
$On
ST1
homing_active
C20
SET
CALL Call Subroutine
Subroutine Z_Hom
z_home_dn
C113
CALL Call Subroutine
Subroutine X1_Hom
CALL Call Subroutine
Subroutine X2_Hom
CALL Call Subroutine
Subroutine Y_Hom
Local:S0:iPT8
x1_sw
X8
Local S0:iPT9
x2_sw
X9
Local S0:iPT10
y_sw
X10
Local S0:iPT11
z_sw
X11
JMP Transition From This Stage
Transition To S2
SG Stage
Stage S2
$On
ST1
Local S0:oPT2
white
Y2
OUT
home_stg_dn
C27
OUT
home_stg_dn
C27
C546
Local S0:oPT3
green
Y3
OUT
NOP
C676
EXIT Exit This Program
NOP

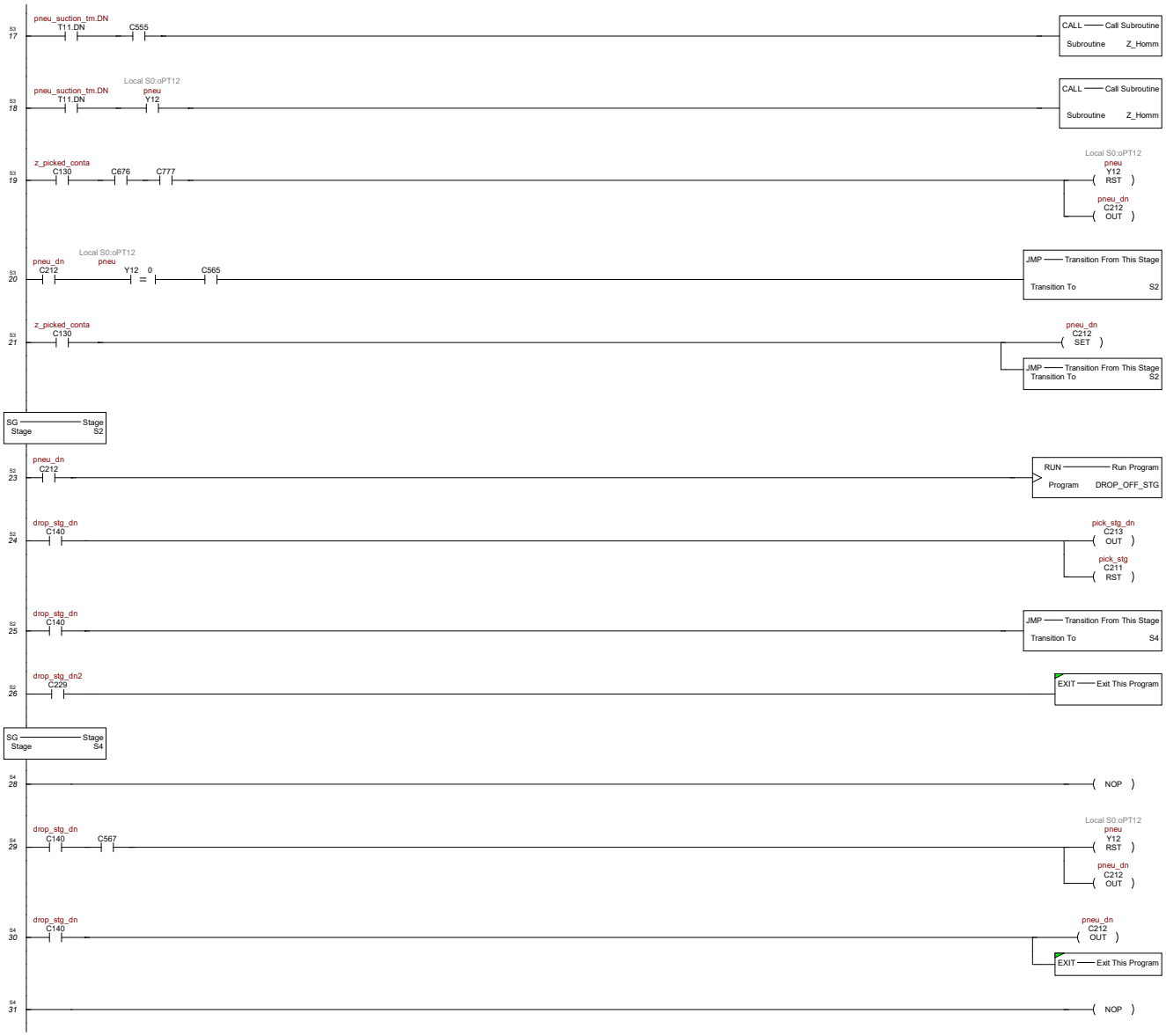

$On
ST1
recv_stggg
C38
OUT
$On
ST1
CNT
Up Counter
Counter1
Counter Struct
CT0
UP
server_ready
C203
Preset
0
RST
SG
Stage
Stage
S0
dropped_count
D172
0
C905
recv_stggg
C38
RST
recv_stggg
C38
JMP
Transition From This Stage
Transition To
S1
SG
Stage
Stage
S1
true_calib_stg_d
C252
MOTION_BUSY
C217
fault
C6
pi_hb_alive
C175
plc_ready
C201
SET
dropoff_done
C215
readyyy
C228
SET
server_ready
C203
1
demo_ready
C204
RST
server_ready
C203
ext_node_conectd
C208
OUT
demo_ready
C204
ready_to_move
C207
OUT
dav_ready
C205
SERVER_READY_REG
D214
1
ext_node_conectd
C208
OUT
start_stggg
C17
RST
ext_node_conectd
C208
server_snd_move
C206
OUT
start_stggg
C17
RST
ready_to_move
C207
MWX
Modbus Network Write
Device
@Gantry_Mod_Da
IP Address
192.168.3.6
TCP Port Number
502
Unit ID
255
Function Code
5 - Write Single Coil
Modbus Address 0... + offset value entered below
To Modbus Offset Address
1
From Do-more Memory Address
plc_ready
On Success, Set bit
C620
On Error, Set bit
C621
samp_data
C209
OUT
server_snd_move
C206
C905
X980
COPY
Copy Data
Source Destination Count
1 GntYAxisMmArr[V150] y_pickup_loc_mm 1
samp_data
C209
C905
X980
COPY
Copy Data
Source Destination Count
1 69 x_pickup_loc_mm 1
2 150 y_pickup_loc_mm 1
3 1200 x_dropof_loc_mm 1
4 150 y_dropof_loc_mm 1
RUN
Run Program
Program
PICK_UP_STG
server_snd_move
C206
C989
X980
COPY
Copy Data
Source Destination Count
1 MHR50 pickup_x_row 1
2 MHR51 pickup_y_row 1
3 MHR52 pickup_z_row 1
4 MHR53 dropoff_x_row 1
5 MHR54 dropoff_y_row 1
6 MHR55 dropoff_z_row 1
RUN
Run Program
Program
PICK_UP_STG
dropped_count
D172
0
CALL
Call Subroutine
Subroutine
Reset_SR
server_snd_move
C206
pi_hb_alive
C175
CMD_ID_REG
D211
LAST_CMD_ID
D218
MOTION_BUSY
C217
fault
C6
RUN
Run Program
Program
PICK_UP_STG
C567
JMP
Transition From This Stage
Transition To
S6

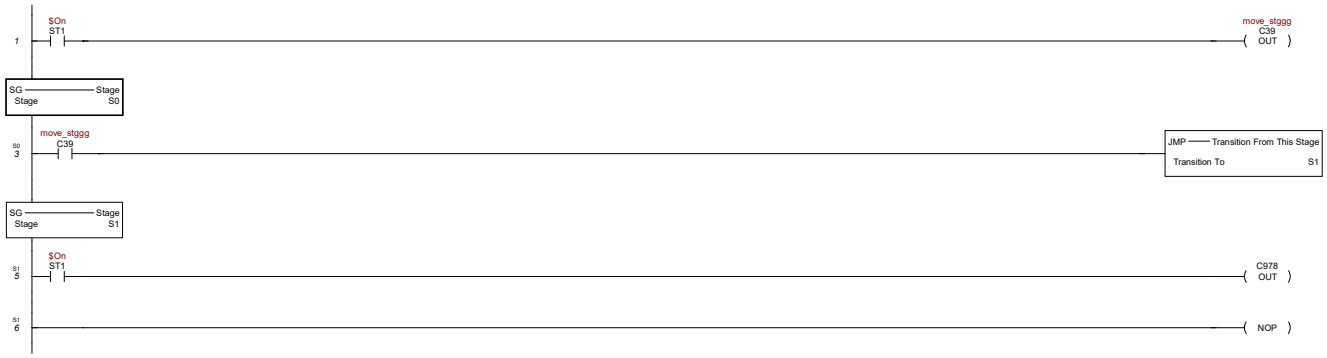

S1
19
pick_stg_dn
C213
JMP Transition From This Stage
Transition To S2
SG Stage
Stage S2
S2
21
$On
ST1
MATH Calculate Expression
dropped_count
Result D172
Expression Counter1.Acc + 1
S2
22
dropped_count
D172 > 0
pick_stg_dn
C213
RST
JMP Transition From This Stage
Transition To S3
SG Stage
Stage S3
S3
24
$On
ST1
JMP Transition From This Stage
Transition To S0
S3
25
container_droped
C141
X989
JMP Transition From This Stage
Transition To S0
S3
26
NOP

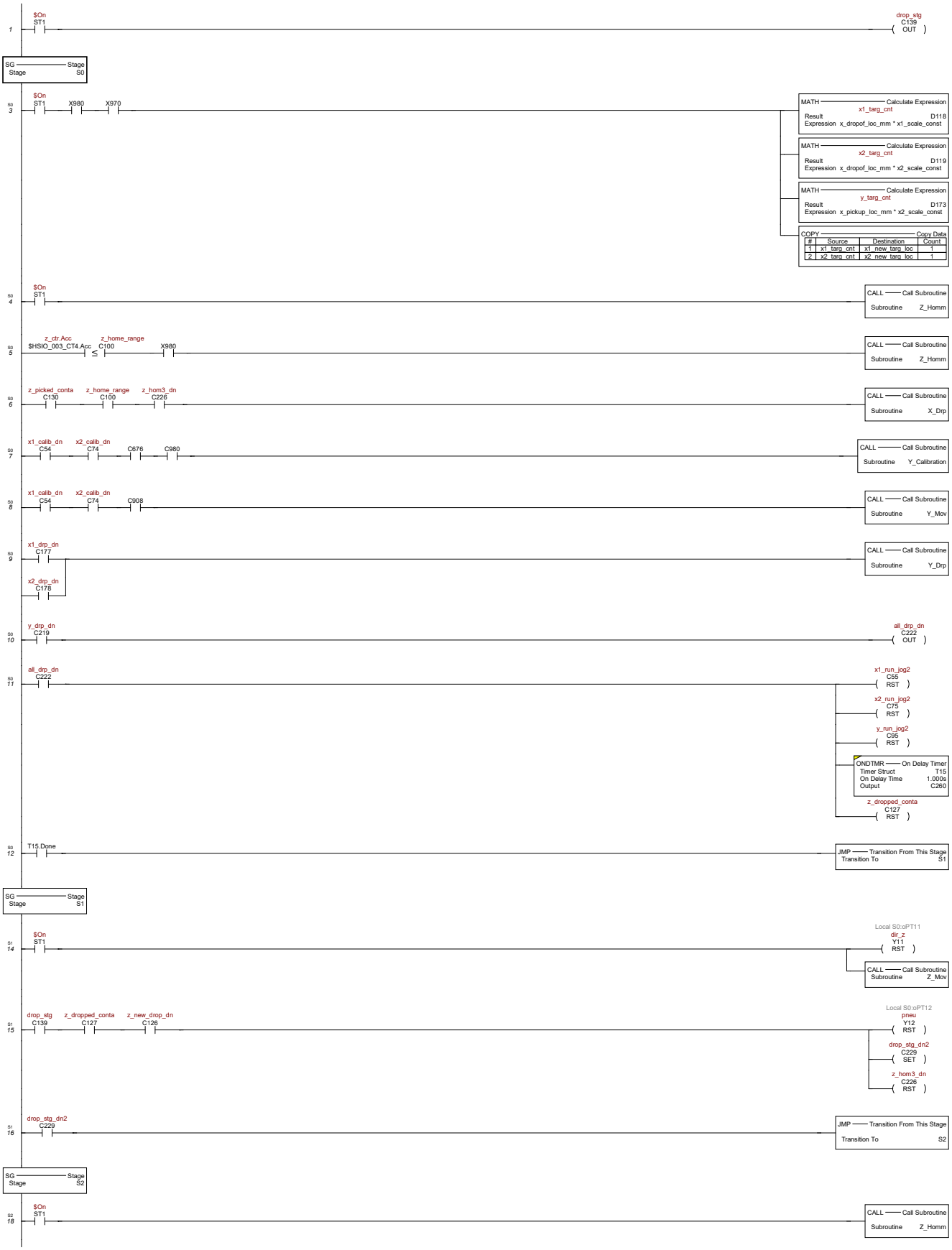

SG Stage
Stage S0
$On
ST1
true_calib_stggg
C250
SET
true_calib_stggg
C250
JMP Transition From This Stage
Transition To S1
SG Stage
Stage S1
$On
ST1
CALL Call Subroutine
Subroutine Z_Hom
z_home_dn
C113
CALL Call Subroutine
Subroutine X1_Hom
CALL Call Subroutine
Subroutine X2_Hom
CALL Call Subroutine
Subroutine Y_Hom
z_home_dn C113
y_home_dn C93
x1_home_dn C53
x2_home_dn C73
JMP Transition From This Stage
Transition To S2
SG Stage
Stage S2
y_home_dn C93
true_calib_stggg C250
ONDTMR On Delay Timer
Timer Struct T4
On Delay Time 1.000s
Output C256
y_home_dn C93
true_calib_stggg C250
x1_home_dn C53
C667
C960
CALL Call Subroutine
Subroutine Z_Calibration
y_home_dn C93
x1_home_dn C53
true_calib_stggg C250
CALL Call Subroutine
Subroutine Z_Calibration
z_calib_dn C114
true_calib_stggg C250
y_home_dn C93
x1_home_dn C53
x2_home_dn C73
CALL Call Subroutine
Subroutine Y_Calibration
y_calib_dn C94
true_calib_stggg C250
x1_home_dn C53
x2_home_dn C73
CALL Call Subroutine
Subroutine X_Calibration
x1_calib_dn C54
x2_calib_dn C74
y_calib_dn C94
all_calib_dnn
C251
OUT
all_calib_dnn
C251
JMP Transition From This Stage
Transition To S3
SG Stage
Stage S3
$On
ST1
CALL Call Subroutine
Subroutine Z_Homm
z_hom2_dn
C118
CALL Call Subroutine
Subroutine Y_Homm
Local S0:oPT8
dir_x1
Y8
RST
Local S0:oPT9
dir_x2
Y9
RST
z_hom2_dn
C118
CALL Call Subroutine
Subroutine X1_Homm
CALL Call Subroutine
Subroutine X2_Homm
true_calib_stggg C250
y_hom2_dn C98
x1_hom2_dn C58
x2_hom2_dn C78
z_hom2_dn C118
JMP Transition From This Stage
Transition To S4
SG Stage
Stage S4
$On
ST1
true_calib_stg_d
C252
SET

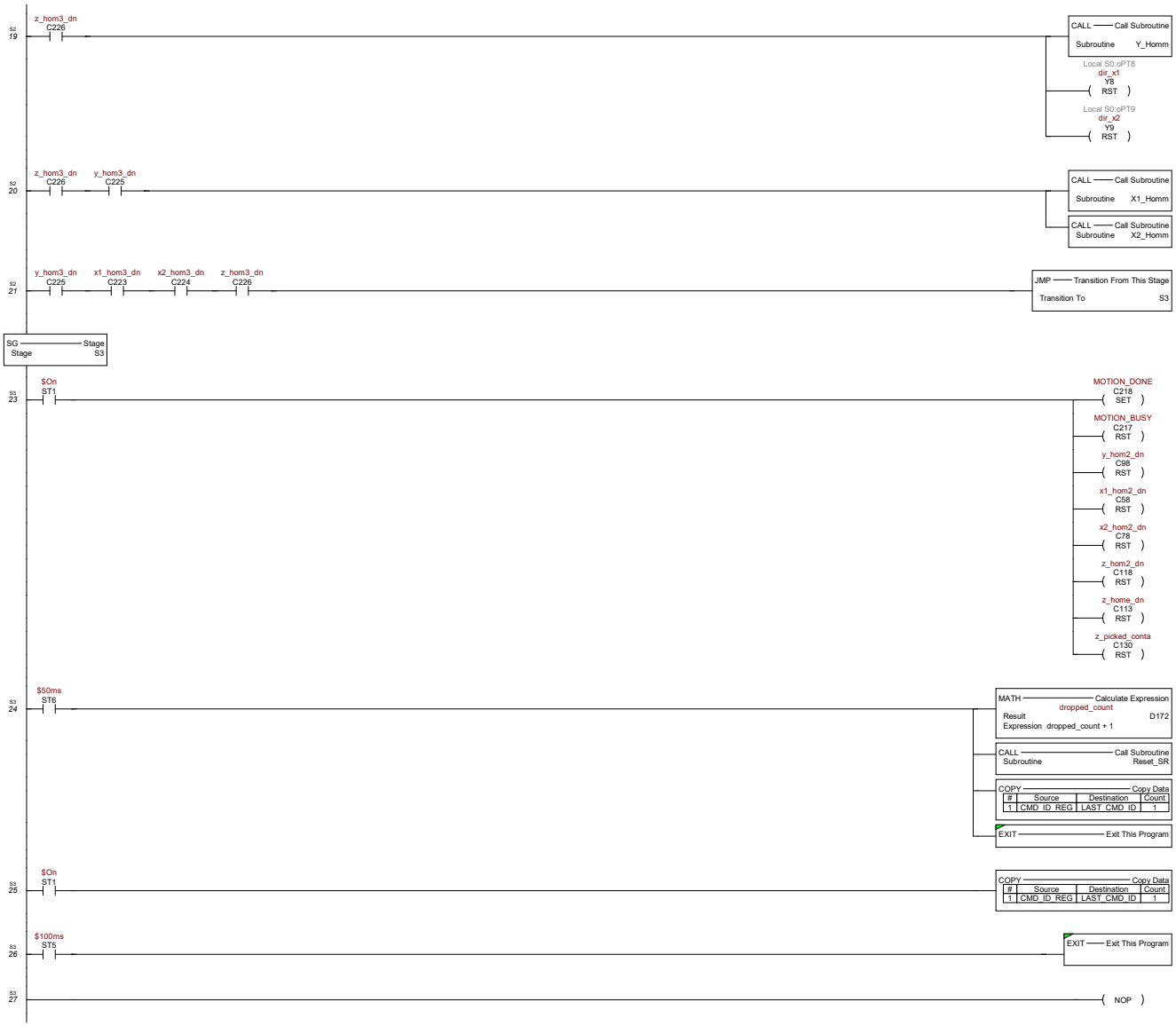

$On
ST1
EXIT Exit This Program
NOP


1
$On
ST1
pick_stg
C211
OUT
MOTION_BUSY
C217
SET
SG
Stage
Stage
S0
S0
3
pick_stg
C211
JMP
Transition From This Stage
Transition To
S1
SG
Stage
Stage
S1
S1
5
$On
ST1
C900
C906
MATH
Calculate Expression
x1_targ_cnt
Result
D118
Expression
x_pickup_loc_mm * x1_scale_const
MATH
Calculate Expression
x2_targ_cnt
Result
D119
Expression
x_pickup_loc_mm * x2_scale_const
MATH
Calculate Expression
y_targ_cnt
Result
D173
Expression
x_pickup_loc_mm * x2_scale_const
COPY
Copy Data
Source Destination Count
1 x1_targ_cnt x1_new_targ_loc 1
2 x2_targ_cnt x2_new_targ_loc 1
S1
6
$On
ST1
CMD_ID_REG
D211
LAST_CMD_ID
D218
fault
C6
COPY
Copy Data
Source Destination Count
1 GntXOneCntArr[pickup_x_row] x1_new_targ_loc 1
2 GntXTwoCntArr[pickup_x_row] x2_new_targ_loc 1
3 GntYAxisMmArr[pickup_y_row] y_new_targ_loc 1
4 GntXOneCntArr[pickup_x_row] pickup_X1_cnt 1
5 GntXTwoCntArr[pickup_x_row] pickup_X2_cnt 1
6 GntXOneCntArr[dropoff_x_row] dropoff_X1_cnt 1
7 GntXTwoCntArr[dropoff_x_row] dropoff_X2_cnt 1
8 GntYAxisMmArr[pickup_y_row] pickup_Y_cnt 1
9 GntYAxisMmArr[dropoff_y_row] dropoff_Y_cnt 1
MOTION_BUSY
C217
SET
MOTION_DONE
C218
RST
plc_ready
C201
RST
pick_stg_dn
C213
RST
S1
7
Local:S0:iPT11
$On
ST1
z_sw
X11
CALL
Call Subroutine
Subroutine
Z_Homm
S1
8
z_home_dn
C113
CALL
Call Subroutine
Subroutine
X_Mov
S1
9
x1_new_loc_dn
C170
x2_new_loc_dn
C171
CALL
Call Subroutine
Subroutine
Y_Mov
S1
10
y_new_loc_dn
C172
CALL
Call Subroutine
Subroutine
Z_Mov
S1
11
z_new_loc_dn
C173
all_new_loc_dn
C174
SET
JMP
Transition From This Stage
Transition To
S3
S1
12
all_new_loc_dn
C174
C455
z_new_loc_dn
C173
RST
JMP
Transition From This Stage
Transition To
S3
SG
Stage
Stage
S3
S3
14
all_new_loc_dn
C174
C676
OUT
S3
15
all_new_loc_dn
C174
Local S0:oPT12
pneu
Y12
SET
S3
16
Local S0:oPT12
pneu
Y12
ONDTMR
On Delay Timer
pneu_suction_tm
Timer Struct
T11
On Delay Time
5.000s
Output
pneu_tim_acc


S3 17
pneu_suction_tm.DN
T11.DN
C555
CALL Call Subroutine
Subroutine Z_Homm
S3 18
pneu_suction_tm.DN
T11.DN
Local S0:oPT12
pneu
Y12
CALL Call Subroutine
Subroutine Z_Homm
S3 19
z_picked_conta
C130
C676
C777
Local S0:oPT12
pneu
Y12
RST
pneu_dn
C212
OUT
S3 20
pneu_dn
C212
Local S0:oPT12
pneu
Y12 = 0
C565
JMP Transition From This Stage
Transition To S2
S3 21
z_picked_conta
C130
pneu_dn
C212
SET
JMP Transition From This Stage
Transition To S2
SG Stage
Stage S2
S2 23
pneu_dn
C212
RUN Run Program
Program DROP_OFF_STG
S2 24
drop_stg_dn
C140
pick_stg_dn
C213
OUT
pick_stg
C211
RST
S2 25
drop_stg_dn
C140
JMP Transition From This Stage
Transition To S4
S2 26
drop_stg_dn2
C229
EXIT Exit This Program
SG Stage
Stage S4
S4 28
NOP
S4 29
drop_stg_dn
C140
C567
Local S0:oPT12
pneu
Y12
RST
pneu_dn
C212
OUT
S4 30
drop_stg_dn
C140
pneu_dn
C212
OUT
EXIT Exit This Program
S4 31
NOP


$On
ST1
1
move_stggg
C39
OUT
SG
Stage
Stage
S0
move_stggg
C39
S0
3
JMP
Transition From This Stage
Transition To
S1
SG
Stage
Stage
S1
$On
ST1
S1
5
C978
OUT
S1
6
NOP


1
$On
ST1
drop_stg
C139
OUT
SG
Stage
Stage
S0
S0
3
$On
ST1
X980
X970
MATH
Calculate Expression
x1_targ_cnt
Result
D118
Expression
x_dropof_loc_mm * x1_scale_const
MATH
Calculate Expression
x2_targ_cnt
Result
D119
Expression
x_dropof_loc_mm * x2_scale_const
MATH
Calculate Expression
y_targ_cnt
Result
D173
Expression
x_pickup_loc_mm * x2_scale_const
COPY
Copy Data
Source Destination Count
1 x1_targ_cnt x1_new_targ_loc 1
2 x2_targ_cnt x2_new_targ_loc 1
S0
4
$On
ST1
CALL
Call Subroutine
Subroutine
Z_Homm
S0
5
z_ctr.Acc
$HSIO_003_CT4.Acc
≤
z_home_range
C100
X980
CALL
Call Subroutine
Subroutine
Z_Homm
S0
6
z_picked_conta
C130
z_home_range
C100
z_hom3_dn
C226
CALL
Call Subroutine
Subroutine
X_Drp
S0
7
x1_calib_dn
C54
x2_calib_dn
C74
C676
C980
CALL
Call Subroutine
Subroutine
Y_Calibration
S0
8
x1_calib_dn
C54
x2_calib_dn
C74
C908
CALL
Call Subroutine
Subroutine
Y_Mov
S0
9
x1_drp_dn
C177
x2_drp_dn
C178
CALL
Call Subroutine
Subroutine
Y_Drp
S0
10
y_drp_dn
C219
all_drp_dn
C222
OUT
S0
11
all_drp_dn
C222
x1_run_jog2
C55
RST
x2_run_jog2
C75
RST
y_run_jog2
C95
RST
ONDTMR
On Delay Timer
Timer Struct
T15
On Delay Time
1.000s
Output
C260
z_dropped_conta
C127
RST
S0
12
T15.Done
JMP
Transition From This Stage
Transition To
S1
SG
Stage
Stage
S1
S1
14
$On
ST1
Local S0:oPT11
dir_z
Y11
RST
CALL
Call Subroutine
Subroutine
Z_Mov
S1
15
drop_stg
C139
z_dropped_conta
C127
z_new_drop_dn
C126
Local S0:oPT12
pneu
Y12
RST
drop_stg_dn2
C229
SET
z_hom3_dn
C226
RST
S1
16
drop_stg_dn2
C229
JMP
Transition From This Stage
Transition To
S2
SG
Stage
Stage
S2
S2
18
$On
ST1
CALL
Call Subroutine
Subroutine
Z_Homm


S2 19
z_hom3_dn C226
CALL Call Subroutine
Subroutine Y_Homm
Local S0:o.PT8
dir_x1
Y8
RST
Local S0:o.PT9
dir_x2
Y9
RST
S2 20
z_hom3_dn C226
y_hom3_dn C225
CALL Call Subroutine
Subroutine X1_Homm
CALL Call Subroutine
Subroutine X2_Homm
S2 21
y_hom3_dn C225
x1_hom3_dn C223
x2_hom3_dn C224
z_hom3_dn C226
JMP Transition From This Stage
Transition To S3
SG Stage
Stage S3
S3 23
$On
ST1
MOTION_DONE C218 SET
MOTION_BUSY C217 RST
y_hom2_dn C98 RST
x1_hom2_dn C58 RST
x2_hom2_dn C78 RST
z_hom2_dn C118 RST
z_home_dn C113 RST
z_picked_conta C130 RST
S3 24
$50ms
ST6
MATH Calculate Expression
dropped_count
Result D172
Expression dropped_count + 1
CALL Call Subroutine
Subroutine Reset_SR
COPY Copy Data
Source Destination Count
1 CMD_ID_REG LAST_CMD_ID 1
EXIT Exit This Program
S3 25
$On
ST1
COPY Copy Data
Source Destination Count
1 CMD_ID_REG LAST_CMD_ID 1
S3 26
$100ms
ST5
EXIT Exit This Program
S3 27
NOP